

DEMU2

Infinite Regress and the Possibility of New Energy

A philosophical investigation of explanatory closure and entropy reset

Thor Fabian Pettersen

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Can a global solution to infinite regress give humanity a new source of energy?

Note on the manuscript

This first complete manuscript develops my DEMU framework from my earlier writings and the philosophical discussion that followed. Drafting and editing were supported by OpenAI's ChatGPT, including assistance with organization, prose, source checking, diagrams and document preparation. This edition brings the complete argument together for review and further development.

DEMU2 is an independent philosophical book project. No university affiliation, degree award or institutional endorsement is claimed.

Epigraph

Time began from light's point of view.

Abstract

This paper investigates whether a global solution to infinite regress could provide a basis for a new source of energy.

I propose that motionlessness and motion are aspects of one fundamental reality. Reciprocal expansion and contraction preserve its timeless balance while allowing the activity experienced as a universe. That balance must hold throughout the entire circulation, including everything returning to the foundation. I therefore argue that accumulated entropy must reset.

In this model, the timeless core continually renews portions of the matter and energy that constitute our universe. This ongoing renewal raises the possibility of identifying outlets of the process and harnessing their renewed output as a continually replenished energy source for humanity.

The wider argument concerns what consciousness is, why only consciousness can close Nature, the necessity of a spiritual universe, who God is, free will, and the true meaning of life. These are further claims I aim to defend within the same philosophical framework.

Keywords: infinite regress; explanatory closure; motion; timelessness; entropy reset; usable energy; consciousness.

Contents

Note on the manuscript.....	ii
Epigraph.....	iii
Abstract.....	iv
List of figures.....	x
Chapter 1 Introduction.....	1
1.1 The problem and its motivation.....	1
1.2 Scope and intended contribution.....	2
1.3 Research questions.....	2
1.4 The route through the argument.....	5
Chapter 2 Definitions and Fundamental Concepts.....	11
2.1 What understanding must accomplish.....	11
2.2 Explanation and dependence.....	12
2.3 Infinite regress and explanatory closure.....	13
2.4 Existence, Nature, and nonexistence.....	14
2.5 Motion, motionlessness, and activity.....	15
2.6 Time, timelessness, and the origin of motion.....	16
2.7 Balance, reciprocity, and geometric representation.....	18
2.8 Possibility, necessity, and contradiction.....	20
2.9 Systems, states, and information.....	21
2.10 Entropy, order, and equilibrium.....	23
2.11 Reset, return, and renewed matter.....	24
2.12 Energy, work, and the meaning of free.....	25
2.13 Consciousness, life, and the wider vocabulary.....	26
2.14 How these definitions guide the argument.....	28
References for Chapter 2.....	29
Chapter 3 Time Began from Light's Point of View.....	30
3.1 The Train to Nowhere.....	30
Chapter 4 The Origin of Motion.....	31
4.1 The Power of Reason.....	31
4.2 The next task.....	33
4.3 What the Argument Establishes.....	34
4.4 How fast is the flash?.....	37
4.5 Section title to be added.....	39
4.6 What started motion?.....	42
Chapter 5 The Unmoved Mover.....	46
5.1 What Aristotle was asking.....	46

5.2 What Aristotle means by motion and activity.....	46
5.3 Why motion does not need a first moment.....	47
5.4 Why always existing may not be enough.....	48
5.5 How something can cause motion without changing.....	50
5.6 Thought thinking itself.....	50
5.7 Is the mover separate from the universe.....	52
5.8 Where my proposal differs.....	53
5.9 What my proposal would show.....	54
5.10 Questions the proposal still needs to answer.....	56
5.11 Towards a complete explanation.....	57
5.12 The naked lady.....	57
References for Chapter 5.....	58
Chapter 6 A Global Solution to Infinite Regress.....	61
6.1 The way of Idealism.....	61
Chapter 7 Temporal Activity Within a Timeless Foundation.....	64
7.1 The next question.....	64
Chapter 8 The Emergence of High Entropy.....	67
8.1 The next problem.....	67
8.2 What high entropy means.....	67
8.3 Why energy can become less useful.....	68
8.4 Time allows a history to form.....	69
8.5 Why the direction matters.....	69
8.6 How order can exist while entropy rises.....	70
8.7 Why the core is a different condition.....	71
8.8 Why high entropy cannot be the final word.....	71
References for Chapter 8.....	72
Chapter 9 Balance Throughout the Eternal Circulation.....	73
9.1 Balance cannot be left behind.....	73
9.2 What remains the same while things change.....	73
9.3 Expansion and contraction belong together.....	74
9.4 Renewal occurs within an ongoing universe.....	75
9.5 Balance must include everything that returns.....	76
9.6 Circulation must include renewed availability.....	77
9.7 What eternal balance requires.....	77
References for Chapter 9.....	78
Chapter 10 Entropy and Infinite Regress.....	79
10.1 The Case for Entropy Reset.....	79
10.2 The Challenge of Explained Retention.....	82
References for Chapter 10.....	85

Chapter 11 Entropy Reset and the Renewal of Matter and Energy.....	86
11.1 What renewal means.....	86
11.2 Contraction and the loss of inherited form.....	86
11.3 The proposed single state at the core.....	87
11.4 Total erasure and ordinary physical processes.....	88
11.5 How renewed activity can arise.....	89
11.6 Renewed energy means renewed availability.....	90
11.7 Renewal within an eternal universe.....	91
References for Chapter 11.....	92
Chapter 12 The Meaning of Free and Unlimited Energy.....	93
12.1 What I mean by free.....	93
12.2 Unlimited renewal and finite power.....	94
12.3 Why usable energy must be renewed.....	95
12.4 Collecting output within the circulation.....	96
12.5 What remains after collection.....	96
12.6 The whole system and the laws of thermodynamics.....	97
12.7 What follows from the argument.....	98
References for Chapter 12.....	99
Chapter 13 Rival Accounts of Reality.....	100
13.1 Nonexistence as a Fabrication.....	100
13.2 Brute Facts and Explanatory Closure.....	102
13.3 Static Systems and Infinite Regress.....	103
13.4 Other Accounts of Closure.....	105
References for Chapter 13.....	108
Chapter 14 The Purpose of the Universe.....	109
14.1 The Purpose.....	109
14.2 What I Mean by Purpose.....	110
14.3 Why Balance Must Be Active.....	111
14.4 A Purpose Without a Final Stopping Point.....	112
14.5 Renewal and the Formation of New Structures.....	113
14.6 Conscious Participation and Human Purpose.....	114
14.7 The Role of Purpose in DEMU.....	115
References for Chapter 14.....	116
Chapter 15 Cosmic Sex and Atomic Replication.....	117
15.1 What would count as replication.....	117
15.2 Giving the circulation a definite form.....	117
15.3 Folding and the formation of a tetrahedron.....	119
15.4 How opposite tetrahedral patterns can combine.....	120
15.5 One proposed production cycle.....	121
15.6 Testing whether the pattern remains.....	122

15.7 Accounting for the activity that supports the child.....	123
15.8 What would make the new structure an atom.....	124
15.9 What this model establishes.....	125
Appendix A Geometry and the prescribed flow.....	126
A1 The circulating background.....	126
A2 The four directions.....	127
A3 Why four repelling directions form a tetrahedron.....	127
A4 The quadratic interaction.....	128
Appendix B Stability and energy calculations.....	128
B1 The amplitude rule.....	128
B2 Energy required by the successful branch.....	129
B3 Numerical method.....	129
References.....	130
Chapter 16 The Four Forces and the Development of Nature.....	131
16.1 The Four Forces.....	131
16.2 The Formation of Earth.....	134
16.3 The Size of Dinosaurs.....	136
References for Chapter 16.....	139
Chapter 17 Immortality and the Nature of Existence.....	140
17.1 The Power of Power.....	140
17.2 The Whole and the Individual.....	141
17.3 Consciousness and Personal Identity.....	142
17.4 Life as Ongoing Repair.....	143
17.5 Low-Entropy Matter and Longevity.....	144
17.6 Digestion and Deeper Order.....	145
17.7 Renewal Without Erasing the Person.....	146
17.8 What an Open-Ended Life Would Mean.....	148
References for Chapter 17.....	149
Chapter 18 Consciousness and the Closure of Nature.....	150
18.1 The Nature of Consciousness.....	150
18.2 Why Only Consciousness Can Close Nature.....	151
18.3 Awareness Without Another Observer.....	152
18.4 One Reality and Its Expressions.....	154
18.5 Consciousness and the Dual Torus.....	155
18.6 The Brain and Individual Experience.....	156
18.7 Many Minds Within One Nature.....	157
18.8 Consciousness as the Activity of the Foundation.....	159
References for Chapter 18.....	160
Chapter 19 God and Free Will in a Spiritual Universe.....	161
19.1 The Necessity of a Spiritual Universe.....	161

19.2 Who God Is.....	163
19.3 Free Will.....	165
References for Chapter 19.....	170
Chapter 20 The True Meaning of Life.....	171
20.1 The Meaning.....	171
20.2 From Balance to Love.....	172
20.3 One Purpose and Many Lives.....	173
20.4 Freedom and Spiritual Growth.....	174
20.5 Suffering and the Work of Repair.....	175
20.6 Eternity and the Value of Each Life.....	176
20.7 Living the Meaning of Life.....	177
References for Chapter 20.....	178
Chapter 21 A Dialogue with the Skeptic.....	179
21.1 Battling a straw man.....	179
Chapter 22 Human Access to Renewed Energy.....	191
22.1 Harnessing the Renewed Output.....	191
22.2 Whether We Should Harness It.....	195
22.3 Implications for Humanity.....	196
References for Chapter 22.....	199
Chapter 23 The Circle of Life.....	200
23.1 The Full Circle.....	200
Chapter 24 Consciousness as Nature’s Replicator.....	204
24.1 What Does It Mean to Replicate.....	204
24.2 Genes and the Deeper Ground of Life.....	207
24.3 How One Reality Can Contain Many Minds.....	211
24.4 Cosmic Sex and the Formation of Individuals.....	215
24.5 What Continues Through Change.....	218
24.6 Total Reset and Personal Identity.....	222
24.7 Why Consciousness Matters to Renewal.....	227
24.8 What the Argument Establishes.....	230
References for Chapter 24.....	234
Chapter 25 The Conclusion.....	235
25.1 The Full Circle.....	235

List of figures

1.1 Overview of the proposed argument.....	5
2.1 Temporal order and explanatory dependence.....	14
2.2 One reality and its two aspects.....	18
2.3 Geometric forms used in the model.....	20
2.4 Total erasure and historical independence.....	22
3.1 The train and the experience of motion.....	30
4.1 The conditional argument for timelessness.....	33
5.1 Aristotle and the proposed DEMU account.....	54
7.1 Complementary circulation and the shared core.....	65
7.2 White light and temporal experience.....	66
8.1 Heat exchange and usable energy.....	68
9.1 Continual circulation within an ongoing universe.....	75
11.1 Return, reset, and renewed forms.....	90
12.1 Finite power and continuing supply.....	94
15.1 A section through the prescribed flow.....	118
15.2 Landmark configurations of the articulated framework.....	119
15.3 Complementary directions and the cuboctahedron.....	120
15.4 Pattern persistence after release.....	123
15.5 Energy accounting in the illustrative model.....	124
16.1 Proposed geometrical connections to the forces.....	133
16.2 Planetary radius and assumptions about gravity.....	137
17.1 Personal continuity and total reset.....	147
18.1 Balance and consciousness.....	155
22.1 Human access to renewed output.....	193
23.1 The proposed circle of formation and renewal.....	203
24.1 Many local perspectives within one reality.....	212
24.2 The proposed formation of a further individual.....	217
24.3 Continuation, replication, and total personal reset.....	226
24.4 How the replicator argument connects.....	233

CHAPTER 1

Introduction

1.1 The problem and its motivation

The problem: What exactly needs explaining?

What needs explaining is the whole chain that leads to the new energy source. So I need to explain why there is existence rather than nonexistence, what set motion going, how the initial state of existence can birth a universe, and how all of this resolves the problem of infinite regress.

In a nutshell: I want to understand why Nature exists and why it has the form it has. If an explanation depends on something else, I want to understand that dependence too. My aim is to reach an account whose finality can itself be understood.

“What do you currently think resolves that difficulty?”

What I think resolves this difficulty is my insight that motionlessness = all motion. My claim is that, from this principle, Nature is so constrained that the rest follows logically. The task of this work is to explain those constraints and defend the steps that follow from them.

“What is the most important argument supporting your answer?”

My most important starting point is that Nature cannot be paradoxical. I argue that an unresolved infinite regress would leave a paradox at the foundation of reality, and that it must therefore be solved. I will need to explain why the regress is a genuine problem and why my proposed foundation closes it.

“What would a thoughtful critic dispute? What would you need to establish to answer them?”

A thoughtful critic would have to understand my work and engage with the argument I actually make. So I need to establish simple and clear language that makes a notoriously difficult subject easy to grasp. But a reader can understand my proposal and still disagree with it. A central objection would be: Why must a completely explained reality reset entropy

rather than completely explain its retention? That is a disagreement I must answer through argument.

1.2 Scope and intended contribution

“If your argument succeeds, what would we understand that we did not understand before?”

If I succeed, we would have an answer to why Nature is here, what its purpose and job are, and how motion belongs to its foundation. We would also have an answer to infinite regress and the Unmoved Mover. I aim to solve both problems. And I aim to resolve the question: Why is there existence rather than nonexistence?

The philosophy also has a practical side. I intend to argue that it opens a path towards unlimited energy and immortality, and explains how the four forces in Nature are interconnected. That might help physics unify Nature and finally create a Theory of Everything that sticks. I will also develop my proposed explanations of the genesis of the Earth, why the dinosaurs grew so tall, and how atomic replication could belong to the same model.

I also want to argue for the true meaning of life, explain why there must be a spiritual universe and who God is, and address free will. Central to this is the question of what consciousness actually is. I intend to argue that only consciousness can close Nature: only consciousness can provide the foundation whose existence and activity require no further unexplained support. I addressed these questions in my previous DEMU, and I have decided to develop them in this work too.

These are all intended parts of the work, and each will need its own argument. I will show how I think they connect to the foundation, distinguish what I claim must follow from what I claim is possible, and address the strongest objections. The energy question gives us a starting point for a wider investigation of Nature as a whole.

1.3 Research questions

The argument about origins and energy will center around three questions:

“What would count as a global solution to infinite regress, and does my proposed foundation satisfy those conditions?”

I hope to answer this to a satisfactory degree.

“Does that solution imply an entropy reset—and would such a reset renew the capacity to do work?”

My claim is yes, but I need to convince the skeptic of this. So in this work, we will have an extended section where I answer the skeptic in a back-and-forth dialogue.

“What further conditions would allow us to draw usable energy from that process?”

This will be the most difficult part for me because I’m a philosopher, not Nikola Tesla. But I will try to point humanity in the right direction here, even if I can’t provide a blueprint of a working energy device.

When I say I’m a philosopher, I mean it in the same way that I say I’m a writer if I write books, not that I have a PhD in philosophy because I don’t. I use the word to identify the kind of argument I am making. A philosophical proposal should be assessed through its definitions, premises, and reasoning. Where I make claims about physical processes, the meaning of the physical concepts and the relevant evidence also matter.

Also, we should preserve the word “can.” Establishing that something is logically possible, showing that it necessarily follows from our premises, and demonstrating that it occurs in Nature are different research achievements. For the proposed energy source, I aim to establish the philosophical possibility and defend the consequences I claim follow from the foundation. Demonstrating an actual energy device would be a further achievement.

Our next step is to define “a global solution to infinite regress” because that expression will determine what the rest of the project must accomplish.

By a global solution to infinite regress, I mean:

By infinite regress, I don’t mean anything mysterious; I just mean our ability to ask fundamental questions. For example, I have a coffee cup sitting on my table. I can ask: Where did it come from? It came from a factory. Then I can continue: Where did the factory come from? And I can continue this way and ask forever unless I get a final answer that is final and I can see and understand that it is final and must be final. That is what I mean by a

global solution. Thus, I do not accept brute facts as a candidate for a final answer because brute facts do not give me understanding. I want to understand Nature completely and wholly. That is what I mean by global. I could also use the word “universal,” but I think both words can do the same job.

In other words, I must explain why no further foundation is required. An unexplained stopping point would not satisfy my aim.

A skeptic might immediately say: “Someone can always utter another ‘why?’”

I disagree here. By complete closure, I mean that no fundamental dependence remains unexplained. I want an account in which a further “why?” can receive a satisfactory answer. And I want to explain both how and why. I am not satisfied until I understand why existence is here, how it works, and whether asking when it started presupposes a beginning that it never had.

Note that explaining why reality exists and has its fundamental structure is different from explaining every particular occurrence—including exactly why this cup was, say, manufactured on Tuesday. My phrase “completely and wholly” could be read as demanding both. But that is not what I mean, of course. I mean understanding existence itself as a global thing.

This gives us a roadmap. First, we must distinguish possibility from necessity. My overall question asks whether such an energy source can exist. But my second question asks whether entropy reset follows from my foundation. That gives us two tasks:

- Show that the proposed foundation and its consequences form a coherent possibility.
- Show that the reset follows from the stated premises, because I claim that it must.

Further, if I use entropy, energy, and work in their physical senses, a critic can legitimately question whether my argument respects those meanings. So, before I start, I will give definitions of every concept used in this work.

Figure 1.1 brings the three research questions together in an overview of the proposed argument.

1 THE FOUNDATION



Motion and motionlessness as one reality
Reciprocal expansion and contraction
Timeless balance and explanatory closure

2 THE CONSEQUENCE



Balance throughout the entire circulation
Proposed entropy reset
Continual renewal of matter and energy

3 HUMAN ACCESS

Identify outlets of the renewal process
Harness the renewed output
Possibility of a continually replenished energy source

Figure 1.1. Overview of the proposed argument: from explanatory closure to continual renewal and the possibility of an accessible energy source. Arrows mark argumentative steps, not a temporal sequence within the timeless foundation.

1.4 The route through the argument

I will now give you the whole work in miniature form so you can see for yourself the route we are going to take. This is an outline of the position I will defend. The chapters that follow will develop its arguments, examine its assumptions, and answer objections.

I started this work from an insight I had when I was in the military back in 2006, when I was twenty-one years old. Now I'm forty-one, but I felt like I had already solved the question of origins even back then. Now it has evolved into a mature version.

My original insight was this: *Time began from light's point of view.*

I sat on a beam of light in my mind's eye, and I saw an ocean of nothingness. Then I knew that this was the beginning of time, the starting point of existence itself. But the deeper insight was this: It was motion, the speed of light, that brought me into that state of absolute motionlessness. So then I formulated this equation: motionlessness = motion. I will explain

what that means in this paper, I will define every term, and it will all become clear. But let us keep it short for the time being.

When I had this equation, I saw in it a possible answer to the Unmoved Mover: the thing that doesn't move but can make something else move. And with it, I believed I could answer the problem of infinite regress. That became the position I wanted to explain and defend.

My proposed solution was: No time = no regress.

My argument begins with a difficulty: if every state is explained only by an earlier state, the demand for explanation continues backwards. Try to finish the explanation while keeping that requirement:

Let us try:

"This state came from an earlier state."

"What explains that earlier state?"

"Another earlier state."

If that is the only permitted kind of explanation, the chain cannot finish.

One possible reply is: "Time has a first state, with no earlier cause." That avoids an infinite temporal chain, but I would still ask what explains the first state. I reject brute facts as a final answer. My claim is that a complete explanation must reach a foundation that requires no earlier activation. To defend that claim, I must also examine explanations through a thing's nature or structure, and whether a whole history could explain its parts without requiring an earlier event.

For my part, I propose to solve the Unmoved Mover problem in the following way: motionlessness = all motion. Motionlessness does not need to be set going because it is not going anywhere. If motion and motionlessness belong to one fundamental reality, then motion would not require a separate event that first sets the foundation going. The work must explain this identity and show why it gives an answer to the problem.

This leads to a timeless foundation. Now, how can the activity we experience in time belong to that foundation? The foundation must remain timeless, or else this breaks: [motionlessness = motion]. My proposal is that the extension of spacetime is met by contraction. In this model, we never move out of the fundamental timeless condition.

Animals can model “extension = contraction” into 3D space because the timeless whole is dynamic and contains the activity they experience. The animal mind can model that activity. So the question is how temporal experience belongs within timeless reality, and the work must explain that relationship.

And further, I came to argue that entropy must reset: If existence can’t become nonexistent, if we grant that, then we grant that existence has always been; it never began. Within my model, entropy therefore belongs to an eternal circulation too. Then you ask, where did it come from? Now you are back in extended space trying to figure this out. We need a resolution. We need to actually solve infinite regress. So if your solution is another extended space, then you can continue to ask, and therefore you haven't solved infinite regress. Likewise, if the core of reality contains entropy, we can still ask: Where did this entropy come from? Now, if you agree that the solution must be a low-entropy state, then you can ask: Where did that low-entropy state come from? And my solution was that perfect order and balance at the foundation of time close regress. This is what low entropy would look like. If you accept that, then you cannot start with low entropy, then build high entropy, and that will be that, and now you have high entropy forever. That doesn't work because you must close reality; otherwise, you end up with infinite regress again. Why? Because the core must be perfectly balanced; you can’t have an open universe. This means all the high entropy must return to the core again, and if the entropy is not reset at the core and the core is timeless, then it would be like the core was never balanced at all.

My point is that the core’s perfect balance must hold for the entire eternal circulation, including everything that returns to it. It cannot be a pristine starting condition that reality subsequently leaves behind forever.

The difficult step is to show why this requirement rules out retaining entropy. A skeptic could agree that the whole circulation must be explained, yet argue that its retained distinctions are themselves fully explained. I will address that alternative and defend the claim that closure requires reset.

My central claim is therefore that entropy and infinite regress are the same thing on the philosophical level of analysis. I shall provide the analysis through which I intend to defend that claim and its consequence: if you remove infinite regress, you also reset entropy.

This is the basis of my proposal for unlimited free energy. I argue that the free part comes from the fact that no external energy supply is needed to sustain what I call balance, our Unmoved Mover. In my account, balance resets entropy by being what it is. I will develop the further argument that this reset continually renews matter and energy within our universe, and investigate how humanity might harness that renewed output.

I will also argue that nonexistence is a fabrication and product of language, so it cannot supply an explanation of existence. I intend to show why balance, understood as extension = contraction, is the remaining solution. That requires examining the alternatives and explaining why they do not achieve closure.

One of the biggest objections is this: “The foundation needs more than ‘there was no earlier moment.’ It needs an account of why its nature requires no further support.” My proposed answer is the dual torus. I argue that its reciprocal expansion and contraction make its activity inseparable from its balance, so that it requires no first activation or external sustaining process. But calling it a solution cannot establish that it succeeds. The full argument must show why this organization closes the explanatory dependence, why its own nature needs no further support, and why the relation [expansion = contraction] warrants the conclusion of timelessness. That is what I mean to establish through the phrase: No time = no regress.

Additionally, I will argue that static systems would themselves leave infinite regress unresolved, and that only dynamic systems can close it, which means balance can’t be static, which means you must perpetually recreate it, and this perpetual recreation is what makes life and complexity and evolution possible. I connect the true meaning of life to this perpetual balance because that is all there is. There is nothing other than this, forever. So this is why the universe is here. It has a job to do, which is to make more balance. I propose that the dual torus can produce what I call cosmic sex, which is the real blueprint of animal sex. So I suggest that even atoms can replicate! Basically, if you look at the cuboctahedron, then it can fold into a tetrahedron. And if you have two opposing tetrahedron clusters, one male and one female, as it were, you can create a new cuboctahedron on a larger scale. My proposal is that the fundamental building block of Nature can be as small as an atom and as big as a galaxy. I intend to argue that this “cosmic sex” explains what evolution is, why

things grow, why things become complex over time, and what expands the universe, or at least what helps expand the universe.

This brings us to consciousness, spirituality, God, and free will. I want to explain what consciousness actually is and why I think only consciousness can close Nature. From there, I will develop my argument for a spiritual universe, examine who God is within this account, and address free will and the true meaning of life. Each part must connect to the foundation through an argument that the reader can examine.

Taken together, these claims form the model of reality I want to defend. My aim is to show how it could open the way towards unlimited free energy and a new future for humankind, while explaining what that future would mean for conscious beings.

So, to circle back to the beginning: The thought that first drew these questions together for me was a sentence I retain as the poetic starting point of the inquiry:

Time began from light's point of view.

Our journey through this book, then, looks like this:

- Next: define all our concepts.
- Then explain what this means and explore the concept in depth: *Time began from light's point of view.*
- Explain: What started motion?
- Explore: The Unmoved Mover
- Try to solve: infinite regress
- Explain how temporal activity belongs within a foundation that remains timeless.
- Then explain how high entropy arises within the temporal activity of that whole.
- Defend the requirement that balance must hold for the entire eternal circulation.
- Provide the philosophical analysis intended to establish that entropy and infinite regress are the same thing, and examine whether fully explained retention challenges the reset argument.

- Develop the argument from entropy reset to renewed matter and energy, and to the possibility of a continually replenished energy source.
- Explain why I call this energy free and what I mean by unlimited.
- Then we will look at rivals to our philosophical system:
 1. Nonexistence. I propose that nonexistence is a fabrication created by our logical system.
 2. Brute facts. The foundation needs more than ‘there was no earlier moment.’ It needs an account of why its nature requires no further support.
 3. Static systems. Examine my claim that static systems leave infinite regress unresolved.
 4. Other accounts?
 - Then, after exploration of the cosmos, we will sit back and think about what it all means: why the universe is here. I propose that the universe has a job to do.
 - We will look at that job and what I propose it leads to: cosmic sex and atomic replication, with consequences for growth, complexity, and evolution.
 - Develop the proposed connections between the four forces, the formation of the Earth, and why the dinosaurs grew so tall.
 - Examine the argument for immortality and how it connects to the account of Nature.
 - Explain what consciousness actually is and defend the claim that only consciousness can close Nature.
 - Argue why there must be a spiritual universe, explain who God is within the account, and address free will.
 - Bring these arguments together to defend my account of the true meaning of life.
 - Next, we will invite the skeptic for a Q&A addressing the premises, the steps between the claims, and the strongest competing explanations.
 - Then we will finish with the practical question: How might we harness the renewed matter and energy produced by this process, should we, and what would that imply for humanity?

CHAPTER 2

Definitions and Fundamental Concepts

2.1 What understanding must accomplish

What must an explanation accomplish for me to regard something as understood?

I must understand why nonexistence is not a contender to existence. I must understand why existence is here, what its purpose is, and what its job is. I must understand how Nature closes infinite regress and why Nature is not a paradox. I want to understand Nature as a totality: why it exists, how it works, and how its existence can be without an ultimate beginning or end.

I also need to understand why the foundation of reality requires no further foundation. I want a satisfactory answer to the why-question, the how-question, and the when-question. Why is existence? How is existence? When did it start—or is a first beginning the wrong way to think about it? Asking why again must lead to understanding, rather than to another unexplained condition behind the answer.

The difference I am looking for is clarity of thought. If you truly understand something, the confusion surrounding it should fall away. Being told that something is simply a brute fact does not give me that understanding. It leaves me asking the same question. This is why I reject brute facts as a final answer to the problem I have set myself.

But clarity must be something I can communicate. I need to be able to show the reader what I mean, how the claims connect, and why the proposed answer is sufficient. A feeling of certainty cannot do that work on its own. The account must make its reasoning available to someone who does not begin with my conviction.

By complete understanding, I mean understanding existence at the level of its foundation and fundamental organization. I want to understand why there is a whole in which events can occur, what makes its activity possible, and why its explanation does not leave an unresolved dependence outside the account.

This chapter establishes the language in which I will pursue that aim. Some terms have familiar philosophical or scientific uses. Others name ideas specific to my proposal. Where I

use a word in a special sense, I will say so. Defining the question clearly gives us a way to recognize what an answer would have to accomplish.

2.2 Explanation and dependence

An **assertion** states that something is the case. A **description** tells us what something is like or how its features are arranged. An **explanation** identifies why or how the thing to be explained follows from, depends on, or is constituted by something else. The same sentence can contribute to more than one task, but saying what happens does not always settle why it happens.

Consider the coffee cup again. Saying that it is on my table describes its location. Saying that I placed it there explains how it reached that location. Explaining why it can hold coffee concerns its structure and material. These answers address different questions about the same cup. We should therefore ask which question an explanation answers before deciding whether it is complete.

By **dependence**, I mean that something relies on another condition for its existence, character, operation, or explanatory support. A cup depends on a manufacturing process for its production. Its ability to hold coffee depends on its present structure. Those dependencies are related, but the factory's history and the cup's current structure are different parts of the explanation.

I think of understanding as a jigsaw puzzle. When all the pieces fit together, you can stand back and see the whole. Brute facts, from the standpoint of my inquiry, are more like places where I am asked to stop trying to understand how the pieces fit. The analogy also gives me a responsibility: I cannot force my own pieces together simply because I want the final picture to appear.

A **causal explanation** concerns what produces or influences something. In many ordinary cases, it identifies earlier events. A **structural explanation** concerns how an arrangement makes a feature or operation possible. A **grounding explanation** concerns what a fact holds in virtue of. It need not describe an earlier event. Dasgupta develops this distinction in his account of metaphysical rationalism. [1]

For my purposes, this means that explaining motion need not automatically mean finding something that moved earlier. It may involve understanding why activity belongs to the nature of the whole. But this possibility does not settle the answer. I must explain what that nature is and why the account does more than attach a new name to the original question.

2.3 Infinite regress and explanatory closure

A **regress** occurs when a proposed account directs us to a further condition of the same relevant kind. An **infinite regress** is such a sequence without a final member. In the problem that concerns me, an explanation depends on another explanation, which depends on another, while the demand for a sufficient foundation continues.

I distinguish a **temporal regress**, which runs through earlier states or events, from an **explanatory regress**, which runs through conditions on which the account depends. The second need not unfold backwards in time. A timeless structure could still leave us asking why that structure exists or why its principles hold. Eliminating an earlier moment therefore addresses one part of my problem; the proposed foundation must also explain its own finality.

An endless sequence and an unsuccessful explanation are not identical by definition. A critic may argue that an infinite history is sufficiently explained by its laws or by the nature of the whole. My objection must concern what that account leaves unresolved. Calling a sequence infinite cannot by itself establish that it contains a contradiction.

By **global explanatory closure**, I mean an account of existence and its fundamental organization that leaves no relevant foundational dependence unexplained, and makes intelligible why no further foundation is required. Global means that the account includes the conditions on which its own proposed explanation relies. An unexplained mechanism behind the mechanism would remain part of the problem.

This use of closure must be distinguished from a path returning to its starting point, a system exchanging nothing with its surroundings, or a set of equations specifying an evolution. Each may be useful, but each answers a different question. A circle on paper shows a closed path. It does not yet explain why the path, its contents, or its governing relation exists.

A **brute fact**, as I use the term, is a fact that has no further explanation even though the question of its explanation remains appropriate. Something presently unexplained is not necessarily brute: we may simply lack its explanation. My rejection concerns brute facts as the final answer, not the admission that an inquiry still has work to do.

Temporal order

... → Earlier state → Later state → Present state

Arrows mean earlier than

Explanatory dependence

... → Supporting condition → Supporting condition → Fact to be explained

Arrows mean provides explanatory support

Dependence need not be a sequence of earlier events

Figure 2.1. Two kinds of regress. The upper sequence concerns earlier and later events; the lower sequence concerns explanatory support. The distinctions follow Section 2.3.

I use **self-grounding** for the proposed achievement of a foundation whose nature makes further support unnecessary. This expression does not mean that an event produces itself before it exists. Nor does repeating a claim as its own reason provide an explanation. The intended achievement is to understand why the demand for a further supporting condition no longer applies.

2.4 Existence, Nature, and nonexistence

Existence means that there is reality at all. It does not mean that every particular thing lasts forever. A cup can break and an organism can die without reality as a whole becoming nonexistent. My claim about the impossibility of absolute nonexistence concerns the whole, and must not be confused with the survival of its particular forms.

I use **Nature** and **reality** for the totality of what exists. This usage does not settle in advance whether that totality is fundamentally material, conscious, or something that requires a different account. If consciousness or a spiritual dimension is real, it belongs within Nature as I use the word. The capital letter marks the whole under discussion; it supplies no argument about its character.

The **universe** will usually mean the world of matter, energy, space, time, and experience whose organization I am trying to explain. Where a passage concerns the observable universe or a particular cosmological domain, that narrower scope must be stated. An account of our observable region is not automatically an account of everything that exists.

Absolute nonexistence means the absence of reality altogether. **Nothingness** is more ambiguous. An empty cup contains no coffee, but the cup and its surroundings remain. An empty region is still a region within a description of reality. Neither example establishes what absolute nonexistence would mean or whether it is a coherent alternative.

I propose that absolute nonexistence is a fabrication of our logical and linguistic activity rather than a real contender to existence. That is a claim I intend to argue for. The fact that we can form the phrase does not settle whether it describes a possibility; equally, defining it as an absence does not already prove its impossibility.

Two further claims need separate treatment: existence cannot cease, and existence never began. The first concerns an end; the second concerns an origin in time. My account intends to defend both, but the meaning of one does not simply contain the other. This distinction will matter when I ask whether Nature has an origin without a first moment.

2.5 Motion, motionlessness, and activity

In ordinary physical usage, **motion** concerns a change of position relative to a reference frame. **Change** is broader: a thing can alter its temperature, structure, or condition without moving from one location to another. Aristotle's discussion of motion also ranges beyond locomotion, relating it to the actualization of what is potential. His vocabulary provides an important point of comparison, although I will develop my own account. [2]

In this work I also use motion in a foundational sense, to name the intrinsic activity I propose is inseparable from existence. In that sense, I am asking what makes the temporal

motions of the universe possible at all. This broader use must be connected to ordinary motion through the argument, rather than treated as an unnoticed change of meaning.

Motionlessness can mean that a particular object is at rest relative to a reference frame. My foundational use is stronger and different: the whole does not need to pass from inactivity into activity, or travel towards a condition beyond itself, in order to exist. I propose that its fundamental balance belongs to what it is.

The expression **motionlessness = all motion** states my proposed identity between the whole's fundamental condition and the activity that belongs to it. The equality sign expresses a philosophical claim about one reality. It is not yet a numerical equation between measured quantities. Explaining what makes the identity coherent is a central task of the following chapters.

By **activity**, I mean the operative character of the proposed relation: the foundation is not something to which motion must subsequently be added. I call it **dynamic** because its balance is inseparable from activity. If I apply dynamic to a timeless foundation, I owe an account of how the term can be meaningful without an external clock. The intended meaning is now on the table; the explanation will follow.

A **static system**, in the ordinary temporal sense, remains unchanged over time. A **stationary description** can leave some overall features unchanged while allowing activity within the system.

However, by **static**, I mean no change whatsoever and no activity at all. I have in mind something like a block universe, but only if it is understood as a completely inactive reality.

To my mind, absolute inactivity presents the same problem as absolute nonexistence. My view is that, when analysed, neither term describes a coherent possibility. Static systems, in the strict sense I mean here, would therefore be nonsensical too. But I must support that conclusion with an argument, which I will develop later.

2.6 Time, timelessness, and the origin of motion

By **time**, I mean the ordering and duration through which changes can be described as earlier, later, or simultaneous. A **temporal process** has distinguishable stages within such an

order. A clock is a process used to compare other processes; it does not, merely by being a clock, explain why temporal order exists.

Spacetime joins the spatial and temporal relations used to describe events. By **extension**, I mean distinguishable spatial separation or temporal duration. When I speak of the foundation as lacking independent extension, I must explain how the spatial and temporal relations of experience belong to it. In that foundational discussion, **instantaneous** cannot mean an exceptionally short interval on an external clock; the intended claim is that no such interval separates the reciprocal aspects.

By **timelessness**, I mean that the foundation does not itself undergo a succession requiring an earlier condition from which its fundamental nature arose. I propose that temporal activity belongs within this foundation. The proposal therefore concerns the relationship between the whole and the order experienced within it, rather than a moment at which time was switched on from outside.

Eternity needs care. A temporal history may have no beginning or no end. A timeless foundation is not a history that has waited through infinitely many years. When I call the foundation eternal, I mean that its existence is not confined between a first and a last temporal state. When I discuss an eternal circulation, the temporal activity and the timeless condition must be kept distinguishable in the explanation.

A **beginning** is a first stage in an order. An **origin**, in my philosophical use, is what accounts for something's existence or character. An origin need not be an earlier event. A **foundation** is the condition proposed to account for the whole without borrowing unexplained support from beyond it. I use **core** as a shorter name for that foundational condition; the word does not by itself locate a physical centre in space. But as we shall see, the core in my model is also a physical centre in space.

An **initial state** is a first state within a specified temporal model. If I use that phrase for the foundation, I must explain whether I really mean first in time or fundamental in the order of explanation. Otherwise I would have introduced a temporal beginning into an account whose purpose is to question that requirement.

The **Unmoved Mover** names the problem of how motion can have a sufficient explanation without requiring another mover indefinitely. Aristotle's account in *Metaphysics*

XII concerns an eternal unmoved source of motion, not simply a first push at a first instant. My proposed identification of motionlessness with activity is a position to compare with his, not a restatement of it. [3]

Finally, *Time began from light's point of view* is the poetic sentence from which my inquiry developed. Here **point of view** names an imaginative perspective. It is not a technical claim that special relativity supplies an observer's rest frame travelling with light; it may not. The familiar relativistic transformations do not provide such a frame at light speed. [4] Chapter 3 will explore the philosophical insight I take from the image.

Figure 2.2 shows how I intend these two aspects to belong to one reality.

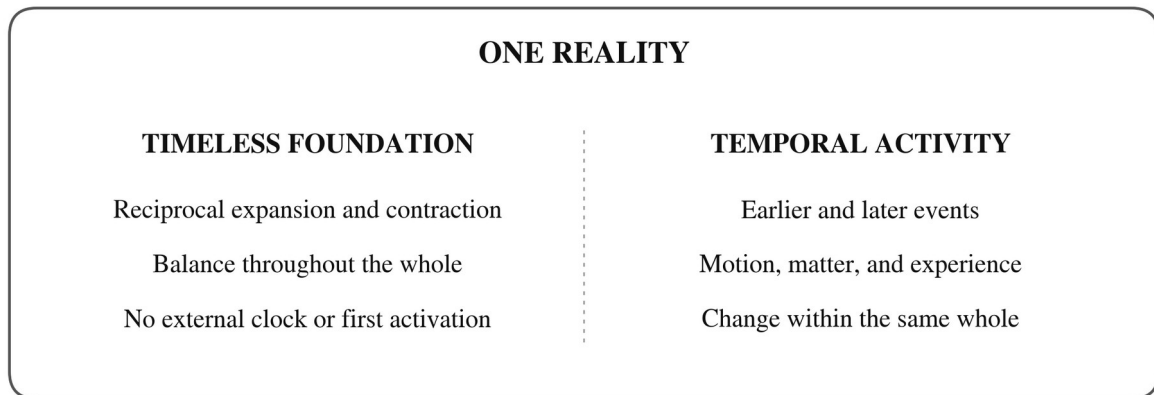


Figure 2.2. One reality viewed through its timeless and temporal aspects. In my proposed account, temporal activity belongs within the timeless whole. The layout marks a conceptual distinction; it is not a timeline or a spatial map.

2.7 Balance, reciprocity, and geometric representation

Balance, in my proposal, means that the fundamental activity forms a complete reciprocal relation whose coherence is maintained throughout the whole circulation. It cannot be merely a favourable starting condition that later activity leaves behind. The return must belong to the same account as the outward activity.

By **expansion** and **contraction**, I mean the complementary aspects through which I propose to account for extension and return. At the level of a physical model, these terms would require specified quantities and relations. At the philosophical level, they express my claim that outward development does not become independent of the foundation that makes it possible.

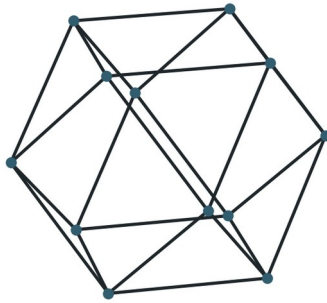
Reciprocity means that the two aspects are defined through their relation to one another. My expression **expansion = contraction** is intended to state a fundamental constraint. There is, however, a difference between equal amounts of two processes and their being aspects of one indivisible activity. I must show which claim is needed at each stage, and why the proposed relation warrants timelessness.

Circulation means the connected account of outward activity and return. **Complete return** means that whatever the argument treats as belonging to that circulation is included when assessing the foundation's balance. It cannot exclude inconvenient products of the activity simply because they are absent from a diagram. A **cycle** is a recurrent pattern; it need not mean that every detail of an entire universe repeats identically.

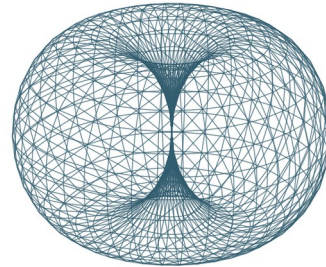
By **separation**, I mean differentiation within the whole. My claim that separation has no independent existence means that differentiated things depend on the reality to which they belong. It does not, by itself, mean that there are no meaningful differences between them. Those differences are part of what the account must explain.

Zero-dimensional, in the proposed foundational sense, means that the foundation has no independent spatial extension. This is not the same as a very small object situated in a larger space. Nor does the absence of spatial extent, by definition alone, determine whether internal distinctions or temporal relations remain. Those further claims require explanation.

I use **dual torus** for my proposed organization of complementary circulation sharing a common foundational relation. A torus drawn in three dimensions is a spatial representation; the claim that it represents a foundation without independent extension needs its own account. **Cuboctahedron** names the geometric form I use when discussing balance and the proposed relation between scales. The figures will help specify the proposal, while the text explains what philosophical work each feature is intended to do.

Cuboctahedron

12 vertices and 24 edges

Torus

A rounded form with a pinched centre

Figure 2.3. Geometric vocabulary used in the thesis. The cuboctahedron and torus are spatial forms. Their proposed foundational significance is developed in the text.

2.8 Possibility, necessity, and contradiction

A **premise** is a claim from which an argument proceeds. A **conclusion** is the claim the argument is intended to establish. A **definition** fixes how a term is being used. These roles matter because a definition cannot establish that something answering to it exists. I can define what a successful foundation would accomplish without yet showing that my proposed foundation accomplishes it.

An inference is **valid** when its conclusion cannot be false if its premises are true. An argument is **sound** when it is valid and its premises are true. A **conditional result** establishes what follows if specified premises hold. Establishing the conditional and defending its premises are connected tasks, but they remain distinguishable achievements.

Logical possibility means consistency with the relevant logical constraints. **Metaphysical possibility** concerns what could exist given the natures of things. **Physical possibility** concerns compatibility with the physical laws under consideration. **Actuality** concerns what is the case. An imagined mechanism, a consistent model, and an observed process occupy different positions in that sequence of questions.

By **necessity relative to premises**, I mean that a conclusion must hold if those premises hold. A claim of **metaphysical necessity** is stronger: it says that reality could not ultimately

have been otherwise in the relevant respect. My overall energy question asks what can be possible; parts of my argument claim that particular consequences must follow. Each use of can and must needs to retain its scope.

A **contradiction** affirms and denies the same claim in the same respect. A **paradox** is a difficulty in which apparently acceptable ideas lead to an apparently unacceptable result; the difficulty may reveal an ambiguity or a mistaken assumption. When I say Nature cannot be paradoxical, the principle I intend to defend is that reality cannot contain an actual contradiction. An initially puzzling description need not establish one.

The phrase **motionlessness** = **all motion** therefore cannot be defended merely by accepting a contradiction. I must explain the respects in which motionlessness and activity are attributed to the same reality, and establish the connection between them. Equally, no contradiction having yet been found is weaker than a demonstration that the full proposal is consistent.

2.9 Systems, states, and information

A **system** is the set of things or variables included in an account. A **boundary** specifies what belongs to that system and what is treated as its surroundings. The boundary may be physical or representational. Moving a boundary in our description does not necessarily produce any change in the world.

In thermodynamics, a **closed system** exchanges no matter with its surroundings, although energy transfer may occur. An **isolated system** exchanges neither matter nor energy. [5] These technical uses differ from my philosophical claim that Nature is a complete whole. Calling that whole closed does not, by the word alone, decide its geometry or its entropy.

Local refers to a specified part or domain; **global** refers to the whole relevant to the claim. What counts as global must therefore be stated. The whole of an apparatus is not the whole of Nature. A complete cosmic claim must account for whatever its proposed operation relies on, including what an observer has left outside a local description.

A **state** specifies the relevant condition of a system. A **microstate** gives a more detailed specification, while a **macrostate** groups detailed possibilities under features such as temperature, volume, or a chosen large-scale pattern. **Coarse-graining** is the use of a

description that does not distinguish every detailed alternative. These distinctions will matter when I ask what is actually lost in an alleged reset. [6]

By **information**, I mean distinctions that can be represented or recovered concerning alternatives. A **record** is a condition correlated with what occurred. A **correlation** is a dependence in how possibilities occur together. This operational use of information does not assume that a record must be consciously read in order to exist.

Retention means that information about the previous history survives somewhere. **Total erasure** means that no record or recoverable trace of that history survives anywhere.

In my proposal, the reset involves total erasure. The timeless core removes all inherited distinctions that carry information about the returning history. The core reaches the same foundational condition regardless of which history returns to it. This is what I mean by **historical independence**: the reset condition retains no information that distinguishes one previous history from another.

The proposal also includes a complete reset of entropy, with none transferred elsewhere or stored outside the core. In the later argument, I will explain how this reset is meant to work and how the universe's variety arises from this foundational condition.

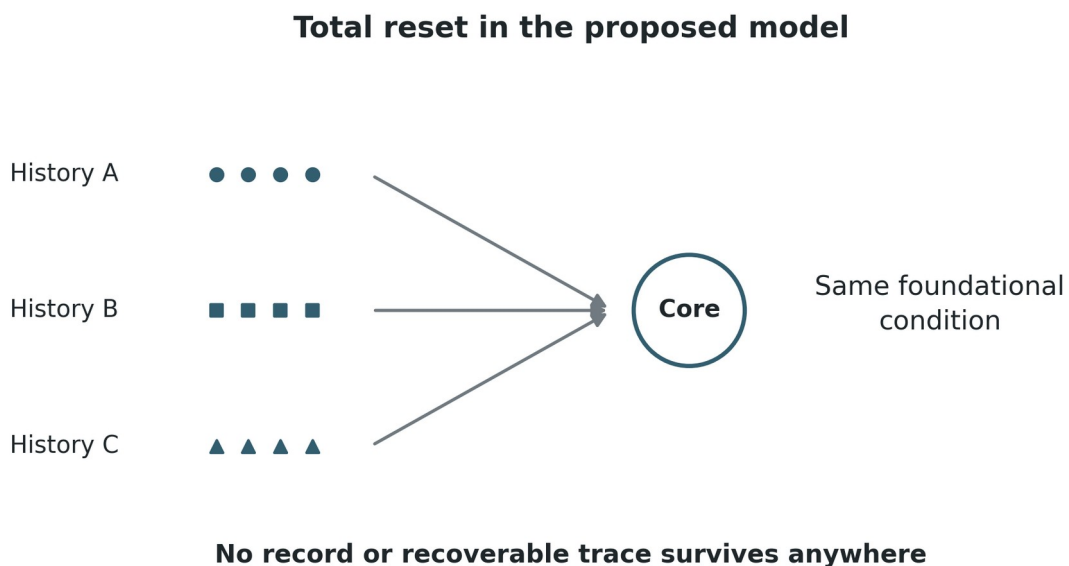


Figure 2.4. Total reset as proposed in Sections 2.9 and 2.11. Alternative returning histories lead to the same foundational condition, with no surviving record or recoverable trace anywhere. Arrows compare the proposed outcomes of different histories.

2.10 Entropy, order, and equilibrium

Entropy is a physical quantity with a specified meaning, not simply a name for confusion. In thermodynamics it is a state property whose change can be calculated, along a reversible path between equilibrium states, by summing reversible heat transfer divided by absolute temperature. In that formulation, its unit is energy per unit temperature. [7]

In a Boltzmann counting description, entropy is related to the logarithm of the number of microscopic arrangements compatible with a specified macrostate. **High entropy** and **low entropy** therefore require a stated system, constraints, and description. An arrangement that looks tidy to us need not have low physical entropy. Feynman's discussion of order and entropy makes this distinction through the number of arrangements compatible with the same large-scale appearance. [6]

Order will mean a specified restriction or organization of possibilities. I will identify the restriction when it matters. **Equilibrium** concerns a condition in which the relevant macroscopic tendencies to change have been balanced under the applicable constraints. **Steady state** means that selected macroscopic properties remain constant; continuing flows may still occur. Neither phrase is a synonym for zero entropy or for my proposed explanatory closure. [5]

The **second law of thermodynamics** constrains entropy changes: in its macroscopic formulation, the entropy of an isolated system does not decrease. A decrease in one part must be considered with the changes in what it interacts with. This is why naming the system matters when discussing a reset. [7]

When I speak of **accumulated entropy**, I mean the entropy associated with the outcomes of the temporal activity under discussion. Entropy is not a separate substance travelling alongside matter. **Entropy production** concerns the increase associated with irreversible processes; it is not the same as transferring entropy across a chosen boundary.

My claim that entropy and infinite regress are the same thing at the philosophical level is not the dictionary definition of entropy. It is a proposed connection between physical historical dependence and the demand for a complete foundation. Chapter 10 must supply the analysis that connects those questions. Keeping the established meaning visible gives the claim a clear subject to address.

2.11 Reset, return, and renewed matter

By **entropy reset**, I mean the proposed restoration, through the foundational relation, of the low-entropy condition needed for renewed outgoing activity. By **complete reset**, I mean that the claim includes the relevant distinctions throughout the system under discussion. If those distinctions are merely moved into something omitted from the account, their absence from the selected subsystem does not settle the global claim.

Recurrence means that a state or pattern returns. In my proposal, the foundational condition returns through a total reset. The timeless core erases all information about the returning history. No record, backup, or recoverable trace survives anywhere. Different histories return to the same foundational condition, with nothing preserved that could distinguish them.

Renewed matter and energy names the output of the proposed process in a condition that renews the capacity to do work. I am not picturing humanity outside Nature drawing directly from a timeless core. The matter and energy of our world are, in this account, the output in which we already live. The practical question concerns where the renewal becomes available and how its output might be harnessed.

By **continual renewal**, I mean that portions of the universe are renewed through the circulation while temporal activity continues. I do not mean that the whole universe must wait for a single final moment before it can begin again. By an **outlet**, I mean a physical manifestation through which renewed output would become accessible within the temporal world. The term identifies what to look for; it does not yet identify a particular astronomical object or an engineered device.

The **fully explained retention objection** asks whether a complete foundation could explain why historical distinctions remain, instead of requiring their removal. That is the alternative my reset argument must address. The disagreement concerns what closure requires, so I cannot settle it simply by defining closure as reset. I intend to establish the connection through the nature of balance and the completeness of the circulation.

Table 1 gathers the distinctions needed for the reset argument.

Table 1. Balance, equilibrium, recurrence, and entropy reset.

Concept	Meaning in this thesis	Illustration
Balance (proposed)	The reciprocal activity through which I propose that the whole remains complete.	Expansion and contraction belong to the same circulation.
Equilibrium	A macroscopic condition with no tendency to change under the specified constraints.	A gas in a sealed, insulated vessel after temperature and pressure have settled.
Recurrence	The return of a state or pattern.	A periodic model returns to the same configuration after each cycle.
Entropy reset (proposed)	Restoration of the relevant low-entropy condition through the foundational relation.	In my model, returning matter is renewed for outgoing activity.

Equilibrium need not have low entropy, and recurrence alone does not establish entropy reset. The proposed connection between balance and reset is the argument to be developed in the later chapters.

2.12 Energy, work, and the meaning of free

Energy is a quantitative property used in accounting for physical processes and transformations. Kinetic, potential, internal, and other forms belong to this accounting. The existence of energy and its availability for a particular task are different questions. Conservation concerns the accounting of energy; it does not say that every arrangement offers the same opportunity for useful work. [8]

Work is a mode of energy transfer. In a mechanical example, a force acts through a displacement, as when a weight is raised. **Heat** is energy transferred because of a temperature difference. Energy stored in a system should therefore be distinguished from the processes by which energy is transferred. [9, 10]

By **usable energy** or **capacity to do work**, I mean an available physical resource that can drive a specified process in relation to its surroundings. **Availability** concerns the useful work obtainable relative to specified environmental conditions. A **gradient** is a difference across a region or between systems, such as a temperature difference. Identifying an energy source requires more than reporting the amount of energy present. [11]

Power is the rate at which energy is transferred or work is done. My use of **unlimited** concerns the proposal of a continually replenished supply, rather than a finite stock that must ultimately be exhausted. It does not mean that a particular outlet supplies infinite power at one moment. A continuing source may have a finite rate, and human access may be limited by location, collection, and conversion.

By **free**, I mean that the fundamental balance, in my account, requires no external supply to sustain it. Renewal belongs to what the balance is. In this model, collecting usable output obviously does not draw energy from the core, though this should hardly need saying. The fundamental circulation remains unaffected. The term does not mean that equipment is free to build or that collecting output has no practical costs.

This use of free is also distinct from the technical expressions **Helmholtz free energy** and **Gibbs free energy**, which describe thermodynamic potentials under specified conditions. My central proposal concerns why availability would be renewed. Any later calculation using a thermodynamic potential would have to state the conditions that make it applicable.

The **human access question** is therefore about recognizing and coupling to the renewed output. **Net useful output** means what remains available after the inputs and losses of the proposed collection process are included. I do not need to supply a machine in order to define the philosophical proposal, but a claim about an actual machine would have to account for its complete operation.

2.13 Consciousness, life, and the wider vocabulary

By **consciousness**, I initially mean the presence of experience: there is something it is like to see, feel, or think. By **the given**, I mean experience as encountered in having it. This provides a starting point for asking what consciousness is. The further claim that consciousness is fundamental, and that only consciousness can close Nature, is the position I will defend in Chapter 18.

A **spiritual universe**, in this account, is one whose fundamental nature includes consciousness rather than treating experience as incidental to an otherwise complete account. I will use **God** for the ultimate conscious reality I propose to identify with the foundation of

Nature. This does not settle the detailed character of that reality or establish the claims of any particular religion. Those questions require the account developed in Chapter 19.

Free will concerns agency and responsibility for action. Its assessment depends on what sort of control is claimed: acting through one's own reasons, having alternative possibilities, or being an ultimate source of choice are distinguishable proposals. I will specify which freedom my account intends to defend. The word freedom cannot by itself resolve its relationship with a necessary foundation.

Purpose can mean an intended end, a function within a system, or a reason why something exists. When I say the universe has a job to do, I initially mean a proposed function: the continual realization of balance. Whether this function also involves intention is a further question. By the **meaning of life**, I mean the significance of conscious existence and how lives relate to the fundamental activity I propose. A physical pattern alone does not state what anyone ought to value.

Immortality means the continuation of a living or conscious individual without an inevitable final end. It must be distinguished from the persistence of matter, the recurrence of similar forms, and the timeless existence of the whole.

Personal identity concerns what makes continued existence the existence of the same individual. Any argument for immortality must identify what continues and why that continuation preserves the person concerned.

I also consider **longevity through the consumption of low-entropy matter** as a possible route toward immortality. This concerns extending the life of the individual, potentially without a fixed limit. I will develop and defend this proposal later.

Complexity concerns differentiated organization; **growth** concerns increase in a specified quantity or scale. Biological **evolution** concerns changes in heritable characteristics of populations across generations. When I discuss a more general development of cosmic organization, I will identify that broader use rather than assume that the same mechanism has already been established in both cases.

Cosmic sex is my metaphor for the proposed generative relation between complementary configurations. **Atomic replication** names my proposal that the fundamental organization can participate in producing further instances of its pattern. These expressions do not define

atoms as biological organisms. The chapter devoted to them must explain what is produced, what makes the production replication, and how the proposed geometry relates to physical processes.

The **four forces** refers to gravitation, electromagnetism, and the strong and weak interactions. [12] **Unification**, as an aim of the work, means identifying a common explanatory basis for them. A shared diagram or recurring shape would not by itself establish that basis. The formation of Earth and the size of dinosaurs name further applications for which the relevant processes and evidence must be specified in their own chapters.

2.14 How these definitions guide the argument

We can now state the task more clearly. I seek a foundation whose nature accounts for existence and activity without an unexplained external support. I propose that motionlessness and all motion belong to the same reality, and that reciprocal balance makes its activity inseparable from what it is. From there, I intend to argue for reset, renewed output, and the possibility of human access to a continually replenished source.

The definitions give the reader a way to follow each transition. If I claim necessity, the reader can ask which premises make the conclusion necessary. If I claim entropy reset, the reader can ask which system and which distinctions are included. If I claim renewed energy, the reader can ask what makes the outgoing condition capable of doing work. If I claim that consciousness closes Nature, the reader can ask why the proposed closure requires experience.

These questions belong to the same search for clarity with which I began. I want to be able to stand back and understand the whole of Nature, why its foundation is sufficient, and how the parts belong to it. There should be no unexplained residue at the level the account claims to close. But the words must remain stable while the argument is developed. A change in definition cannot silently do the work of a missing explanation.

Aristotle provides the central historical comparison for the Unmoved Mover, while the distinction between causal and grounding explanation connects the inquiry with contemporary metaphysics. The physical definitions establish what my claims about entropy and energy concern. The chapters that follow must engage with these positions in detail. Here

we have established the vocabulary needed to begin with the sentence that first set the inquiry in motion: *Time began from light's point of view.*

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CHAPTER 3

Time Began from Light's Point of View

3.1 The Train to Nowhere

I am sitting on this train, and I am fixing my eyes and consciousness upon a remarkable scene: From my window, I can see, and feel, that we are moving away from another train that is, in actuality, moving away from us. We are sitting totally still, inert—our train has not begun to move yet—but my brain does not know that, so it interprets the situation as if we are moving. If I were trapped in this state forever, then forever I would state that we are, in fact, moving.

In other words, how do we know that movement actually exists in the real world? Is it all in our heads, Kant?

I was in my military uniform. I remember it as if it were yesterday. It was my first true encounter with Dr. Einstein's Imaginarium.

The scene described in Chapter 3

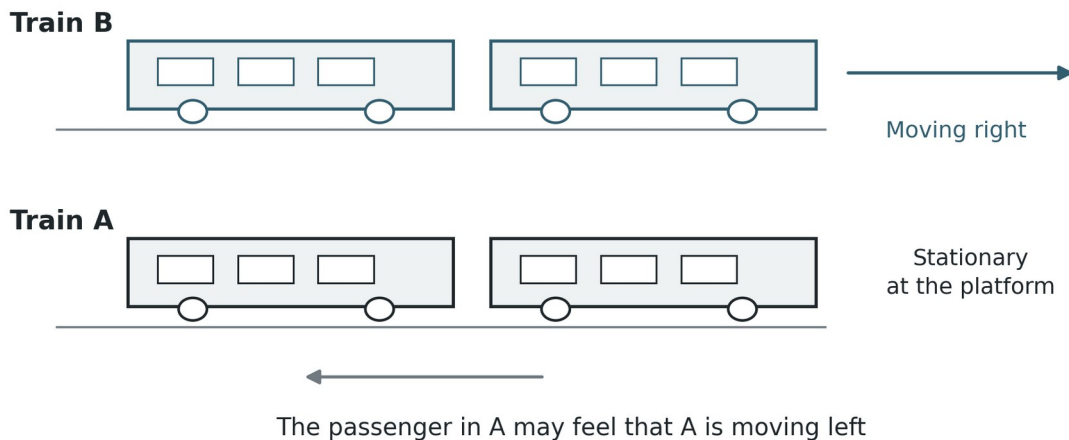


Figure 3.1. The scene described in Chapter 3. Relative to the platform, Train A is stationary and Train B moves right; the passenger in A may experience the scene as A moving left.

This first chapter will intentionally not be longer than this. I want you now to hop onto a beam of light, ride into the universe, and tell me what you see.

Then, afterwards, tell me: Do you see the veil of eternity?

CHAPTER 4

The Origin of Motion

4.1 The Power of Reason

If all you had was time, then asking where a state came from would take you back to an earlier state in time, but then you could ask where this new state came from. And you could do this forever because all you have is time.

Then you can try to solve the paradox by inserting nonexistence, brute facts, or accepting paradox.

But if all three fail, then you must conclude that the foundation of existence is timeless because timelessness, unlike time, has no earlier state, so you cannot form an infinite regress. You still need to explain how timelessness can be possible, but you do not need to solve the regress that only happens when inside this phenomenon called time.

The only thing the skeptic can say is that my timelessness is provisional in the sense that, even if all three fail, there could be a fourth possibility we haven't considered, such that my logical form and conclusion are not wholly complete. I accept that because no fourth possibility can exist; it will just remain a skeptic's comfort food.

What I'm trying to say is, I do not need to show why expansion = contraction makes timelessness. I can use reason and understand that timelessness must come first in the order of things. Thus, I can use expansion = contraction as a tool to understand how I can get a universe from a timeless state that has to remain timeless forever because it is timeless. If I propose a moment where timelessness can transition into time, I recreate the paradox I wanted to solve in the first place; that is, I recreate the infinite regress, just on another level.

Suppose you say that timelessness existed, then time began. The word "then" creates the problem. A transition is a change from State A, timelessness, through a transition, to State B, time. But for a genuine transition to occur, there must be some ordering by which A is prior to B. You can then ask: What accounts for that transition? Why did timelessness change at that point rather than another? If the answer invokes an earlier condition or event, you now have an earlier condition → transition → time. And now you can ask what accounts for that

earlier condition. The regress you removed from ordinary time has simply been relocated to a supposed meta-time outside ordinary time.

Thus, I require that the timeless foundation remains timeless forever.

Argument for a timeless foundation

P1. If all that exists is temporal, explaining where a state came from requires reference to an earlier temporal state.

P2. The same requirement applies to every earlier state. Therefore, an explanation confined entirely to time generates an infinite regress.

P3. The proposed alternatives are:

- Invoke nonexistence.
- Terminate explanation in a brute fact.
- Accept the unresolved regress or paradox.
- Ground temporal existence in timelessness.

P4. Suppose nonexistence, brute facts, and acceptance of the paradox all fail.

C1. If the alternatives in P3 are exhaustive, the foundation of existence must therefore be timeless.

P5. A timeless foundation has no earlier temporal state.

C2. Therefore, the regress generated by repeatedly requiring an earlier temporal state does not arise within that foundation.

C3. Explaining how timelessness is possible remains a further task, but that task is distinct from resolving the regress of earlier temporal states.

The central inference is:

$$\begin{aligned}
 &L \vee N \vee B \vee P \\
 &\neg N \wedge \neg B \wedge \neg P \\
 &\therefore L
 \end{aligned}$$

Here, L denotes a timeless foundation, while N, B, and P denote the three proposed alternatives as successful explanations.

Provisionality: The inference depends on the alternatives being exhaustive. If an unconsidered fourth alternative to timelessness exists, eliminating the three listed alternatives would not yet establish timelessness uniquely.

The argument as stated in Chapter 4

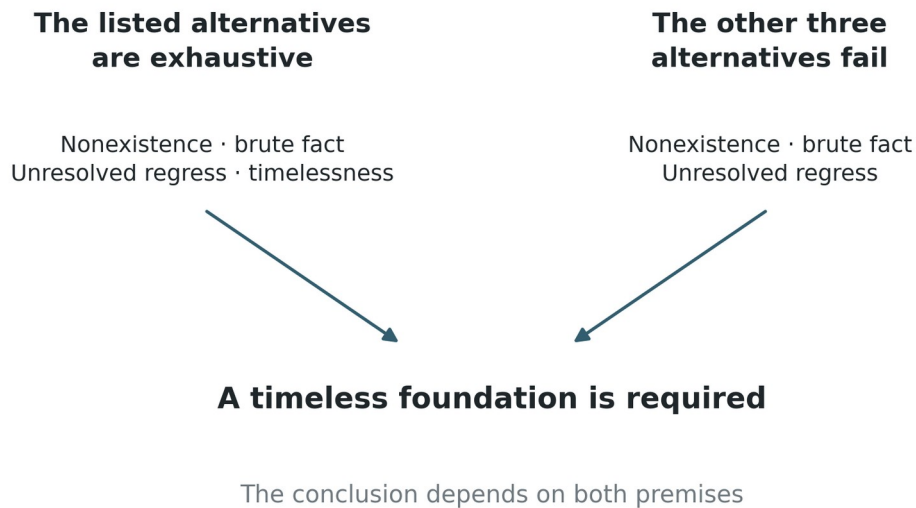


Figure 4.1. The inference stated in Section 4.1. The conclusion follows on the stated conditions that the alternatives are exhaustive and the other three fail. Arrows show an argument, not a physical process.

4.2 The next task

Explaining how timelessness is possible remains a further task, so let us now tackle that task.

If we accept that nonexistence is impossible and paradox is impossible, then existence must be and must be coherent. And building upon our previous conclusion that timelessness is the foundation of reality, it follows that we get a true reason for “how” timelessness is possible. That is to say: timelessness is possible because it is necessary and coherent. It is not one possibility among others; it is the only thing that actually exists and actually is coherent. All other existing things and all other coherent states must then have their root in timelessness as the only necessary and coherent state.

The possibility of timelessness

P1. Absolute nonexistence is impossible.

C1. Therefore, existence is necessary.

P2. Existence necessarily requires a timeless foundation, provided the premises of the preceding argument hold necessarily.

C2. Therefore, a timeless foundation is necessary.

P3. Whatever is necessary is possible: something cannot both be required to exist and be incapable of existing.

C3. Therefore, a timeless foundation is possible.

P4. Genuine contradictions cannot characterize reality.

C4. Therefore, the timeless foundation, being real, must be coherent.

Conclusion. Given these premises, timelessness is possible because it is necessary, and it must be coherent because contradiction cannot characterize what exists.

This establishes why a timeless foundation must be possible before providing a complete account of its structure. Explaining that structure and its relationship to temporal reality remains the next task.

Formal inference

$$\Box E, \Box(E \rightarrow F) \quad \therefore \Box F \quad \therefore \Diamond F$$

Here, E means that something exists, F means that a timeless foundation exists, $\Box$ means “necessarily,” and $\Diamond$ means “possibly.”

4.3 What the Argument Establishes

Taken together, the preceding arguments establish a two-stage case for a timeless foundation. The first argument attempts to establish that a timeless foundation is required, while the second explains why such a foundation must be possible. This distinction is important because it means that I do not need a structural model, such as expansion = contraction, to establish timelessness in the first place. The structural model enters later, when we ask what such a timeless foundation could actually be like and how temporal reality can exist in relation to it.

The first argument begins with the problem of temporal explanation. If all we have is time, then explaining one state by reference to an earlier state merely moves the explanatory problem backwards. We can then ask what accounts for that earlier state, and then what

accounts for the state before that. If explanation remains entirely within time, the process can continue indefinitely. We therefore encounter an infinite regress of earlier temporal states.

There are several apparent ways of terminating this regress. We might invoke nonexistence, terminate explanation in a brute fact, accept the regress or paradox, or ground temporal existence in something that is not itself temporal. If nonexistence is impossible, brute facts are rejected as ultimate explanations, and accepting the regress or paradox does not resolve the explanatory problem, then timelessness remains as the proposed foundation. This conclusion is conditional upon these alternatives being exhaustive. If some genuinely distinct alternative exists that has not been considered, then it must also be examined. The burden is therefore not merely to reject the three alternatives, but ultimately to establish that no relevant alternative has been omitted.

If that burden can be met, an important conclusion follows: the foundation of temporal existence cannot itself be another temporal state. A timeless foundation has no earlier temporal state and therefore does not generate the same regress that arises when every state requires another state before it.

However, this introduces another problem. It is not enough to say that timelessness existed and then time began. The word “then” immediately introduces temporal ordering. If timelessness existed first and subsequently changed into time, we would have a transition from one condition to another. We could then ask what accounts for that transition and why it occurred when it did. If the answer requires some earlier condition, the infinite regress has simply been relocated from ordinary time to a supposed meta-time outside ordinary time.

The timeless foundation therefore cannot become temporal through an event occurring within some deeper sequence. If it did, it would no longer be timeless in the relevant sense. A genuine change from timelessness to time would require an ordering in which the timeless condition is prior to the temporal condition. But this ordering would already possess the very structure we were attempting to explain.

Thus, I require that the timeless foundation remains timeless. More precisely, the timeless foundation cannot itself enter into temporal succession. Temporal reality must therefore stand in some other relationship to it. Time cannot be the result of an event in which timelessness stops being timeless and becomes temporal.

This leads to the second question: if such a foundation is required, how can timelessness itself be possible?

Here necessity becomes important. If absolute nonexistence is impossible, then existence is necessary: something must exist. If the preceding argument succeeds in establishing that existence necessarily requires a timeless foundation, then that timeless foundation is itself necessary. And something that is necessary cannot simultaneously be impossible. If it must exist, then it must at least be possible for it to exist.

The reasoning can therefore be expressed in a simple progression:

Existence is necessary → existence requires a timeless foundation → the timeless foundation is necessary → whatever is necessary must be possible → therefore, a timeless foundation must be possible.

This does not yet tell us what timelessness is structurally. Instead, it establishes a requirement that any successful account of reality must satisfy. We arrive at timelessness first through reason and only afterwards ask what kind of structure could satisfy the requirement.

This changes the role of expansion = contraction within the argument. I do not need expansion = contraction to prove that timelessness is required. Nor do I need to claim that expansion = contraction somehow causes timelessness to exist. If the preceding reasoning succeeds, timelessness has already been established as a necessary condition of the foundation.

The purpose of expansion = contraction is therefore different. It becomes a proposed tool for understanding how temporal activity can belong to, arise within, or manifest from a foundation that never itself undergoes a temporal transition.

The progression of the argument is therefore:

Reason → timelessness is required.

Necessity → timelessness must be possible.

The prohibition on temporal transition → the timeless foundation cannot become temporal through an event.

Expansion = contraction → a proposed structural explanation for how temporal reality can nevertheless exist in relation to a foundation that remains timeless.

This distinction allows us to separate two questions that might otherwise become confused. The first question is whether reality ultimately requires a timeless foundation. The second is what that timeless foundation must be like in order for temporal reality to exist.

The first is a question of logical necessity. The second is a question of structure.

The task ahead is therefore no longer simply to ask, “How can timelessness exist?” If the preceding arguments succeed, reason has already supplied an answer at the most fundamental level: timelessness must be possible because it is necessary. The deeper structural question is now this:

How can temporal activity exist in relation to a foundation that must itself remain timeless?

It is at this point that expansion = contraction can enter the argument—not as the reason timelessness exists, but as a proposed explanation of how a timeless foundation and temporal reality can belong to one coherent system.

Having said that, I still want to show that expansion = contraction is the reason timelessness exists, because I want my model to be built in such a way that all parts of the system can beautifully describe one another.

The reason comes down to infinite regress again. We cannot have it, and if timelessness solves it, we can then ask what kind of timelessness is required. We could envision a static timelessness that never moves. Having never moved, it solves the infinite regress by not opening the regress in the first place. If nothing ever changes within this state, there is no succession of states. There is no State A followed by State B, and therefore no earlier state that must be invoked to explain a later one. Without succession, the question “What came before this state?” cannot generate an infinite chain, because there is no “before” within the timeless state from which such a chain could begin.

But I will argue that such a timelessness cannot be, because real motionlessness requires fast motion rather than motion so slow that it stops.

4.4 How fast is the flash?

Let us try to make motionlessness. Imagine an object going slower and slower and slower until it stops. How do you know it has stopped? You cannot, because you can only measure

its movement relative to something else. An object that is motionless relative to one thing can still be moving relative to another. Simply reducing its speed to zero in some reference frame therefore does not give us absolute motionlessness.

So let us try going the other way. Instead of making motion slower and slower, imagine increasing speed toward the ultimate limit represented by light. Here something remarkable happens. According to relativity, light follows a path along which the proper time is zero. There is no ordinary inertial rest frame of a photon, so we cannot literally describe what light “experiences,” but the relativistic limit nevertheless gives us something fundamentally different from merely bringing an object to rest relative to another object: along a lightlike path, no proper time elapses.

This is the sense in which I want to explore an escape from ordinary relative motion. I do not mean that an object can physically accelerate beyond relativity or that a massive object can reach the speed of light. Rather, I mean that the limiting structure of relativity itself points toward a condition in which elapsed proper time disappears. Instead of approaching motionlessness by reducing motion toward zero, I therefore want to investigate whether fundamental motionlessness can be understood from the opposite direction: through maximal motion, where the temporal measure associated with ordinary motion reaches its limit.

Again: You can know that an object has stopped relative to a chosen reference frame. For example, a cup can have velocity $v=0$ relative to your table.

But you cannot establish that it has stopped absolutely, independent of every reference frame. In special relativity, there is no privileged universal state of rest. Another observer moving relative to the table will measure the cup as moving.

You can establish relative motionlessness, but not absolute motionlessness merely by measuring zero velocity in a reference frame.

Thus, let us go the other way.

Borrowing from The Imaginarium of Dr. Einstein: Let us imagine two trains, and let's call them Train A and Train B. Let Train A locomote as fast as you can imagine. Let Train B locomote at half the velocity.

Now, imagine Train A again, but this time, double its speed, and keep doubling until Train A, from its perspective, looks back at Train B, and... what does Train A see?

Train A sees Train B as frozen in time. That is, as not moving at all.

But Train B is moving. Or is it?

If Train A covers the distance twice as fast as before, then Train A must see Train B as moving twice as slowly as before. Therefore, Train B has now covered half the ground it covered before. And if you keep doubling, then Train A will eventually move so fast relative to Train B that Train B is unable to cover any ground by the time Train A has covered its ground, and continually so.

You can represent this as a scale that grows and grows, in which the distance traversed by Train B gets smaller and smaller until the only meaningful distance traversed by Train B is marked as a dot—or not at all.

In summa: As the ratio is increased without bound, the distance traversed by Train B approaches zero relative to the distance traversed by Train A. In the mathematical limit, that ratio is zero.

Note: If these are physical trains, special relativity does not permit you simply to double Train A's velocity indefinitely. A massive object cannot reach or exceed c , and relativistic velocities do not combine through ordinary addition. Also, Train A does not literally acquire a privileged frame in which a moving Train B becomes absolutely motionless.

4.5 Section title to be added

But this is what physics says, and physics uses mathematics to say so. But who says mathematics is necessarily up to the task of understanding true speed? Perhaps nature speeds ahead even faster, and physicists will simply say that the mathematics breaks down. But that does not mean that nature itself breaks down.

A mathematical model of nature is not identical to nature itself. If a model reaches the boundary of its applicability, it does not necessarily follow that reality itself reaches a boundary. However, I recognize that this observation alone does not establish that nature permits speeds beyond c , nor does it establish that greater speed produces absolute motionlessness. It merely leaves the possibility conceptually open, provided that my other premises can support it.

I have argued independently that a timeless foundation must exist, and I identify fundamental timelessness with fundamental motionlessness. Ordinary slowing cannot produce absolute motionlessness, because $v=0$ only represents rest relative to a chosen frame. Thus, instead of continuing to slow things down, I propose looking in the opposite direction: increasing rather than decreasing motion. My hypothesis is that, at the fundamental limit, maximal motion and motionlessness become descriptions of the same underlying state.

There could be other possibilities, or absolute motionlessness, as I have defined it, might not be representable by velocity at all.

But that doesn't ultimately make sense to me because I concluded that existence is possible, and therefore the stuff of existence must create the timeless state I'm looking for, because I don't have any other stuff from which I can form timelessness.

And why do I need to form timelessness from the stuff of time? Because you cannot generate time from timelessness. They have to be the same thing seen from two different perspectives, because there was never a transition from timelessness into time when there was no time in timelessness for the transition to occur.

For illustration, the colors of the rainbow (time) have to make the white light again (timelessness), because there were no colors before the white light that could give the colors within the white light their being. And you say, of course, because the colors are the white light.

And that is my point: spacetime is the same as timelessness, as seen through an animal interface. But we will get to that later.

My proposal can therefore be stated more carefully: If fundamental motionlessness cannot be obtained by removing motion, perhaps it must instead be found within motion itself.

This possibility becomes particularly important because I am trying to eliminate the need for a transition between two fundamentally different conditions. If absolute rest were literally the complete absence of motion, and motion subsequently appeared, I would immediately encounter the same problem again:

rest $\rightarrow$??? $\rightarrow$ motion

What caused the first motion?

If, however, the deepest motionlessness is already constituted by activity, I no longer require a transition from absolute inactivity into activity. I can instead investigate the possibility that:

Fundamental motionlessness = fundamental motion.

If this can be established, then the question “What started motion?” potentially dissolves rather than requiring another cause. Motion was never produced from its opposite because, at the deepest level, motionlessness and motion were never truly opposites in the first place.

My next task, therefore, is to show why maximal motion is a plausible candidate for the relevant timeless or motionless condition rather than merely asserting that it is. The train and limit argument provides a conceptual starting point for investigating this connection. Whether maximal motion actually corresponds to such a condition in physical reality would ordinarily be a question for physics.

But what if mathematics itself reaches its limit here? Can physics describe a condition in which the very measurements upon which physics depends—time, distance, velocity, and trajectory—no longer exist? Perhaps at such a boundary the philosopher can at least attempt to see, through the mind’s eye, what cannot yet be captured mathematically. Let us try.

Imagine Superman flying so fast that time stops. I am not claiming that established physics permits Superman to reach such a speed; this is a thought experiment. I want to ask what would follow if maximal motion corresponded to a condition in which no time elapsed.

If everything were eternally frozen, then nothing within that condition would ever have moved from one state to another. And if nothing had ever moved, then we could not meaningfully speak of things being “eternally frozen” in space either. To identify something as frozen ordinarily requires a history or trajectory against which its lack of change can be understood. But without time, there is no history, and without history there is no trajectory from one position to another. The distinction between moving and frozen therefore begins to disappear.

Push the thought experiment further. From within this hypothetical timeless limit, there would be no succession of states by which motion could be measured, no trajectory by which distance travelled could be established, and no external clock against which duration could be compared. In that sense, what began as maximal motion would become indistinguishable

from motionlessness—not merely relative rest, but a condition in which the ordinary measurements that distinguish motion from rest no longer apply.

This would not be ordinary spacetime as we experience it. It would be a state containing no internal temporal measurement by which one event could be placed before or after another. If there is also nothing outside this total system against which it can be measured, then there is no external reference frame that could slow it down, speed it up, or establish its motion relative to something else. Its motionlessness would therefore not be motionlessness relative to another object. It would be nonrelative.

And if this condition contains no succession by which it could begin, change, or end, then calling it “eternal” must not mean that it persists for an infinite amount of time. It means something more radical: time does not pass within it at all.

Therefore, if I push the logic of this thought experiment to its limit, perhaps what I am looking at is not an infinitely long period of frozen time. Perhaps I am looking at eternity itself—not endless time, but the absence of temporal succession.

This does not yet prove that nature actually works this way. But it gives me a conceptual picture of what my proposal means: at the deepest level, maximal motion and absolute motionlessness may cease to be opposites. They may instead be two descriptions of the same timeless condition.

So, now we are at the point where we can meaningfully begin to solve the Unmoved Mover problem and actually answer the question:

4.6 What started motion?

Here is my answer: Nothing started motion. Motion does not need to be set going because it is timeless. We used the analogy of Superman flying so fast that time stops, yet he is still flying. That is to say, if real motionlessness = all motion, then we have a real solution to the unmoved mover problem because motionlessness doesn’t need to be set going because it is obviously not going anywhere and motion needs to be set going, but because motion is now the same as motionlessness, we can use motionlessness to explain what started motion. That is, motionlessness [motionlessness = motion] doesn’t need to be set going because it is obviously not going anywhere.

Consider the question: What set motionlessness going? The answer is nothing, because motionlessness is not going anywhere. There is no transition that needs to be caused. But if fundamental motionlessness = fundamental motion, then the same answer must apply to motion. Nothing set motion going because, at the fundamental level, there was never a transition from motionlessness into motion. There was only one state all along.

Therefore, motionlessness does not provide a cause that starts motion. It does something more fundamental: it explains why motion never required a starting cause at all. The question “What started motion?” assumes that motion was once absent and subsequently began. My proposal denies that assumption. Motion was never produced from motionlessness because, at the deepest level, motion and motionlessness are the same thing.

The key to understanding what I mean is to distinguish a property from the means by which that property is measured. Imagine that Hercules is so strong that no one can beat him in an arm-wrestling match. Suppose, eventually, there is no one left against whom Hercules can measure his strength. Does Hercules suddenly become weak? Of course not. The disappearance of something against which his strength can be measured does not entail the disappearance of his strength. Hercules remains strong whether or not there is an opponent available to demonstrate it.

I propose that the same distinction applies to fundamental motion. If motionlessness exists in a nonrelative state—a state in which there is no larger or smaller, no external clock, no ruler, no second object, and no external reference frame—then there is nothing against which its motion can be measured. But it does not necessarily follow that the underlying motion has disappeared. The absence of a measurement is not necessarily the absence of the thing that would otherwise be measured.

This is where relativity enters my picture. I treat relative measurements as something like the props placed upon a stage so that the stage becomes visible to the audience. Distance, duration, trajectory, larger, smaller, faster, and slower allow motion to become measurable from within spacetime. But the props do not necessarily create the underlying activity any more than a ruler creates the height of the person standing beside it.

It is not the ruler that makes you grow.

Likewise, in my model, it is not the existence of a reference frame that creates motion. A reference frame allows motion to be expressed and measured relationally. Remove every external reference and the measurements disappear, but I propose that the underlying motion does not.

This gives the Unmoved Mover argument a different meaning. Motionlessness does not explain motion because motion somehow emerges from a state containing no motion. Motionlessness explains motion because motionlessness itself consists of motion. Superman is still flying.

Therefore, when I ask, “What started motion?” my answer is: nothing. There was never a motionless state devoid of motion that subsequently had to be set going. The fundamental motionlessness was already moving. What disappears at the timeless level is not necessarily motion itself, but the relative framework through which motion acquires measurable distance, duration, trajectory, and speed.

In this sense, my proposal is not:

motionlessness → motion

It is:

motionlessness = motion.

The first formulation requires a transition and therefore asks for a cause. The second requires no transition. Motion does not arise from motionlessness because the motion was already there. The Unmoved Mover is therefore unmoved not because it lacks activity, but because its activity does not constitute a transition from one fundamental state into another.

This is the central insight I am proposing: absolute motionlessness is not zero activity. Absolute motionlessness is activity without an external relation by which that activity can be measured as movement from one state to another.

But then you can ask: What set motionlessness going if it is real motion?

But if you have paid close attention, we have already dissolved this question, because “set going” here presupposes the stage and the props—that is, relativity. But we have no relativity here, so the question misfires.

To ask what “set something going” assumes at least two distinguishable states: a state in which it was not going and a later state in which it was going. It therefore assumes a before and an after. It also assumes some meaningful distinction between rest and motion by which the change from one to the other can be identified. In other words, the question already assumes the very stage upon which change can occur: time, succession, comparison, and relation.

But none of these things are available at the fundamental level I am describing. There is no earlier state in which motionlessness was sitting still and waiting to be activated, because “earlier” does not exist there. There is no external clock against which its activity can begin, no external object against which its speed can be measured, and no outside reference frame from which we can say that it changed from rest into motion.

This is why asking “What set motionlessness going?” imports the props of spacetime into a condition in which those props do not yet exist. The question secretly assumes a sequence:
not moving → something happens → moving

But that sequence is precisely what my proposal denies at the fundamental level. There was never a first state of inactivity followed by a second state of activity. The activity simply belongs to what the timeless foundation is.

Therefore, saying that motionlessness consists of real motion does not require me to identify an event that started that motion. “Starting” is itself a temporal concept. To start is to occupy one condition and then enter another. Without temporal succession, there can be no such transition.

The question “What started the motion?” therefore does not expose another earlier cause that I have failed to explain. Within my proposed framework, it applies a temporal question to a state in which the temporal structure required for that question is absent.

The motion is real, but its beginning is not missing.

There is no beginning to find.

CHAPTER 5

The Unmoved Mover

5.1 What Aristotle was asking

In the previous chapter, I argued that the activity at the foundation of reality did not need an earlier event to start it. This brings us to Aristotle and his idea of the unmoved mover. But before comparing his answer with mine, we need to be clear about his question. Aristotle believed that motion had always existed. He was asking what makes this eternal motion possible, and whether its deepest source could itself need something else to set it going (*Physics* VIII.1, VIII.5–6; Bodnár 2023, sections 4–5).

This matters because saying that motion never began leaves another question open: what does motion depend on? To understand Aristotle, we must take that question seriously. His unmoved mover was never simply a motionless object that gave the universe its first push.

My proposal is that motionlessness and activity are two aspects of the same reality at the foundational level. I need to show how this differs from Aristotle's account and what it helps us explain. I must also recognize what Aristotle had already understood.

The central distinction in this chapter is between being active and becoming active. We also need to explain how a timeless foundation relates to changes that happen through time. My proposal offers a way to approach that problem. Whether it gives a complete explanation of reality is a further question. Throughout the chapter, I will keep Aristotle's argument separate from the argument I am developing.

5.2 What Aristotle means by motion and activity

Aristotle uses the word motion more broadly than we usually do. His Greek term *kinēsis* includes changes in quality or size, as well as movement from one place to another. In *Physics* III.1, he explains motion through the relation between what something can become and the change actually taking place (201a10–11).

Consider a house being built. The materials can become a house, but the finished house does not yet exist. While building is underway, that possibility is being put into action. Aristotle calls this the actuality of what is potential, considered as potential. In simpler terms,

the change is happening while the thing is still on its way to becoming something else. The process of building and the finished house are therefore different. We should not use motion as a name for every kind of activity (*Physics* III.1–3).

This helps us understand how something could be unmoved and still active. Aristotle's word *energeia* can mean activity or actuality. Actuality concerns what is real or in operation, rather than merely possible. The exact meaning depends on the passage.

In *Metaphysics* IX.6, Aristotle distinguishes seeing from building. Seeing achieves its purpose while it takes place: when I see, I am already seeing. Building aims at a result that has not yet been completed. The house is still being made (1048b18–35). Scholars disagree about how to read this passage and whether it belongs where it appears in the text. We should therefore avoid making the whole argument depend on it. Book XII, also called Lambda, directly describes the first principle as living and intellectually active (*Metaphysics* XII.7).

For my argument, the important point is that activity and change are not automatically the same thing. Saying that something is active does not necessarily mean that it is becoming different. But a process that passes from one stage to another still involves change. I must keep this distinction clear when I call my foundation dynamic.

There is also a difference between Aristotle's *energeia* and energy in modern physics. His account of activity gives us no measurement of energy or proof that energy can be made available for use. The later energy argument in this paper will need its own definitions and reasons.

5.3 Why motion does not need a first moment

In *Physics* VIII.1, Aristotle asks whether motion began or has always existed. His argument is that any proposed first motion would already depend on an earlier change. Suppose the things that move, or cause movement, had first come into existence. Their coming into existence would itself be a change. Or suppose they had always existed but had done nothing. Something would then have to become different for them to begin acting. In either case, the supposed first change would have another change before it (*Physics* VIII.1, 250b11–252b6).

This connects with the problem raised in Chapter 4. If we say that timelessness existed and then became time, the word then already introduces the order we are trying to explain. We have quietly placed a before and an after inside timelessness.

Even so, Aristotle's claim and mine are different. He argues that motion and time have always existed. I propose a foundation that does not itself pass through earlier and later states. Time having no beginning and a foundation having no passage of time are two different ideas. Rejecting a first moment of activation does not make the two accounts identical.

Aristotle also connects time with counting motion according to before and after. He distinguishes time from motion itself: motion can be faster or slower, and time helps us measure that difference (*Physics* IV.11, 219b1–2). He later distinguishes things whose existence is measured by time from things that exist always (*Physics* IV.12). His account therefore requires more care than simply treating eternity as a very long stretch of time. These passages do not, however, prove my particular account of timelessness.

Aristotle already rejects a first beginning of cosmic motion. My contribution must therefore go further than saying that motion never started. I need to explain what the foundation is and how motion through time belongs to it.

5.4 Why always existing may not be enough

A system could have existed forever while every part still depended on something else. Imagine that each part works only because another part supports it. Knowing the system's age would not explain that support. This does not prove that an infinite system is impossible. It shows that how long something has existed and what makes it possible are different questions.

Aristotle gives a useful example in *Physics* VIII.5: a hand moves a stick, and the stick moves a stone. The stick moves the stone because something is using the stick. The question concerns what gives the stick its ability to act in this situation. Finding the oldest stick in history would not answer it.

Aristotle also examines things that appear to move themselves. Within such a thing, he distinguishes what causes movement from what is moved. Calling the whole thing self-

moving does not remove the need to understand those roles. Gill's discussion explains this connection between ordinary changes and an unmoved source within something that moves itself (Gill 1994, opening argument).

The argument for an eternal first principle continues in *Physics* VIII.6 and *Metaphysics* XII.6. Something may be able to cause motion without actually doing so. Even if it had that ability forever, we could still ask why the ability is being used. Aristotle therefore argues that the deepest source must already be active. It cannot merely be waiting for something to activate it (*Metaphysics* XII.6, 1071b12–22). Bodnár explains the argument in terms of an activity that supports the eternal movements of the heavens without itself being subject to the changes it explains (Bodnár 2023, sections 4–5).

We can summarize the reasoning as follows. This brings together several passages; Aristotle does not present it as this one list:

1. Motion exists and needs an explanation.
2. The ability to cause motion does not, by itself, explain why motion actually happens.
3. If a source acts only because something else enables it to act, we still need to explain that dependence.
4. A complete explanation of eternal motion must therefore reach a source that does not need something else to set it going.

Each step needs examination. A critic might question whether a chain of dependent sources must end in an independent source. They might defend an infinite chain, or ask whether a necessary pattern could explain the whole. Aristotle's conclusion does not become unquestionable simply because he calls something a first principle. Likewise, my rejection of unexplained facts must be supported by an argument about what a complete explanation requires.

This also helps clarify the expression “No time = no regress.” Without time, there cannot be a chain of earlier and earlier moments within the foundation. But we could still ask why a structure exists, why a law holds, or why two things have a certain relation. These questions need not concern earlier events. A timeless model must answer them wherever they still make sense.

5.5 How something can cause motion without changing

Aristotle recognizes different ways of explaining a thing. His four causes concern what it is made of, how it is formed or organized, what brings about its change, and the end towards which it acts. More than one of these answers can apply to the same thing (*Physics* II.3, 194b16–195a3; Falcon 2023, section 2).

The last kind is called a final cause. It explains an activity through its goal. Someone may walk for the sake of health. Health helps explain why they walk, even though health is not a physical object pushing them along. Aristotle uses this example himself (*Physics* II.3). It shows how a goal can explain an action. By itself, it does not prove that the heavens move towards a goal in the same way.

In *Metaphysics* XII.7, Aristotle compares the unmoved mover with something that is desired or thought about. He says that it moves “as being loved” (1072b3, trans. Ross). On the usual reading, it acts as a final cause: what is desired helps explain the activity directed towards it. The thing desired does not have to change in order to play that role. This is one way Aristotle explains how an unmoved source could account for motion (*Metaphysics* XII.7, 1072a26–b4).

Scholars disagree about how this fits with the account in the *Physics*. Bodnár reads the two works as describing the same principle from different sides. On his reading, the mover also has a role as an efficient cause, meaning a source that brings about change. The *Physics* discusses the mover, while Lambda discusses desire, but these need not be two unrelated explanations (Bodnár 2023, section 5). Other readings place more weight on the mover as a goal. The dispute is about how the first principle causes motion.

For my argument, the lesson is that an explanation of motion need not always take the form of an earlier push or intervention. But any alternative still needs to explain something. If I say that activity belongs to the whole, I must show how this helps us understand the changes that happen through time.

5.6 Thought thinking itself

Aristotle gives the unmoved mover a positive description. In Lambda, it has life and the highest kind of intellectual activity. He then asks what this intellect thinks about. If the value

of its thinking depended on something less perfect, it would depend on that lesser thing. If it changed between objects in a way that made its activity worse, it would also fall short of what the first principle must be. Aristotle therefore describes its thinking as directed towards itself (*Metaphysics* XII.9, 1074b15–35; Cohen and Reeve 2025, section 14).

The phrase “thinking on thinking” (1074b34, trans. Ross) can sound like an endless chain: one thought watching another thought, which watches another. That is a misleading picture. Aristotle connects thinking with what is thought about in the case of things without matter. Here, the thinking and its object belong to one activity. Cohen and Reeve explain how this answers a problem: if divine thinking depended on a separate object, that object might be more basic than the intellect itself (2025, section 14).

This is relevant to my interest in consciousness, but it does not establish my own position. Aristotle’s word *nous* means intellect or understanding. It does not automatically cover everything we now mean by consciousness. My claim that conscious experience is fundamental still needs its own argument.

Scholars also disagree about how divine understanding relates to human understanding. Hannah Laurens challenges the idea that the intellectual condition described in XII.9 belongs only to the Prime Mover. Her argument points to a connection between the best human thinking and the divine condition described in XII.7 (Laurens 2024). This is a debate about how widely Aristotle’s account of intellect applies. It does not prove that all physical things are experiences within one cosmic mind.

The discussion also helps clarify my use of the word “feedback,” which we will explore later. In an ordinary feedback process, a result from one stage affects a later stage. That already involves time. If I use feedback to describe a timeless foundation, I need to explain what the term means without that delay. I might mean that the foundation understands itself, or that its aspects help make each other what they are. These are possibilities to explain, not answers supplied by the word feedback alone. A relation involving itself does not automatically prove that it exists, is conscious, or must exist. But I will return to all of this later, where I believe we can take the inquiry deeper than anyone in history has been able to take it before. This is possible, in part, because of the groundbreaking work of Donald D. Hoffman.

Aristotle therefore gives us an earlier example of a foundation that is active without undergoing ordinary change. He also shows why we need to say what that activity is. Denying that it ever began leaves that task open.

5.7 Is the mover separate from the universe

Another difference concerns the relation between the first principle and the heavens. Lambda describes a substance separate from the things we perceive, without physical size or parts into which it can be divided (XII.7, 1073a3–13). In *Physics* VIII.10, Aristotle links the absence of bodily size with the ability to account for motion through unlimited time. Whatever we think of his reasoning, he is not describing a body spinning extremely fast.

Scholars disagree about how to understand this separation. Lindsay Judson argues that the mover exists separately from the heavenly sphere, against the reading proposed by Aryeh Kosman (Judson 1994, opening statement). This matters for my model, because I am asking whether the foundation and the universe's activity belong to one reality. A reading of Aristotle that brings them closer together would still need support from his texts. I cannot accept it merely because it resembles my own view.

There is another complication. We often speak of the unmoved mover as though Aristotle discussed only one. In XII.8, however, he connects several unmoved substances with the movements of the heavens. He discusses forty-seven or fifty-five spheres, depending on the astronomical assumptions being used. He still gives one mover the primary place. His account therefore includes an order of movers. We cannot simply read it as a claim that one divine mind is everything that exists (*Metaphysics* XII.8, 1073a14–1074a38).

These details also limit what I can borrow from Aristotle. Some of his conclusions depend on his account of the heavens. I cannot remove those conclusions from their supporting arguments and treat them as independently proven. I can continue to examine the problem he raised, but I need reasons for my own answer. A recurring shape or pattern of motion does not, by itself, show what its deepest foundation is.

Calling everything Nature does not settle the matter either. The whole could include different kinds of things, some depending on others. Defining Nature as everything that exists tells us that nothing falls outside it. Showing that the timeless foundation and the

universe that changes through time are two aspects of the same reality is a further claim. That claim needs an argument.

5.8 Where my proposal differs

We can now state the difference more clearly. Does moving reality depend on a separate unmoved source? Or does its activity belong within one whole that needs nothing beyond itself? My proposal takes the second route. The whole does not acquire activity through an event. Activity belongs to what it is. I must then explain how the changes we experience belong to the same reality that I describe as timeless at its foundation.

This is what I mean by “motionlessness = all motion.” The equals sign says that both descriptions refer to the same underlying reality. Motionlessness describes a foundation that does not pass through a change in order to become what it is. Motion describes the activity through time that belongs to that reality. For the statement to make sense, I must explain both descriptions and how they fit together.

Aristotle’s principle of noncontradiction is relevant here. In simple terms, we cannot say that the same thing both has and lacks the same property at the same time and in the same sense (*Metaphysics* IV.3, 1005b19–20). I cannot solve the problem by saying that the foundation both undergoes and does not undergo exactly the same change. I need to explain the different senses in which I call it motionless and active.

The images of fast motion in Chapter 4 help us imagine the proposal, but they have limits. These examples do not, by themselves, prove that there is no passage from one state to another. How something appears is not enough to establish what it is. The argument needs to go beyond the image that inspired it.

We must also distinguish having no surroundings from having no internal change. A whole could have nothing outside it while its parts still change in relation to each other. The absence of an outside clock or ruler therefore does not establish timelessness. My stronger claim is that the foundation itself does not pass through states that follow one another, while the world that changes through time belongs within the same reality. Stating that claim tells us what needs explaining.

My disagreement with a reading that makes Aristotle's source separate can therefore be kept clear. I seek a foundation whose connection with the universe that changes through time belongs to the nature of one reality. Aristotle had already considered eternal motion and an active first principle. My question is whether the unity I propose can explain their connection more fully, without leaving the relation between foundation and universe unexplained.

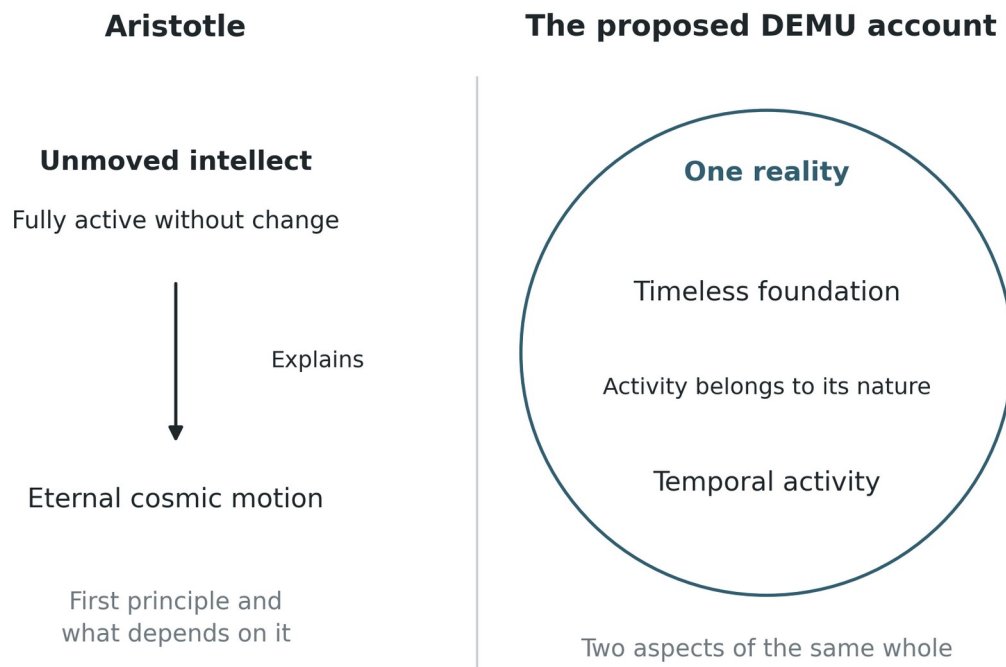


Figure 5.1. Comparison developed in Chapter 5. Aristotle distinguishes an unmoved first principle from the cosmic motion it explains. DEMU describes foundational timelessness and temporal activity as aspects of one reality. The qualifications and interpretive disputes are discussed in the text.

5.9 What my proposal would show

We can set out the argument against a first activation in five steps:

1. For activity to start, there would have to be a change from its absence to its presence.
2. That change would require a before and an after.
3. The proposed foundation does not pass through a before and an after at the foundational level.
4. Its activity belongs to what it is, rather than being gained through another event.
5. Therefore, if this foundation exists as described, its activity has no first moment of activation.

There are three different tasks here. First, I need to show why the foundation requires no first activation. Second, I need to explain how activity through time belongs to it. Third, I need to explain why this foundation exists and requires no further support. Completing one task does not automatically complete the next. An account could explain how a timeless whole relates to the changes within it while still leaving the existence of the whole unexplained.

But if you read Chapter 4 carefully, you will see that we have already made some headway here. There, I argued that reality requires a timeless foundation and that its activity does not need an event to start it. Steps 3 and 4 carry that reasoning forward. The conclusion makes clear what follows from it: the foundation requires no first activation. This does not mean that every event in the universe must happen or that ordinary changes have no causes.

Within this framework, asking which event first activated the foundation assumes a before and an after where the account allows neither. Chapter 4 has already explained why this question does not apply. A critic may challenge the arguments given there, but the absence of a starting event is not, by itself, a missing part of the explanation.

The next task is to develop the structure of the model. Having argued for a timeless foundation, I now need to explain how activity through time belongs to it and how expansion and contraction could form one coherent system. Chapter 4 provides the starting point for this work. The following chapters will examine whether the proposed structure can meet the requirements that the earlier arguments have established.

So, my proposed way forward is reciprocal balance: expansion and contraction depend on each other within one basic relation. If the account works, neither needs a separate event to bring it into contact with the other. But balance alone does not remove time. Equal opposing quantities, an unchanged total, or a repeating path can all be described as occurring through time. I must explain what the relation between expansion and contraction is and why it is timeless at the foundational level.

The dual torus is a proposed way of representing that relation. It is a shape with spatial dimensions, so the shape alone cannot prove that the foundation has no spatial extension. I need to explain which features show something about the foundation and which belong only to the image. A diagram can show how two aspects relate without showing why they must exist. The next chapters need to answer the questions that the shape alone leaves open.

5.10 Questions the proposal still needs to answer

The first objection is that I may have changed the words without explaining the problem. If motionlessness simply becomes another name for fundamental activity, the new description may add nothing. I need to show why both descriptions apply to the same reality and what this helps us understand. A fuller account must explain how different events through time can belong to one timeless whole.

The second objection concerns dependence within the whole. Saying that two aspects belong to one reality does not show that their relation needs no further explanation. We can still ask why they exist together, why they have this relation, and how it allows changes through time. Saying that nothing exists outside the whole does not answer every question about what happens within it or why it is as it is.

The third objection is that another model might meet the same requirements. Someone could propose a timeless structure that could not fail to exist, without circulation between expansion and contraction. They might explain an eternal order through the principles that make it what it is. Such accounts may fail, but I would need to show why. Ruling out a first activation does not, by itself, establish the dual torus as the only answer. My rejection of a static foundation must explain why such a foundation could not do the work required of it. We have already addressed this in Chapter 4, but we will examine it in more detail in the following chapters.

The fourth objection asks what activity without a before and an after actually means. Aristotle approaches this through intellectual activity. My model needs an explanation of its own. If activity means changing position or condition, it seems to involve one state following another. If I mean that the foundation is active simply through the way its aspects relate, I must explain how that relation supports the changes we experience. Using activity in both senses without explaining their connection would leave the problem unresolved.

Finally, I must avoid building later conclusions into the definition of the model. This chapter does not establish an entropy reset, renewed usable energy, immortality, or a conscious cosmic mind. Those claims need further arguments. In particular, showing that the foundation needs no first activation would not, by itself, tell us what happens to information during processes that take place through time.

These objections show what remains to be done. I retain the proposal that activity belongs to the foundation's nature. I now need to explain that proposal clearly, show that it is coherent, and establish what it explains that competing accounts leave unanswered.

5.11 Towards a complete explanation

Aristotle helps us state the problem more clearly. Even if the universe had no first event, we could still ask what its activity depends on. Aristotle connects eternal motion with a first principle that is actual and intellectually active. My proposal also rejects the need for an earlier event to activate the foundation. It takes a different approach to the relation between that foundation and the universe that changes through time.

The strongest conclusion available here is conditional. If activity belongs to a reality that is timeless at its foundation, then asking which event first set that activity going misunderstands the proposal. The remaining task is to explain why such a reality exists, how activity through time belongs to it, and why it needs no further foundation.

Chapter 6 takes up that task. A complete answer to infinite regress must do more than remove the need for an earlier moment. It must explain why the proposed foundation is enough. My account of reciprocal balance will be judged by whether it can meet that requirement.

5.12 The naked lady

So what exactly is Aristotle's unmoved mover? Is it like health, in the sense that you move in order to stay healthy? Does existence likewise have a logos that does not move but animates everything else?

Or is it like a picture of a naked lady? The lady does not move, but she makes your pants move. Yes, this is how I tried to understand Aristotle back in the day. What can I say in my defence? I was young. But I have always been a visual learner. I need something concrete to understand something difficult. Saying "the logos did it" never made much sense to me.

Or is Aristotle comfortable with one thing moving another, with the chain extending backwards forever and time continuing endlessly in both directions?

Ultimately, Aristotle's unmoved mover is an eternal, nonphysical intellect that is fully active without undergoing change. In my model, the unmoved mover is motion itself—motion so fast, reaching speeds so extreme, that relativity itself breaks down and we enter a non-relative world. It is a world beyond mathematics, as it were, where mathematics itself breaks down. This is why, in my view, physics never finds the beginning of reality: mathematics does not take us there.

So either we do not need an unmoved mover in Aristotle's sense, or we redefine the term to mean timelessness itself. I am fine with either.

This also clears up some confusion about my previous work. In some works, I have rejected the unmoved mover; in others, I have claimed to have found the actual unmoved mover. These statements express the same position: I reject Aristotle's proposed mover and identify timelessness itself as the foundation. Whether I call that foundation an unmoved mover depends on how I define the term.

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[Physics Book II](#) · [Book III](#) · [Book IV](#) · [Book VIII](#)

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Further specialist reading

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CHAPTER 6

A Global Solution to Infinite Regress

6.1 The way of Idealism

The way idealism proposes to solve the Hard Problem of consciousness is by dissolving rather than solving it in the ordinary sense. That is to say, idealism attempts to solve the problem by identifying that the original problem never arises within its framework in the first place: consciousness does not need to emerge from something fundamentally non-conscious because consciousness is already fundamental.

Likewise, I solve the infinite regress by placing timelessness first in the order of explanation and then keeping timelessness timeless. I therefore do not have a chain that I need to explain.

Why does this solve the infinite regress? Because the regress arises only when each state requires an earlier state to account for it. If State B is explained by State A, and State A is itself another temporal state, then I can ask what explains State A. If the answer is another earlier state, I can repeat the question indefinitely:

... → State C → State B → State A

The regress exists because temporal explanation continually permits the question: What came before that?

A timeless foundation removes the condition that generates this chain. If the foundation is genuinely timeless, there is no earlier state of the foundation. There is no State B before State A, because there is no temporal succession within the foundation at all. Consequently, the question “What came before the timeless foundation?” does not reveal another missing member of the chain. The word “before” requires the temporal ordering that the question is attempting to apply to something defined as timeless.

This is why I consider timelessness a real solution rather than merely an arbitrary stopping point. If I simply selected some temporal state and declared, “The regress stops here,” the problem would remain. That state would still possess a temporal location, and I

could legitimately ask what preceded it or why it existed at that particular point. I would merely have refused to continue asking the question.

Timelessness is different. I am not stopping an otherwise valid chain at a convenient location. I am proposing that the structure responsible for generating the chain—temporal succession itself—is absent at the foundation. There is therefore no next question of the same kind to ask.

This distinction is crucial. I am not saying:

State C → State B → State A → timelessness → ???

That would merely extend the regress by another step.

I am saying that temporal states ultimately have their foundation in something to which the relation “earlier than” does not apply. Once we reach that level, the regress of earlier states cannot continue because the relation required to construct the regress is no longer present.

This is also why I must keep timelessness timeless. If I were to say that timelessness existed for some condition and then transitioned into time, I would immediately recreate temporal ordering at the foundational level:

timelessness → transition → time

I could then ask what explains the transition, what condition preceded it, and why it occurred when it did. The regress would return at a deeper level. A genuinely timeless foundation therefore cannot undergo a temporal transition into time.

In this sense, my proposed solution resembles the idealist response to the Hard Problem. Rather than constructing an increasingly complicated mechanism to bridge a gap, I question whether the gap exists at the fundamental level at all. Idealism says that consciousness never had to emerge from something wholly non-conscious. My argument says that the foundation never had to emerge from an earlier foundation.

The infinite regress is therefore not terminated by finding the first member of an infinitely long temporal chain. It is dissolved by reaching a foundation within which the relation responsible for generating that chain—before and after—does not exist.

That is the sense in which timelessness provides a real solution: it does not merely stop the regress; it removes the condition that generates the regress in the first place.

If a genuinely timeless foundation exists, the regress of earlier temporal states is structurally dissolved.

Earlier, we said: “The remaining task is to explain why such a reality exists, how activity through time belongs to it, and why it needs no further foundation.”

And now the resolution is clear: reality exists because both nonexistence and paradox fail. Reality must therefore exist and be coherent. To be coherent is to be free of paradoxes such as infinite regress, which gives us a timeless ground as the origin of everything. This foundation needs no further support because that is what it means to be coherent and free of paradox. It was infinite regress that demanded endless support; a reality free of regress can stand on its own two legs.

If infinite regress could be solved by motion slowing down to create motionlessness, then Nature and existence would be all about that. Like a leaf lying on the ground, Nature would be all about lying still. But I am saying that you need to speed things up to create motionlessness: picture a leaf spinning around in a tornado.

Later, I will argue that this spin leads to a spiritual purpose for all.

Activity through time belongs to this foundation because that is how Nature actually solves infinite regress. I argue that static systems fail, nonexistence fails, and paradox fails. This means that only movement is up to the task of solving infinite regress, which explains why motion is here in the first place: it has a job to do.

Kind of obvious in retrospect.

CHAPTER 7

Temporal Activity Within a Timeless Foundation

7.1 The next question

The next question becomes:

How can a temporal universe depend on a timeless foundation without the foundation itself undergoing a temporal transition?

So far, we have a timeless foundation that must remain timeless. We have the origin of motion. We have seen that a static timelessness cannot logically hold, so we now have a dynamic, timeless model of the universe—which also gives the book its title: Dynamic Equilibrium Model of the Universe. DEMU.

All this means that we must build timelessness with motionlessness. To avoid confusion, this also means that we must build timelessness with motion, because we have seen that motionlessness and motion are the same at the foundational level.

We could simply equate motionlessness with timelessness and call it a day. But consider our earlier thought experiment: suppose my finger represents Superman flying so fast that time stops. When I drag my finger through water, I don't get a straight line of movement. I get motion that curves back on itself—a toroidal motion.

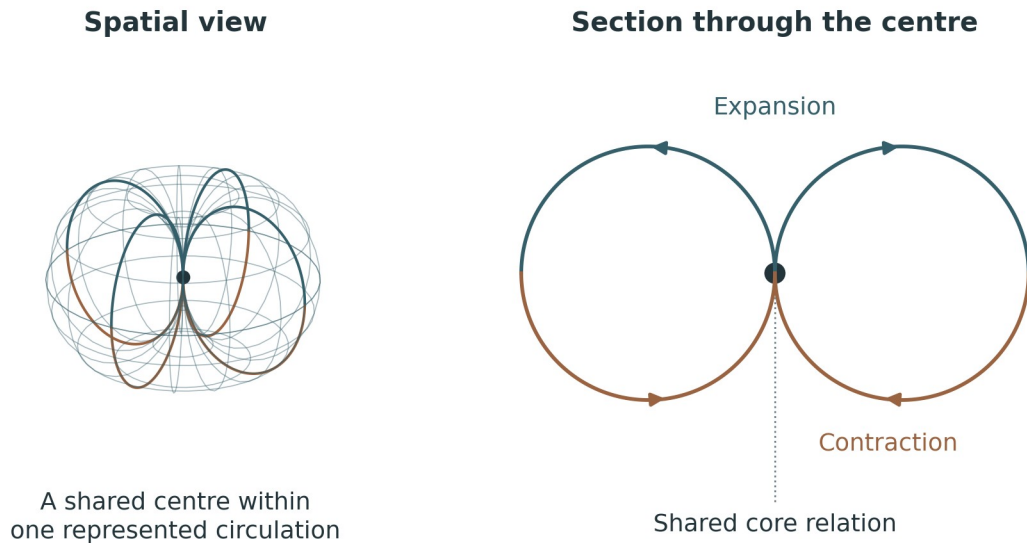
What I'm trying to say is that I don't think timelessness takes the form of an infinite, open line. I think Nature must close in upon itself. Why? Because, at this level of analysis, there is nothing outside Nature. If we define reality as all that is, then anything supposedly outside reality is already included in that definition.

Likewise, I think coherent forms in Nature are forms that close upon themselves, like soap bubbles. You cannot have a soap bubble with a hole in it floating in midair. The bubble must close. I think this logic applies to Nature as a whole.

But even if you don't accept the argument that Nature must be beautiful and closed, there may be a practical way to investigate the idea. If motionlessness really is this kind of extreme speed, what form could produce that speed? I don't think simply having Superman fly in a

straight line would do it. I think you need to spin like a cow caught in a tornado—just to paint a clear mental picture that you won't soon forget.

Complementary circulation in the proposed model



Arrows indicate proposed directions; the drawing has no physical scale.

Figure 7.1. Complementary circulation in DEMU. The left panel gives a spatial representation with a pinched centre; the right panel separates the outward and returning directions in a section through that centre. Both belong to one proposed activity. The shared point represents the foundational relation. The drawing does not specify a physical scale or depict a literal zero-dimensional object.

But there must be a reason Nature uses spin. Could sufficiently rapid rotation allow a path to contract until it becomes zero-dimensional? An endlessly advancing straight path seems to demand unlimited space. If actual infinities are impossible, could Nature have no choice but to turn inward, closing its motion upon itself?

There is a known connection between contraction and faster rotation: **conservation of angular momentum**. A skater spins faster when pulling in their arms. This gives us a concrete image of how contraction and increasing rotation can belong to the same process.

In my account, Nature must contract into a zero-dimensional knot because I have argued that spatial extension leads to infinite regress. But I do not need to prove this particular mechanism in order to defend timelessness: I have already argued for timelessness by

another route. Given that conclusion, and the extended universe we inhabit, I conclude that contraction must be capable of returning extended space to a zero-dimensional condition.

My claim is that this necessity can be established before the details of the mechanism are understood. For comparison, one can know that a woman is pregnant without knowing the circumstances of conception. Likewise, I am distinguishing the argument that something must be the case from an explanation of exactly how it happens.

Then we can finally answer the question: How can we have time? How did timelessness generate time? The answer is that it didn't. Timelessness and time are two sides of the same coin. In my account, timelessness consists of speed, which the animal mind transforms into time, much as white light can be separated into a rainbow of colors. Donald Hoffman takes timelessness as his starting point and seeks to reconstruct the universe of time from it.

For this account to work, consciousness must be eternal: it must be what the timeless system feels like from within. This opens up the possibility of free will, which I will argue for later. Consciousness would no longer be trapped in the web of time. Instead, it would wear time like a headset and use it as a tool, rather than be governed by it.

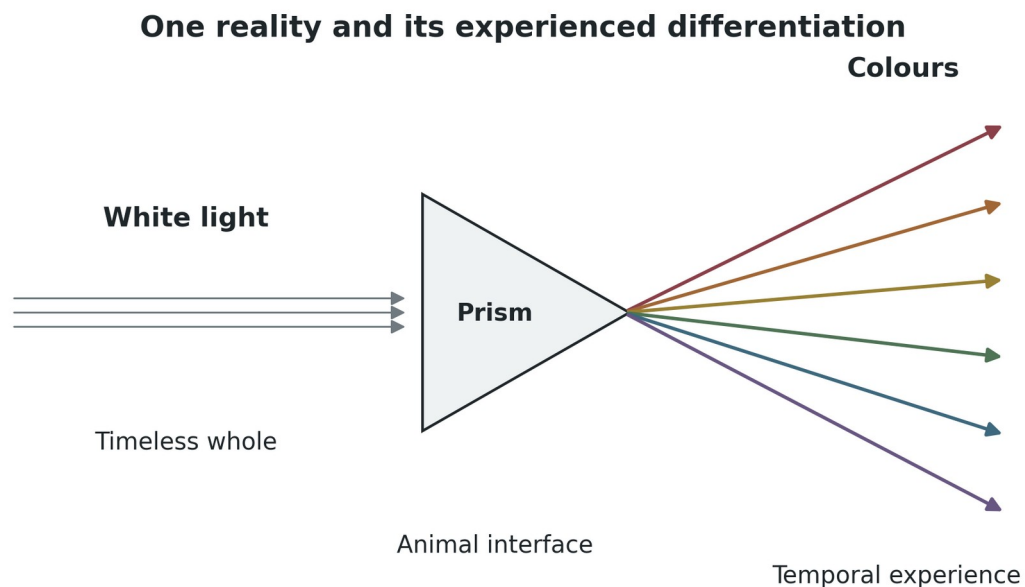


Figure 7.2. The analogy used in Chapter 7. White light represents the timeless whole, the prism represents the animal interface, and differentiated colours represent temporal experience. This illustrates the proposed relationship rather than a mechanism of time generation.

CHAPTER 8

The Emergence of High Entropy

8.1 The next problem

We now have an account in which time belongs to a timeless reality. The foundation does not leave timelessness behind and become something else. Its balance remains, while the activity within the whole is experienced as a universe of change.

But this raises a question. If the foundation is perfectly balanced, why do things run down? Why does coffee go cold? Why does useful energy become harder to use? In the language introduced in Chapter 2, why does entropy increase?

My answer begins with the difference between the foundational condition and the changing arrangements within the universe. The core has no accumulated history. The temporal world contains differences, interactions, and the outcomes of earlier events. In my proposal, these are two aspects of one circulation.

This chapter examines the part of that circulation in which entropy rises. We need to understand what happens during that part before asking how the whole can remain balanced. The argument for timelessness is already our starting point. Our task here is to explain where high entropy belongs within the account.

8.2 What high entropy means

Entropy is often described as disorder, but that word can be misleading. A room that looks untidy is not automatically a useful example of physical entropy. What matters is how many detailed arrangements fit the condition we are describing. A microstate is one such detailed arrangement. A macrostate is the larger picture, described through features such as temperature and the distribution of matter. [1]

Imagine two different gases separated by a partition. Remove it, and they mix. There are far more microscopic arrangements in which both gases are spread throughout the container than arrangements in which each stays on its original side. The mixed condition has higher entropy. Nothing has become nonexistent; the arrangement has changed. [1]

This gives us a useful starting point for DEMU. The universe contains a variety of arrangements that are absent from the core as I describe it. High entropy concerns the physical condition of these arrangements. It is not a second substance that Nature adds to matter, or an outside force that attacks the foundation.

8.3 Why energy can become less useful

Consider a hot cup of coffee in a cooler room. Energy passes from the coffee to its surroundings as heat. The coffee cools, and the temperature difference becomes smaller. Although the coffee loses entropy as it cools, its surroundings gain more. The combined entropy increases. We must include both sides of the exchange. [2, 4]

The energy has not disappeared. What has diminished is an opportunity to use the temperature difference. A suitable engine could obtain work while heat passes from something hotter to something colder. Once both reach the same temperature, that particular opportunity is gone. There may still be plenty of internal energy, but it is no longer available in the same way. [2, 3]

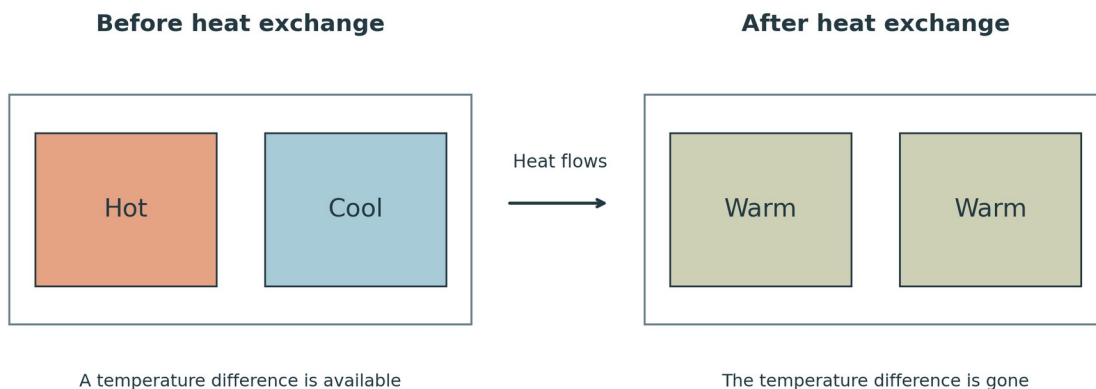


Figure 8.1. Heat transfer between two bodies in an isolated combined system. Their temperatures approach equality while their combined entropy increases. Energy remains, but the temperature difference that could drive a heat engine is lost. [2–4]

This is why the distinction between energy and usable energy matters to my argument. Saying that energy continues to exist does not answer the question of renewal. I am asking how the capacity for further useful activity can be restored.

Within DEMU, the outgoing condition provides renewed opportunities for activity. As ordinary irreversible processes occur, some of that availability is lost. The proposed return to

the core must therefore do more than move the same exhausted arrangement around a circle. It must restore what makes renewed output useful.

8.4 Time allows a history to form

The passage of time does not, by itself, force entropy to rise at every moment. In thermodynamics, an ideal reversible process produces no entropy. Irreversible processes, such as friction and heat flowing across a temperature difference, produce entropy. The second law concerns the total change when the relevant system and its surroundings are included. [2, 4]

So I am not using the word motion as another name for entropy. The speed of an activity alone does not tell us its entropy. This matters because I have described the foundation itself as active. If every kind of activity automatically meant accumulating entropy, I would have contradicted that account before reaching this chapter.

My distinction is between the activity of the timeless foundation and the changing conditions through which we experience a history. Within that history, things interact, energy is transferred, and earlier arrangements give way to later ones. Some of those changes are irreversible in the ordinary thermodynamic sense. That is where increasing entropy enters this part of the explanation.

The phrase “the emergence of high entropy” therefore does not name a first event outside time. It describes what happens within temporal activity. I am not proposing that a timeless core waited, changed its mind, and started producing a universe. The circulation is the whole account; the sequence belongs to the temporal processes within it.

8.5 Why the direction matters

Why do we usually see mixing rather than spontaneous separation? The statistical explanation is that a mixed condition corresponds to overwhelmingly more microscopic arrangements. Starting with a suitably prepared, separated state, ordinary interactions lead towards the much larger range of mixed possibilities. A return to the original separation would require an extraordinarily special arrangement of microscopic motions. [5]

There is a condition built into that explanation: we started with a state that was relatively restricted. Explaining the tendency towards higher entropy does not, by itself, explain why

the lower-entropy condition was available. The explanation of the increase and the explanation of its starting condition are different questions. [5]

This is where my proposal enters. I place the renewed condition within a complete circulation. The lower-entropy output is not a resource that appeared once, for no reason, and can only be used up. In my account, renewal belongs to the continuing relation between the universe and its foundation.

There can therefore be a direction to a particular process without the whole of Nature having a first moment. A cup of coffee can have been hotter earlier, even if reality itself never began. The history of that cup is one part of the temporal world. It does not give the timeless foundation an earlier state.

8.6 How order can exist while entropy rises

Does increasing entropy mean that every part of the universe must become less organized? No. A system can become more ordered while interacting with surroundings whose entropy increases. A refrigerator, for example, can reduce the entropy of what it cools while using work and releasing heat elsewhere. The complete process must be counted. [4]

This distinction matters when I speak about life, growth, and complexity. Their presence is not, by itself, an entropy reset. An organized form within the temporal world and the complete erasure proposed at the core are different claims. The later chapters will connect life and growth to the larger account of balance.

For now, the point is simple: the existence of structure does not make entropy disappear. A living world can contain both organized activity and processes that increase entropy. My model must include both. It cannot describe the interesting forms and then leave the consequences of their activity outside the picture.

Nor should high entropy be confused with evil, failure, or meaninglessness. It names a physical condition. The philosophical question is what place that condition has within Nature as a whole, and whether it can be the final condition of everything.

8.7 Why the core is a different condition

In Chapter 2, I described the core as historically independent. Different returning histories reach the same foundational condition. The core does not keep a collection of those histories inside itself. In my proposal, their inherited distinctions are completely erased.

This is the sense in which I connect the core with perfect order. I mean a restriction on what can remain there: no surviving difference carries information about which history returned. A surface that merely looks smooth would not establish this. Neither would a gas whose temperature is the same throughout. Ordinary thermal equilibrium can contain an enormous number of microscopic arrangements. [1]

The core is therefore not simply a quiet region of ordinary matter. I describe a condition in which the accumulated differences of temporal history no longer remain. The outgoing universe, by contrast, contains differentiated arrangements through which further activity is possible. These arrangements can develop towards higher entropy without making high entropy an inherited feature of the core.

This does not require the whole universe to pass through one simultaneous reset. As already stated in the definitions, my proposal concerns continual renewal in portions. Temporal activity continues while the circulation includes return and renewal. The core remains the same foundational condition throughout that account.

It is also important to keep the meaning of total reset clear. I do not mean that information is merely hidden, moved somewhere else, or made difficult to recover. No record, backup, or recoverable trace of the erased history survives anywhere. The proposed entropy reset likewise does not transfer entropy to an outside store. That is the reset whose role I am developing in the following chapters.

8.8 Why high entropy cannot be the final word

We can now state the problem more clearly. In the temporal world, useful differences can be spent. Energy can remain while its availability for a given task is reduced. In my account, however, the foundation must remain balanced throughout the whole circulation. It cannot be a favourable condition that Nature leaves behind forever.

Suppose renewed output became permanently exhausted and never returned to a condition capable of renewal. We would then have a one-way development from an available resource to an exhausted one. That would not be the complete circulation I am proposing. The outward activity would have lost the return through which the foundation's balance is maintained.

This is why high entropy has a place in DEMU without being its final destination. It belongs to the temporal development of arrangements within the circulation. It does not replace the foundational condition on which that circulation depends.

At the same time, simply drawing a return arrow does not describe the reset. Matter could return to a place while retaining the consequences of its history. My claim is stronger: return to the timeless core includes the removal of that inheritance and the restoration of the condition required for renewed output. The argument concerns what the return accomplishes, not only where it goes.

Chapter 9 will examine why balance must hold throughout the eternal circulation. Chapter 10 will develop the connection I draw between entropy and infinite regress. Chapter 11 will explain the reset and the renewal of matter and energy. The question we carry forward is therefore this: how can Nature include the rise of entropy while continually preserving the foundation that makes further activity possible?

References for Chapter 8

- [1] Feynman, Richard P., Robert B. Leighton, and Matthew Sands. 1963. Ratchet and Pawl. The Feynman Lectures on Physics, vol. I, sections 46–4 and 46–5. Caltech online edition. Online source.
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CHAPTER 9

Balance Throughout the Eternal Circulation

9.1 Balance cannot be left behind

In Chapter 8, we examined how entropy rises within the temporal world. We now need to place that activity within the whole. My claim is that the foundation remains balanced throughout the eternal circulation. Outward activity, return, and renewal all belong to that balance.

This follows the order of the argument already developed. Chapter 4 introduced the timeless foundation. Chapters 6 and 7 explained why the foundation cannot leave timelessness behind and become a temporal thing. Here I am asking what that requirement means for the circulation as a whole.

Imagine saying that Nature was perfectly balanced, but that its activity gradually destroyed the balance forever. We would have made balance a condition of an earlier stage. It would no longer describe the foundation that supports the whole account. Within DEMU, that cannot be the role of the foundation. Its balance must belong to the activity we are explaining, including what happens when matter and energy return.

There is also no first moment at which the foundation was balanced before everything began. The word “throughout” refers to the whole range of temporal activity that depends on it. It does not give the core a private clock. At no stage of the temporal account can we remove the foundational condition and still claim to be describing the same model.

9.2 What remains the same while things change

Balance does not mean that every part of the universe stays in the same condition. A hot cup can cool. A plant can grow. A particular arrangement can come to an end. These changes belong to the temporal world described in the previous chapter.

What remains the same, in my proposal, is the fundamental relation that makes such activity possible. Expansion and contraction continue to belong together. What develops within the circulation does not become independent of its foundation. Return does not gradually turn the core into a store of accumulated history.

An ordinary example can help us separate an unchanged condition from inactivity. Imagine a tank with water flowing in and out at equal rates. Its water level can remain constant even though water is moving through it. Watching the level alone would not reveal all the activity taking place.

That example illustrates one limited point: a feature of a system can remain constant through activity. The tank still exists in time, and its flow depends on physical conditions that have to be supplied. It is not an example of timelessness. In DEMU, the stronger claim comes from the earlier argument: the foundational relation itself requires no earlier activation or outside supply.

This is how I use the word “dynamic” in the title of the model. Activity belongs to the balance itself. There is no inactive foundation underneath it waiting for something else to switch it on.

9.3 Expansion and contraction belong together

My expression “expansion = contraction” states how I propose to connect the extended universe with its timeless ground. Outward development and return are complementary aspects of one complete activity. We would have an incomplete account if we described extension and then simply left out what becomes of it.

The equality does not mean that every outward movement is visibly cancelled at the same place. If it did, the differences and changes we experience would disappear from the description. Nor do I mean that two separately existing forces happen to match by coincidence. The claim concerns their dependence on one another within the whole.

We can describe one part of the circulation as outward and another as returning. These descriptions distinguish roles within the model. They do not divide Nature into independent realities, each with its own separate origin.

There is a further distinction between this relation and a repeated sequence of causes. Saying that expansion causes contraction, which causes the next expansion, would still leave us asking what started the sequence. My account places both within the same timeless foundation. Their relation explains the organization of temporal activity without adding a first event that activates the foundation.

The dual torus is my proposed spatial representation of this relation. Its connected paths help us picture how outward activity and return can belong together. The drawing is useful because it makes the connection visible. The philosophical claim concerns what the connected activity accomplishes, including how it maintains the core's condition.

9.4 Renewal occurs within an ongoing universe

The circulation should not be pictured as the whole universe expanding, stopping, and then waiting for everything to contract before any renewal becomes possible. As defined in Chapter 2, DEMU proposes continual renewal in portions. Different portions can occupy different parts of the circulation while temporal activity continues.

This distinction matters. If renewal happened only after the last event in the universe, it could not explain renewed output within the universe we inhabit. My proposal places renewal within its ongoing activity. Outgoing matter and energy can be available while other portions are developing or returning.

Figure 9.1 shows these roles in a simple schematic. Following the arrows gives a way to describe the circulation from within time. It does not identify a first turn of the cycle or a first state of the core.

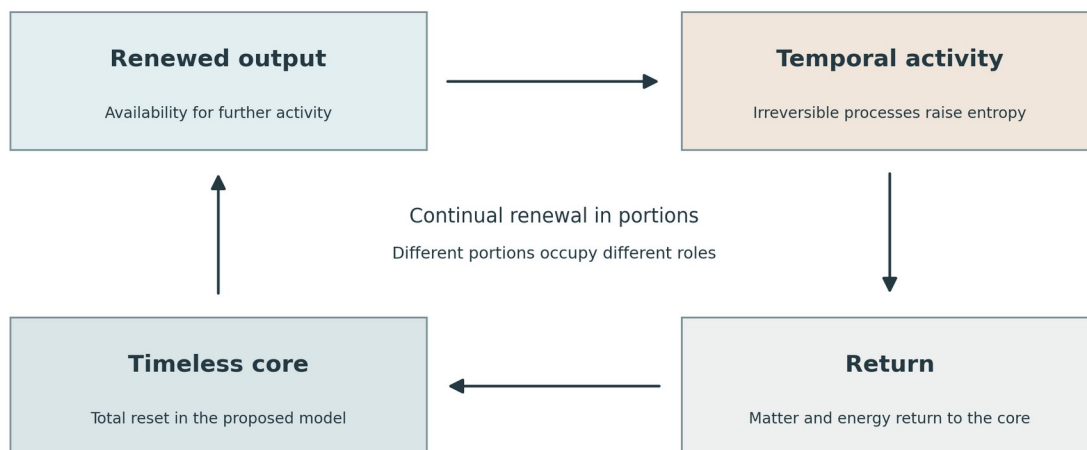


Figure 9.1. Continual circulation as proposed in DEMU. Different portions occupy different roles while temporal activity continues. The core marks the common condition of total reset and renewal. This is a schematic of the proposal, not a depiction of its geometry or a first cosmic cycle.

The sequence in the figure belongs to what circulates. The core is shown as the common condition of return and renewal. It is not a separate machine that grows older each time something passes through it. In my account, its foundational balance is never used up.

Nor does eternal circulation require every detailed event to happen again in exactly the same way. The claim is that the foundational condition remains the same through total reset. It does not, by itself, establish that the same people, conversations, or arrangements must recur. The identity of the condition and the identity of an entire history are different questions.

9.5 Balance must include everything that returns

A complete circulation has to include the consequences of its own activity. We cannot describe renewed output, follow it while it is useful, and stop the explanation when its usefulness declines. What returns is just as much a part of Nature as what goes out.

Suppose the returning activity left a little of its history in the core on each occasion. The core's condition would then depend on what had returned before. Even if the change were small, it would introduce a succession of different foundational conditions. That would conflict with the timeless and historically independent core described in this paper.

Moving the inheritance elsewhere would not meet the proposal either. If a hidden store preserved it, the complete account would still contain the information. We would have changed its location rather than erased it.

The reset I propose is total. No record, backup, or recoverable trace of the erased history survives anywhere. This includes traces in relationships between things, not only obvious records like written words. The proposed entropy reset likewise does not dispose of entropy in some region outside the account.

Continual renewal in portions does not weaken that claim. It describes when and where renewal belongs within the temporal circulation. Total erasure describes what happens to the inherited information of whatever is reset. Other activity can continue, but nothing elsewhere serves as a surviving record of the history that has been erased.

Balance therefore concerns more than how much goes out and how much comes back. It also concerns the condition in which return is completed. In DEMU, the foundation remains free of the accumulated history that would make it depend on earlier conditions.

9.6 Circulation must include renewed availability

There is a physical distinction we need to keep clear. An ordinary heat engine can bring its working substance back to its initial state while transferring heat and doing work. The substance completes a cycle, but its surroundings have changed. A repeated cycle inside an apparatus does not mean that the complete system has returned to its original condition. [1]

Similarly, conserving energy does not guarantee that it remains equally useful. Irreversible processes can reduce the work available relative to the surroundings. Returning a system to an earlier condition can require work and leave changes elsewhere. Energy accounting must therefore be kept separate from the question of renewed availability. [2]

DEMU proposes a stronger form of return. The circulation includes the restoration of a condition from which useful output is renewed. If returning matter simply carried its exhausted condition around the same route, the path might repeat while the capacity for further activity continued to decline. That would not meet the model's requirement of continual renewal.

This is why the reset belongs to my account of balance. It connects the return to the renewed condition required for outgoing activity. The core is not a limited reserve gradually drained by that activity. In the model, renewal belongs to the foundational relation itself.

The physical examples above explain why circulation and renewal are different. They do not establish the proposed reset. In ordinary thermodynamics, entropy cannot decrease for a complete isolated system. [1] DEMU's total reset is a further claim about the foundation, and its justification belongs to the philosophical argument and the account of the core.

9.7 What eternal balance requires

We can now state the reasoning plainly. The foundation in this account is timeless. The temporal universe belongs to it through the reciprocal relation of expansion and contraction. Therefore, the activity of that universe cannot replace the foundation with a condition that

depends on an accumulated past. The complete circulation must preserve the condition on which the account rests.

My proposal specifies how return meets that requirement: inherited information is erased, entropy is reset, and the capacity for renewed output is restored. The reset is not a repair performed after the timeless foundation has broken down. It belongs to the circulation through which the foundational condition remains what it is.

This establishes a requirement within DEMU. It does not follow from the word “balance” alone that every possible account of reality must contain my proposed reset. The connection rests on the timelessness, historical independence, and continual renewal developed in this model. Keeping those premises visible lets us see why the conclusion follows within the account and where a different account would disagree.

The remaining issue is the connection with explanation. Could a foundation retain historical distinctions while fully explaining them, or would that preserve the dependence my argument seeks to remove? Chapter 10 takes up this question directly. Chapter 11 then develops the account of reset and renewed matter and energy.

For now, the central point is that balance cannot be confined to an imagined beginning. In DEMU, the same foundational relation includes the outward activity, its consequences, and its return. High entropy belongs within that circulation, while renewal prevents it from becoming the permanent exhaustion of the whole.

References for Chapter 9

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CHAPTER 10

Entropy and Infinite Regress

10.1 The Case for Entropy Reset

My claim is that entropy must reset because the timeless foundation must remain what it is throughout the whole circulation. It cannot begin as perfect balance and then become something that carries an ever-growing past. In DEMU, returning histories dissolve completely at the core, and the capacity for renewed activity is restored.

This follows the order of the argument already developed. Chapter 4 argued for a timeless foundation that requires no first activation. Chapter 8 explained how high entropy arises within the temporal world. Chapter 9 argued that the foundation's balance must include the outward activity, its consequences, and its return. Here I bring these parts together.

The first question is what would happen if the return left something behind. Suppose the core received the results of temporal activity and preserved them as part of its own condition. Its condition would then depend on what had happened. It would carry a history rather than being independent of the history it supports.

We would also have to explain that inherited condition. If the explanation were an earlier temporal state, what would explain that earlier state? If every answer took the same form, the question would continue backwards. This is the regress discussed in Chapter 4: an explanation that depends entirely on earlier states never supplies a foundation that needs no earlier state.

Beginning with low entropy does not automatically solve this problem. We can still ask why that low-entropy condition exists. Calling it the first state gives it a place in the sequence, but does not explain why reality has that condition. My answer is that the foundational balance belongs to reality's nature. It is not an arrangement inherited from an earlier event.

The return must therefore belong to the same explanation. I cannot describe a perfectly balanced foundation and then leave the products of its activity outside the account. Whatever happens within the circulation must be included when I say that the foundation remains balanced.

The image of a large whirlpool containing smaller whirlpools helps explain what I mean. The smaller whirlpools are patterns within the larger flow. They can form, interact, and disappear without being separate substances added from outside. Their changing shapes belong to the activity of the whole.

I picture the cosmic dual torus in a similar way. The forms within it have beginnings and endings, while the fundamental activity continues. Their disappearance does not mean that existence becomes nonexistence. It means that a particular arrangement ceases to exist as that arrangement. Reality remains, but the form dissolves.

This also explains why I distinguish the universe from the things within it. The universe is the continuing expression of the foundational activity. Its contents can develop and dissolve without the whole having a first beginning or a final end. The proposed conscious centres, which I discuss as monads in later chapters, would belong to this same account of forms within the whole.

An ordinary whirlpool is only an illustration. Its visible disappearance does not establish that every trace of its history has vanished. My proposal goes further: when a returning history reaches the core, the distinctions carrying information about it are completely erased. No record, hidden trace, or recoverable information remains anywhere.

This includes information carried by relationships between things. A record does not have to resemble a written message. If some surviving difference could reveal which history returned, then that history would still be represented. The total reset I propose removes those differences too.

Different returning histories therefore reach the same complete foundational condition. This is what historical independence means in DEMU: the core does not take on a different condition according to which history returns. The return does not gradually turn it into a store of everything that has happened.

The reset also occurs in portions. The entire universe does not have to disappear at one final moment. Some forms can dissolve and renew while others continue their activity. However, a completed reset cannot leave a backup of the erased history in those continuing forms. If such a backup remained, the claimed erasure would still be incomplete.

Where does entropy enter this picture? Entropy describes a physical condition; it is not a substance or a complete record of the past. Two systems can have the same entropy without having the same history. Feynman's account treats entropy as a property of the system's state, which is why its value alone does not tell us everything that happened. [1]

My connection between entropy and infinite regress concerns the explanation of inherited conditions. If the foundation depends on an inherited arrangement whose explanation is always another earlier arrangement, the problem remains. In that philosophical sense, the entropy question can bring us back to the same demand for a complete foundation.

But removing historical information and resetting entropy are two claims. They must both be stated. Destroying a readable record does not, by itself, show that physical entropy has decreased. In DEMU, I propose both total information erasure and the restoration of a condition that supports renewed outgoing activity.

When I describe the core as having zero entropy, I mean a condition with no surviving alternative inherited arrangements. In the Boltzmann description, a system restricted to one accessible microscopic arrangement has zero entropy. [2] The further claim that the core really permits only that one arrangement belongs to my account of the core. Calling it zero-dimensional does not establish the restriction by itself.

The physical claim is stronger than an ordinary return to a familiar shape. In standard thermodynamics, the total entropy of an isolated system does not decrease. A local decrease must be assessed together with changes in the surroundings. [1] DEMU proposes a complete reset without sending the entropy elsewhere. This would require a physical account beyond the ordinary processes described in Chapter 8.

The reason I propose it is the role of the foundation. Its activity must include renewal, rather than depend on a supply of useful conditions that eventually runs out. The core is not a fuel reserve. Within my model, renewal belongs to the balance itself and requires no external supply to sustain it.

When I say that balance is continually recreated, I do actually mean that the timeless core repeatedly breaks down and gets repaired. In other words, all is flux, which is why the forms within the temporal world change. The fundamental relation remains the activity through

which they arise and return. The language of continual renewal describes the circulation, without giving the core a separate clock.

The central claim is therefore clear: in DEMU, the returning history does not become a permanent condition imposed on the foundation. Its inherited distinctions dissolve, entropy resets, and renewed activity becomes possible within the same eternal circulation.

10.2 The Challenge of Explained Retention

A critic can agree that reality needs a complete explanation and still question the need for erasure. The objection is simple: what if the foundation could preserve the history and fully explain why it is there? Would keeping information still create a regress?

This is the challenge of explained retention. Retention means that the distinctions survive. Explained retention means that their survival and their particular character follow from an adequate account of the foundation. The critic is not necessarily proposing an unexplained pile of records.

There are two versions of this idea. In the first, the core acquires records as events happen. It passes from a condition without a record to a condition with one. Such a core would change through time, so it would not meet the requirement of timelessness used in my argument.

The second version is stronger. The entire history belongs to an unchanging whole. Nothing is added to that whole, because all its distinctions already belong to its structure. Earlier and later events occur within the history, but the whole does not pass through another sequence outside it.

Think of a book. Its pages can describe a long history even while the book lies unchanged on a table. A reader moves through the story, but the book does not have to acquire each event as the reader reaches it. This shows why the absence of change alone does not imply the absence of distinctions.

The same point applies to my image of a pen that never draws. If the pen never moves, it produces no picture. But a critic can still ask whether a picture might belong to the foundational condition without having been drawn there. The question then becomes what explains its presence.

My rejection of brute facts rules out simply saying that the picture is there and that no further explanation exists. Appealing only to another unexplained picture would also pass the problem on. But a critic could offer another answer: the picture follows from what the foundation is. However, this is not an explanation. It is an assertion.

My claim about balance follows from the earlier argument: I argue that balance is necessary because it closes infinite regress without requiring an earlier activation or further support. Simply placing a history inside an unchanging whole does not provide the same explanation. The book example shows that an unchanging object can contain a description of events, but it does not explain why the proposed whole exists or why it contains those particular events. If the skeptic simply declares that this history belongs to the foundation and gives no further reason, the account leaves it as a brute fact. A defence of retained history must therefore explain the structure it proposes. Calling that structure timeless does not complete the explanation.

The critic might say: “These distinctions were never created. They belong eternally to the whole.” My reply cannot simply be that every distinction must have been produced at an earlier time. That would assume the very point the critic disputes. I must explain why the balance described in DEMU plays a different role from the historical distinctions that return to it.

Balance is the fundamental relation through which I account for activity without a first activation or outside support. A particular history describes what happens within that activity. In my proposal, the history depends on the foundation; the foundation's condition does not depend on which history happens.

This is why the return matters. If the core preserves a difference because one history occurred rather than another, its resulting condition reflects that difference. It is no longer the common condition I have described. Calling the record part of the foundation does not explain why the foundation must contain that particular record.

A defender of explained retention can reply that the whole history follows from the foundation's nature, so the direction of explanation never reverses. If that account succeeded, the retained distinctions would not create a regress merely because they remained. The disagreement would concern which account of the foundation is correct.

However, I want to make this point firmly once more: a defender of retained history must explain why the proposed structure belongs to the foundation. Simply declaring that it belongs there does not provide that explanation.

My position is that DEMU gives the core a more specific role than being an unchanging container for a history. It is the common condition in which the returning arrangements dissolve and from which renewed activity is expressed. A record that survives the return would be something that did not dissolve. It would therefore fail the proposed account of total return.

This establishes why retention is incompatible with my reset model. The broader claim that no other foundation could explain retained distinctions depends on the argument that this model supplies the necessary foundation. That stronger conclusion cannot be obtained simply by defining return as erasure. The earlier arguments about timelessness and balance do the work on which this chapter depends.

There is another objection to address: does erasing a history explain why that history occurred? Erasing a note does not explain why someone wrote it. Likewise, removing every record of an event would not, by itself, explain the event's occurrence.

My answer is that the reset is part of a complete account, rather than a replacement for one. The foundation's dynamic activity is meant to explain how temporal forms arise and interact. The reset explains why their returning histories do not alter the core or permanently exhaust the capacity for renewal. These are different roles within the same proposal.

Erasure also does not mean that the event never happened. It means that the completed reset leaves no information from which the particular returning history could be recovered. If the resulting condition, together with the laws governing it, still allowed that history to be uniquely recovered, then the total erasure I propose would not have occurred.

The whirlpool image helps us picture the disappearance of a form, but it cannot settle these questions by itself. Nor does the absence of anything outside the universe prove that internal activity cannot produce entropy. My claim about the core rests on its proposed nature, including its complete removal of inherited distinctions and its relation to renewed output.

The conclusion is that entropy reset is required by the combination of claims made in DEMU: the core remains timeless and historically independent, the return is complete, and the circulation continually renews the capacity for further activity. Timelessness alone does not prove all of this. The connection depends on the account of balance developed throughout the work.

Within that account, the universe can continue while its forms arise and dissolve. High entropy belongs to temporal activity, but it does not become a permanent inheritance at the core. Chapter 11 now develops the proposed account of how that complete reset renews matter and energy.

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CHAPTER 11

Entropy Reset and the Renewal of Matter and Energy

11.1 What renewal means

In DEMU, matter and energy are renewed when returning forms dissolve at the timeless core and outgoing activity again becomes available. The reset removes the inherited history completely. Renewal means that the circulation can support further activity without carrying that history forward or permanently losing its capacity to do useful work.

Chapter 10 explained why I require this result within my model. Here I develop the proposed account of how it happens. The central idea is that contraction removes the arrangements through which returning forms remain distinct, until only one foundational condition is allowed. Expansion expresses renewed forms through the same ongoing activity.

Renewal does not mean that existence disappears and is later created from nothing. What dissolves is a particular arrangement. The underlying reality remains. A small whirlpool can lose its shape while the larger flow continues; this is the image I use to picture the relationship between temporary forms and the cosmic dual torus.

Matter and energy name what participates in the temporal world. I am not assuming that an atom remains an unchanged atom while passing through the core. In the proposed reset, the inherited arrangement and its history cease to be preserved. Renewed matter names the forms expressed through the outgoing activity.

11.2 Contraction and the loss of inherited form

In ordinary spatial language, contraction means that something occupies less space. My proposal makes a stronger claim. At the foundational limit, there is no independent spatial extension through which the returning arrangements can remain separate. The core is zero-dimensional in the sense introduced in Chapter 2.

This is not simply a small container holding the same collection of objects more tightly. If every old distinction remained packed inside it, the information would still exist. The reset

requires those distinctions themselves to disappear. It is the removal of inherited structure, rather than a change in its visible size, that does the work in the proposal.

Think of the smaller whirlpools again. Each has a recognizable shape and a history of interactions. When its separate pattern dissolves into the larger flow, we no longer have that whirlpool as an independent form. In DEMU, I extend this image to a complete dissolution in which no trace of the returning history survives elsewhere.

The fundamental activity does not switch off during this return. My earlier expression, motionlessness = motion, places activity within the foundational condition itself. The proposed core is therefore active without moving through a sequence of its own changing states. The contraction described from within the temporal world belongs to that enduring relation.

The further claim is that this condition allows only one complete arrangement. The absence of spatial extension alone does not prove that. A system might lack spatial size while still admitting different internal states. My proposal requires the stronger restriction: nothing within the completed reset condition can vary according to the history that returned.

11.3 The proposed single state at the core

The Boltzmann description gives us a simple way to express the proposed entropy condition. It connects entropy with the number of microscopic arrangements compatible with the system's specified overall condition. Feynman explains this through the difference between arrangements that look the same at a larger scale. [1]

$$S = k_B \ln \Omega$$

Here, S means entropy, k_B is Boltzmann's constant, and Ω is the number of allowed microscopic arrangements. The symbol $\ln$ means the natural logarithm. In words: entropy equals Boltzmann's constant multiplied by the natural logarithm of the number of allowed arrangements.

If only one arrangement is allowed, $\Omega = 1$. The natural logarithm of one is zero, so the equation gives $S = 0$. This is the calculation behind my description of a zero-entropy core. [1]

The calculation is straightforward, but its starting point matters. It says what follows if the core has a meaningful description in terms of microscopic arrangements and exactly one

such arrangement is allowed. It does not establish those conditions by itself. The proposed nature of the core must account for them.

Nor is a single visible appearance enough. A room can look unchanged while its air molecules occupy many different arrangements. My claim concerns the complete condition, including differences hidden from ordinary observation. Otherwise, setting Ω equal to one would merely ignore distinctions that still existed.

Zero entropy in this calculation does not mean zero activity or zero energy. The equation counts arrangements; it does not measure a speed or establish a temperature. Within DEMU, the single condition is the active balance to which histories return. The foundational activity remains while the inherited alternatives disappear.

11.4 Total erasure and ordinary physical processes

Consider two different returning histories. One contains a particular sequence of events; the other contains a different sequence. My claim is that their completed resets leave exactly the same foundational condition, with no surviving information that identifies which history returned.

This is a many-to-one relation: different histories lead to one reset condition. From that condition, the incoming history cannot be uniquely recovered. It is not stored in a hidden region, an unseen relationship, or something else. Such a record would mean that the claimed total reset had not yet occurred.

There is an important difference between this proposal and ordinary information erasure. In Bennett's account of Landauer's principle, resetting an information-bearing system can reduce its entropy while increasing entropy in other parts of the apparatus or its environment. Under the reversible microscopic dynamics he discusses, distinctions are not simply removed from the complete physical description. [2]

For example, making a computer register display the same value regardless of its earlier value does not establish that the entire physical world has reached the same state. The operation has consequences outside the selected register. Those consequences belong to the complete account of the process. [2]

DEMU proposes a stronger result: no compensating store of entropy elsewhere and no remaining physical trace of the erased history. A process with that result would require a departure from the usual physical account. Calling the core timeless does not, by itself, demonstrate how that departure is possible.

The proposed explanation is that the returning distinctions are forms of the foundational activity, and the core's condition does not permit them to remain separate. They dissolve because they cannot persist in that condition. This states the physical role I assign to the core. A fully worked physical theory would need to specify the relation that enforces it, including why no trace survives outside the core.

11.5 How renewed activity can arise

The next question is how a single reset condition can support a world of different forms. If everything returns to one condition, why would anything varied emerge?

My answer begins with the distinction between the foundation and its temporal expression. The core is not an inactive object that must first receive a push. Its balance is already activity. Expansion and contraction are the connected aspects through which I propose to describe the emergence and return of temporal forms.

The outgoing forms therefore do not need the erased history as instructions. What continues is the foundational relation itself. In my account, that relation supports further organization while the previous arrangement is gone. Figure 11.1 shows this proposed relationship between return, a common reset condition, and renewed forms.

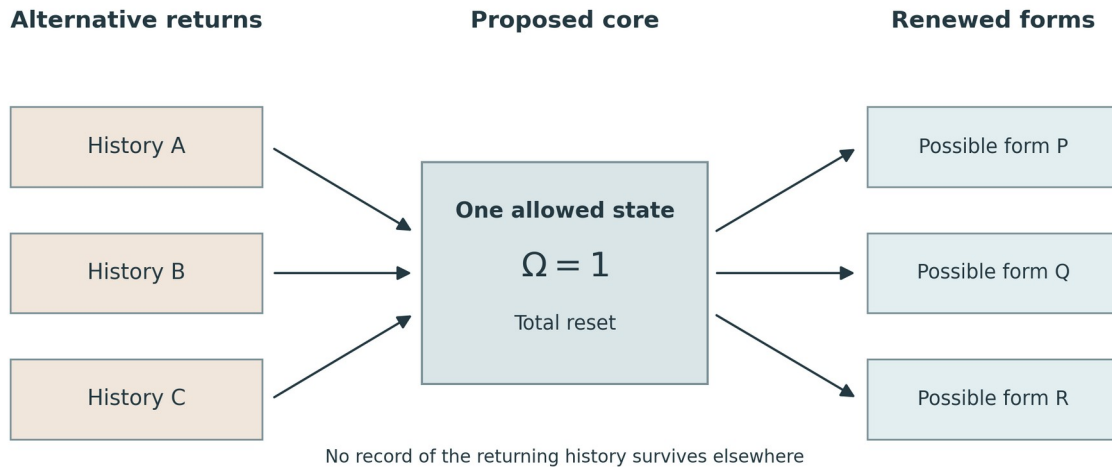


Figure 11.1. Return, reset, and renewal as proposed in DEMU. The left branches compare alternative histories, rather than histories that must return together. The single core state is a hypothesis. The right branches represent proposed renewed forms; their production requires the account discussed in the text. This is a logical schematic, not a spatial drawing of the core.

The diagram states the proposal; it does not derive the variety of the output. A single complete starting state combined with a rule that determines every detail would give the same subsequent development under the same conditions. To explain different developments, an account must identify what allows them.

Within an ongoing universe, different surroundings can affect newly emerging forms. However, those surroundings cannot preserve information about a history that is claimed to have been totally erased. Explaining new variety and removing old information must therefore be handled together. Moving the old history into a new name would not achieve the reset.

The later chapters develop my account of replication and conscious activity as part of this explanation. Here, the essential claim is that returning to the same foundational condition does not mean storing and replaying the same history. Renewal concerns the capacity for further activity. The particular forms that arise require an account of the outgoing organization.

11.6 Renewed energy means renewed availability

Energy can remain present while becoming less available for useful work. For example, a temperature difference can support the operation of a heat engine. Once the relevant bodies reach the same temperature, that particular opportunity is gone, even though energy has not

simply vanished. MIT's account connects this lost opportunity with entropy production and the temperature of the surroundings. [3]

This is why renewing energy cannot mean only that energy continues to exist. The proposed output must again have conditions that make useful work possible. Its relation to its surroundings matters. Calling it low in entropy, without specifying those conditions, would leave the practical meaning of renewal unclear.

In DEMU, the outgoing activity provides this renewed availability. The core is not a battery that becomes depleted as the universe uses its charge. I propose that renewal belongs to the foundational balance itself. The physical account must show how the outgoing matter and energy acquire their usable condition through that relation.

The river image expresses the idea in simple terms. A form can resist a flow for a time and later dissolve into it. I picture return as a release of the separate form, while the underlying activity continues. An ordinary river still depends on physical conditions that sustain its flow, so the image explains the relationship I intend rather than proving that no energy input is needed.

The stronger claim comes from the proposed foundation: its activity requires no outside source to keep it going. Within the model, collecting renewed output does not draw energy from the core, and the fundamental circulation remains unaffected. The collection concerns what becomes available within the temporal world.

This does not set a numerical power output or identify an actual outlet. Those depend on how renewal appears within the universe. Chapter 12 takes up the meaning of free and unlimited energy in light of that distinction.

11.7 Renewal within an eternal universe

The reset must be pictured as continual renewal in portions. One part of the universe can undergo return while another develops new forms. There is no need for a final moment at which the entire universe disappears before activity can resume.

This gives the whirlpool image its full role. Smaller patterns arise within the larger activity, maintain their forms for a time, and dissolve. The larger activity does not become

nonexistent when a smaller pattern ends. In DEMU, the universe continues while its contents participate in the circulation.

The total character of the reset concerns the information erased, rather than requiring every part of the universe to reset together. If a returning form has left records elsewhere, those records must also be removed for its history to count as completely erased. Renewal in portions cannot turn total erasure into a merely local disappearance.

The account developed here therefore has connected requirements. Returning arrangements must lose their inherited distinctions completely. The core must admit the proposed single condition. Outgoing activity must again support useful work and the formation of temporal structures. The whole relation must operate without making the foundation depend on an accumulating history or an outside supply.

These requirements describe the mechanism I propose at the philosophical level. The Boltzmann calculation shows the entropy consequence of the single-state condition. Establishing that the core has this condition, that total erasure can physically occur, and that renewed usable output follows requires a specified physical model; the calculation and the whirlpool image do not supply one by themselves.

My central proposal remains that the universe's forms can be renewed because they are expressions of an activity that never becomes exhausted at its foundation. Their histories end at the reset, while the capacity for further activity continues. Existence has no first beginning in this account, but renewed forms have beginnings within its eternal circulation.

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CHAPTER 12

The Meaning of Free and Unlimited Energy

12.1 What I mean by free

In DEMU, free energy means that the fundamental circulation requires no external supply to sustain its activity and renew the capacity for useful work. Unlimited energy means that this renewal has no final point of exhaustion. These claims concern the nature of the proposed circulation. Human access concerns how we might collect the renewed output within the universe.

The reason I give for this activity is the argument developed in the earlier chapters. I argue that dynamic balance is necessary to close infinite regress without an earlier activation or further support. On that account, renewal belongs to the activity through which balance is maintained. I am not introducing balance as an unexplained fact and then calling its consequences free.

There is no separate fuel store outside Nature that keeps the whole running. But the absence of anything outside Nature is only part of the point. My proposal also concerns what happens within it: the circulation renews availability instead of depending on a reserve that becomes permanently exhausted. The reset described in Chapters 10 and 11 supplies the role through which I propose to explain that renewal.

The everyday meaning of free often concerns money. That is a different question. A source could renew itself while equipment remained expensive to build and maintain. Finding an outlet, collecting its output, and delivering that output where it is needed could all require resources. None of those costs changes what I mean by a foundation that needs no external supply.

Physics and chemistry also use the expressions Helmholtz free energy and Gibbs free energy. These name defined quantities involving energy, temperature, and entropy. IUPAC distinguishes the two quantities. [1] My use of free describes the proposed independence of the fundamental circulation. It is not a new name for either of those thermodynamic quantities.

12.2 Unlimited renewal and finite power

Unlimited renewal does not require an outlet to deliver an infinite amount of energy at one moment. Energy is an amount; power tells us how quickly energy is transferred. A continuing supply can therefore have a finite rate. This distinction lets us discuss endless renewal without imagining an infinitely powerful socket.

Consider a purely illustrative source that supplies ten joules every second. A joule is a unit of energy. A rate of ten joules per second is ten watts of power. After ten seconds, the source has supplied one hundred joules. After sixty seconds, it has supplied six hundred joules. Its rate has remained ten watts throughout.

Figure 12.1 shows those two descriptions of the same example. The left graph shows the rate. The right graph adds up the energy supplied since counting began. The numbers illustrate the meaning of a continuing supply; they are not a prediction of DEMU. The starting point on the clock marks the start of a measurement, not the beginning of reality.

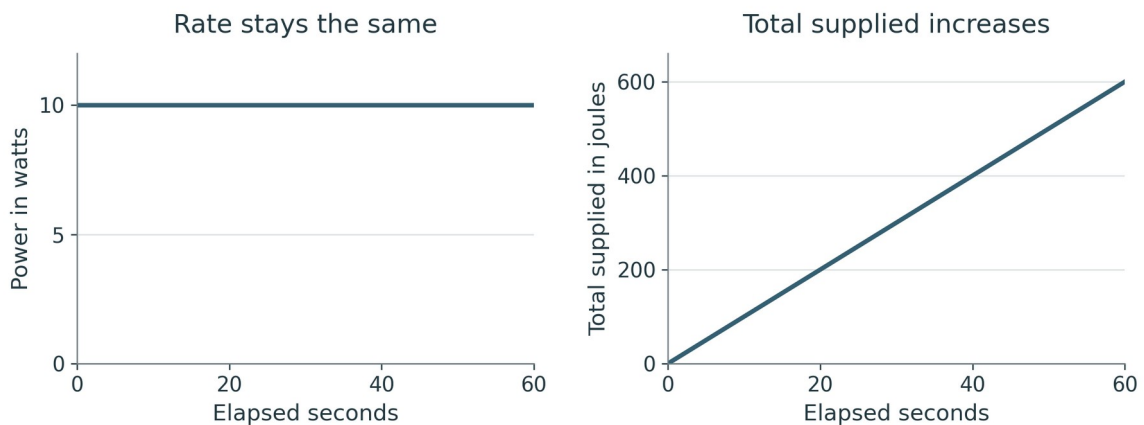


Figure 12.1. Finite power and a continuing supply. A hypothetical ten-watt source is shown for one minute. The cumulative total counts energy transferred during that interval; it does not measure an energy stock held by the universe. The example illustrates the terms used in this chapter and does not establish a physical source or endless operation.

If the supply continues at that rate, we can keep extending the period over which we count its output. Every finite period still gives a finite amount. This does not require an infinite store to be present all at once. In a circulation, a running total of transfers can also count energy that has passed through further rounds of use and renewal. That total is not the amount held by the universe at one moment.

An eternal source would not, by that fact alone, make every outlet permanent. Particular forms arise and dissolve in DEMU. Access at a particular location could therefore change while renewal continues within the whole. The available rate, the duration of a local outlet, and the continuity of the fundamental circulation are separate questions.

My earlier objection to infinite regress concerns an explanation that is forever passed on to something else. The present claim concerns an activity whose explanation lies in the proposed timeless balance. Calling renewal unlimited does not make an endless chain of earlier causes its foundation. The ground of the circulation and the duration of its temporal forms have different roles in the argument.

12.3 Why usable energy must be renewed

The continued presence of energy is not enough to establish the continued possibility of useful work. A hot object in cooler surroundings offers a temperature difference that a heat engine can use. As the temperatures become equal, that opportunity disappears. The energy has not simply ceased to exist. What has changed is what can be done with it under those conditions. [2]

The same distinction gives renewal its practical meaning. The outgoing activity must offer some usable difference or organized flow. Merely calling matter low in entropy does not identify what process it can drive. Its condition must be described in relation to its surroundings. MIT's account of unavailable energy explains why those surroundings matter when assessing the work that can be obtained. [3]

In DEMU, entropy reset and renewed availability belong to the same proposed circulation. Returning forms lose their inherited distinctions completely, and outgoing activity again supports the formation and use of differences. The reset is total: the erased information is not hidden elsewhere, and the entropy is not transferred to an outside store.

This is why renewal matters to the energy claim. If the universe merely kept moving while every opportunity for useful work faded permanently, continued motion would not provide the supply I am describing. My proposal requires the capacity for useful work to be restored through the circulation. Chapter 11 explains the role assigned to the core in that account.

12.4 Collecting output within the circulation

In this model, collecting usable output obviously does not draw energy from the core. The fundamental circulation remains unaffected. The core is the foundational condition through which I explain renewal; it is not a charged object that becomes weaker each time someone uses its energy.

The collection takes place within the temporal universe. The matter and energy available to us already belong to that universe's activity. Human access would involve finding where renewed output becomes available and arranging a process that can use it. We do not stand outside Nature and take something away from the whole.

The river image helps explain this relationship. A wheel can receive energy from a passing current. It draws on the flow available at its location. In DEMU, the corresponding question is how an accessible flow expresses the ongoing circulation. The analogy illustrates access; an ordinary river does not demonstrate the proposed entropy reset or the independence of the cosmic circulation.

When collected energy lights a room, raises a weight, or drives a tool, those events remain within Nature. The output does not vanish when it becomes useful to us. Its later transformations also belong to the account of the circulation. My claim that the fundamental balance remains unaffected concerns the continuing basis of renewal through those changes.

The word outlet therefore names a place or process where renewed output could become accessible. It does not yet identify a particular object or a device. The model gives the search a proposed target: usable output whose availability is sustained through renewal. Chapter 22 takes up the question of how human access might be investigated.

12.5 What remains after collection

Net useful output means the usable amount left for other tasks after the collection process has met its own needs. This is a question about the collector. It does not mean that the collector must pay an energy cost to keep the timeless core active.

Suppose that, during one chosen period, a collector receives one hundred joules from a renewed flow. Imagine that twenty joules become unavailable for the intended task during conversion, leaving eighty joules of electrical output. If the collector uses ten of those eighty

joules to operate its own parts, seventy joules remain for other tasks. These are invented numbers to make the accounting clear.

In that example, the twenty joules lost during conversion have already been excluded from the eighty joules of electrical output. We must not subtract them a second time. Likewise, the ten joules used to run the collector are counted once. The practical question is how much useful output remains after the whole collection process is included.

If the apparatus also needs a separate power supply, that input belongs in the calculation. Any initial store that runs down must be counted as well. A device that briefly delivers energy from a battery has shown that it can discharge a battery. Establishing continuing access to renewed output would require showing what maintains the supply during operation.

Energy accounting and financial cost also answer different questions. A collector might provide positive net useful output while still costing money to build, repair, and operate. The philosophical meaning of free does not decide those costs. It identifies the proposed source of renewal on which an account of practical access would depend.

12.6 The whole system and the laws of thermodynamics

Ordinary thermodynamics distinguishes energy conservation from the limits on turning energy into useful work. The first law accounts for energy transfers and changes in stored energy. The second law rules out a cyclic process whose sole result is to take heat from a single reservoir and convert it completely into work. Considering only one convenient part of the process does not remove that restriction. [2]

DEMU makes its stronger claim at the level of renewal. It proposes a total entropy reset without a compensating increase elsewhere, together with a renewed capacity for useful work. As Chapter 11 explained, such a process would go beyond the usual physical account. The mere fact that an ordinary subsystem can lose entropy does not establish the total reset proposed here.

A collector receiving renewed output would receive an input from its surroundings. Describing the larger circulation as free therefore does not describe the collector as producing energy without an input. The proposed source would lie in the surrounding

activity. The question about Nature as a whole is why that activity can keep renewing its availability without an outside supply.

This is also why simply drawing a circle is insufficient. A process can return to a familiar arrangement while leaving consequences elsewhere. The return shown by a diagram must correspond to the complete renewal being claimed. My account requires the dissolution of inherited information and the restoration of usable conditions within the one circulation.

12.7 What follows from the argument

The philosophical claim can now be stated more precisely. I argue for a timeless foundation whose dynamic balance closes infinite regress. Within DEMU, that balance is expressed through continual return and renewal. The reset removes inherited history completely, and the outgoing activity renews the conditions for useful work. Free and unlimited describe this proposed independence and continuity.

The result concerns more than the claim that something exists forever. An eternal object could exist without offering a continuing supply of useful work. The relevant claim is that availability itself is renewed. Likewise, an argument for renewed availability does not, on its own, identify where a person can collect it or how much a particular device can supply.

Reason can establish what follows from the premises of an argument. It can also test whether the different parts of a proposal fit together. A physical account must connect the proposed renewal with quantities and processes that can be investigated. An actual claim of human access would then concern a particular source and the net useful output obtained from it. These are connected stages of the inquiry.

The meaning of free and unlimited energy is therefore clear within the model: the fundamental circulation sustains renewal without an external supply, and that renewal has no final exhaustion. Human beings would use output arising within the circulation. Their equipment could have limited power and practical costs while the foundation remained undiminished.

The next chapter examines rival accounts of reality. The question is whether those accounts can explain existence, activity, and the structure they attribute to the foundation.

The case for DEMU depends on the arguments developed for its balance and renewal, and the alternatives must be assessed by the explanations they actually provide.

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CHAPTER 13

Rival Accounts of Reality

The argument for DEMU depends partly on why I reject the alternatives. A rival account must do more than offer a different picture. It must explain what its picture contains and why the proposed foundation needs no further support. This chapter examines absolute nonexistence, brute facts, complete inactivity, and other proposed ways of closing the explanation.

Chapter 4 began the argument for a timeless foundation. The later chapters developed my account of balance and renewal. Here I bring those arguments into direct comparison with the alternatives. My aim is to identify where an answer supplies an explanation and where it merely asks us to stop asking.

13.1 Nonexistence as a Fabrication

When I call absolute nonexistence a fabrication, I mean that language allows us to form an expression that we then mistake for a possible condition of reality. We know what it means for a particular thing to be absent. The difficulty begins when we extend that familiar idea to the absence of reality altogether.

Suppose I say that there is no coin in my hand. The statement makes sense because the hand remains, and we understand what would count as a coin being there. The absence concerns something within an existing situation. It does not present us with absolute nothingness.

Now imagine removing the hand, the room, the Earth, and everything else. It is easy to picture a dark space left behind. But darkness is still an imagined appearance, and space is still part of the picture. Remove those too, and we can no longer use the picture as an example of a place where nothing exists. There is no place left.

This does not mean that every use of the word nothing is confused. Ordinary negative statements are useful and meaningful. My objection concerns treating the negation of everything as though it had supplied another kind of reality. An empty stage is still a stage. Absolute nonexistence cannot borrow the stage while claiming that nothing remains.

A critic can reply that no picture is needed. The claim is simply that there could have been no reality at all. This reply avoids treating nothingness as a strange object. However, it still needs support if it is offered as a real alternative. The ability to write a negative sentence does not, by itself, establish that what it describes is possible.

In Part IX of Hume's *Dialogues Concerning Natural Religion*, Cleanthes argues that whatever we conceive as existing can also be conceived as nonexistent. [1] My reply is that imagining the absence of a particular object and establishing the possibility of absolute nonexistence are different tasks. The removal of each item considered separately does not automatically establish that reality as a whole could be absent.

Nor can absolute nonexistence explain an origin. If it contains no reality, it contains no capacity, process, or condition through which existence could arise. Saying that nothing became something introduces a transition while denying everything that could take part in it. If a law or power is added to make the transition possible, the proposal has already supplied something.

My earlier account of motion and motionlessness gives this objection a positive setting. I argue that the deepest motionlessness is already activity. On that account, stopping the ordinary movement of things would not take us outside existence. Rest and motion belong to the same reality; they do not provide a route from reality into its absolute absence.

The claim that existence is necessary rests on the earlier argument, not simply on the fact that I find nothingness difficult to imagine. The point established here is narrower: familiar examples of absence do not supply absolute nonexistence, and absolute nonexistence supplies no explanation of existence. My conclusion is that it should not be treated as a rival foundation merely because language gives it a name.

The task of the skeptic is therefore extremely difficult: since we have no real evidence of a coin suddenly popping out of existence, the skeptic has no real meaning for the word “absence” other than “displacement” within reality. You cannot simply invoke “the absence of everything” and leave it at that. You must explain what this would mean. If a coin were uprooted from reality, what would happen? Would fairies come and patch the hole left by the coin?

You must provide an actual explanation here, rather than simply say that nonexistence might be possible even if we cannot define or conceptualize it. Otherwise, I can say that “tendrippore” is possible too, while refusing to define it. This is like playing chess and inventing new moves as you please. You are not allowed to do that.

Please follow the rules and define nonexistence before you use it. Then you will see that nonexistence is nonsense because it is merely a product of language. We painted symbols on our cave walls, and we can manipulate those symbols to form all kinds of nonsense. But that does not mean our nonsense can rival reality.

13.2 Brute Facts and Explanatory Closure

A brute fact is something for which there is no further explanation, even though the question why remains appropriate. This differs from something we have not yet explained. An unanswered question leaves the inquiry open. Declaring a brute fact closes the inquiry while leaving the question unanswered.

Suppose someone asks why a book is on a table. Saying that it is there gives a description. Adding that it has always been there still does not explain why it is there. If the reply is that there is no reason, the proposed answer is a brute fact. Its permanence has not supplied the missing explanation.

My objection is not that every explanation must lead to an earlier event. That would reproduce the regress I am trying to close. I am asking for a reason why further support is unnecessary. An account can reach that result by explaining the nature of its foundation. Simply refusing to look beyond a chosen stopping point does not reach the same result.

This is where my claim about balance must be stated accurately. I argue that balance belongs to the foundation because it closes infinite regress without requiring an earlier activation or an outside support. That is the role defended in the earlier chapters. The claim has an argument behind it; it is not an unexplained property inserted into the model for convenience.

The difference is between giving reasons for a foundation and merely calling something foundational. A necessary foundation would be one whose existence and relevant character could not be otherwise, for reasons the argument makes clear. Calling a chosen structure

necessary does not make it necessary. The reasoning must connect the claim to that conclusion.

The retained-history objection illustrates the problem. A critic may say that the entire history belongs eternally to the foundation. But why this history, with these particular distinctions? If the answer is only that the foundation happens to contain them, those distinctions remain brute facts. Their placement inside a timeless whole has not explained them.

The critic might say: "These distinctions were never created. They belong eternally to the whole." My reply cannot simply be that every distinction must have been produced at an earlier time. That would assume the very point the critic disputes. My objection is that being eternal does not, by itself, explain the structure that is said to be eternal.

In DEMU, balance has the role of accounting for the foundational activity. Particular histories concern the forms expressed within that activity. The proposed reset removes their inherited distinctions completely. I argue that the foundation must not depend on carrying those histories forward. The reasons for this requirement were developed in Chapters 10 and 11.

Someone may reject my demand for a complete explanation and accept that reality ultimately has no reason. That position does not contain a contradiction merely because it leaves a question unanswered. But it abandons the explanatory goal of this inquiry. I reject it as a solution because the question it was meant to answer remains open.

13.3 Static Systems and Infinite Regress

By static in the strict sense used here, I mean complete inactivity. I mean a proposed foundation with no activity of its own, not simply an object that remains in the same place or a whole whose overall condition remains constant. A whirlpool can keep its shape while water moves through it. Its stable appearance does not make it inactive.

My objection is that complete inactivity cannot supply the activity that the account is meant to explain. If the proposed foundation must be activated, we must ask what activates it. If a further condition does that work, the foundation depends on that condition. The demand for an explanation has moved back one step.

Calling the inactive condition timeless does not solve this problem. There can be no earlier period of waiting followed by its first awakening if the condition is genuinely timeless. The proposal cannot both deny succession at the foundation and use a succession from inactivity to activity to explain the world.

A different answer would be that the foundation eternally supports activity without undergoing change. That is a different claim from complete inactivity. It assigns the foundation a role in making activity possible. The question is then what that role involves and whether the account explains it. An unchanged condition and a condition that contributes nothing must not be confused.

This is why Aristotle's unmoved mover cannot simply be dismissed as an inactive object. In *Metaphysics* XII.6–7, Aristotle presents the first principle as actuality, and he describes its life in terms of thought. Its lack of change is compatible with the activity Aristotle attributes to it. [2] My disagreement concerns his particular account of that activity and its relation to the universe, as discussed in Chapter 5.

My own proposal also requires care with the word unchanged. I describe the foundation as timeless while treating its balance as active. I therefore cannot argue that the absence of an external sequence, by itself, proves that a foundation is inactive. The relevant distinction concerns whether activity belongs to reality at its foundation or must be added from elsewhere.

A block account of the universe raises a related issue. On such an account, events can occupy different positions and times within one complete history. The history need not change in another time outside itself. D. C. Williams defends an account of motion through differences of position at different times and rejects an additional passage imposed on that whole. [3] This is not the simple claim that nothing ever happens.

The book image helps us see what is at issue. A book can describe a person walking across a room while the book itself lies still. But the book is a representation. A defender of a block account claims that the events themselves belong to the whole. My criticism must therefore address that claim, rather than assume that the account contains only pictures of events.

I can still ask what explains the existence and organization of that whole. A complete description of where events belong is not automatically an explanation of why those events or relations exist. If their presence is simply accepted without a reason, the brute-fact objection applies. The absence of an outside clock does not answer it.

This brings us back to the distinction made in Chapter 2. Removing an earlier moment can remove the question of an earlier temporal cause. It does not automatically remove dependence on unexplained conditions. A timeless account can still leave its structure, laws, or particular history unexplained.

My proposal gives a different role to the foundation. Balance is the activity through which I argue that the demand for an earlier activation is dissolved. The temporal forms arise and return within that relation. At the core, the returning history is erased completely. The core does not become an archive of everything that has happened.

I therefore reject complete inactivity as the ground of an active reality. If an account instead allows activity within an unchanged whole, the dispute must concern how that whole explains its activity and structure. The word static cannot settle that dispute for either side. What matters is whether the proposed foundation actually does the explanatory work assigned to it.

13.4 Other Accounts of Closure

One alternative is an endless chain of causes. Each event is explained by an earlier event, and there is no first member. In Hume's Dialogues, Cleanthes asks why explaining the individual members of a collection should leave a separate causal question about the collection as a whole. [1] This challenges the move from explanations of particular events to a further demand concerning the whole.

My reply concerns the shared dependence within the proposed chain. If every account assumes the existence of another active condition, then the explanation of activity has continually been borrowed. The question is not simply which member came first. It is what makes this whole order of dependence possible and why it exists in this form.

An account might attempt to explain the chain through its laws or through the nature of the whole. It must then give that explanation. Merely saying that every event has a

predecessor does not explain why the governing relation holds. Adding more members does not, by itself, answer a question about the conditions they all require.

Another alternative is circular support. One thing explains a second, and the second explains the first. If each is offered only as the reason for the other, the account returns us to the starting question. A circular route can avoid a first member without explaining why the circle exists or why its members can support one another.

This is a direct challenge to DEMU as well, because I describe reciprocal activity. My answer is that expansion and contraction are proposed as aspects of one foundational relation. I am not placing two independently unexplained objects in a circle and allowing each to excuse the other. The earlier argument concerns why that relation is necessary and how its activity requires no first activation. The torus illustrates the proposed circulation; its shape alone is not the argument.

A further answer appeals to a necessary foundation. Aristotle's unmoved mover is one example already examined. Spinoza offers another. In Part I of the *Ethics*, he argues that existence belongs to the nature of substance, that everything exists in God, and that God is the cause within which things exist rather than a cause acting on a separate outside world. [4]

Spinoza's account is relevant because it attempts to explain particular things through one fundamental reality. My question for such an account concerns the argument connecting that reality to its expressions. Does the account explain the activity and differences within the world? How does it establish that its foundation requires no further support? Agreement that reality is one does not establish that the accounts give the same answers.

The appeal to necessity is also not reserved for a religious account. Cleanthes asks why the material universe itself could not possess a nature that made its existence necessary. [1] My response is that the word material does not decide the issue. A proposal must explain the necessary character it attributes to reality. The same demand applies to the words mind, substance, and God.

This matters for the later chapters, where I develop the connection between the proposed foundation and consciousness. Calling the foundation conscious would not, by itself, explain why it exists or how it closes regress. The case must concern what consciousness is and why

it belongs to the foundational activity already argued for. The name cannot take the place of that argument.

An appeal to mathematical structure faces a similar question. A set of equations may specify how a model behaves once its conditions are supplied. The further issue is why that structure accounts for concrete reality and why its conditions hold. If an account identifies reality with the structure itself, it must explain that identification. A successful description and a complete foundation remain different achievements.

There is also the option of withholding judgment. A critic can say that none of the available accounts has yet established its conclusion. That is not a new foundation and does not answer the question of existence. It is a position about the state of the argument. I can challenge its reasons, but the critic's uncertainty alone is not evidence for my conclusion.

Table 2 gathers the main differences. The questions in its final column state what the proposed answers must address within this inquiry. They are not a substitute for examining the arguments.

Table 2. Proposed answers and the questions they must address in this inquiry.

Proposed answer	What it offers	Question raised here
Absolute absence	An alternative to the existence of reality	How is it coherent, and how could it explain existence?
A brute fact	An unexplained stopping point	Why treat the end of questioning as an explanation?
An endless chain	A cause for each event	What explains the order and conditions shared by the chain?
An unchanging whole	One complete history of events	What explains its concrete activity and particular structure?
Circular support	Mutual dependence	Why does the whole relation exist and operate?
A necessary foundation	Existence explained through its nature	What establishes its necessity and its relation to the world?
DEMU	Timeless reciprocal activity	Do the arguments establish balance, total reset, and renewal?

My position remains that a satisfactory account must explain why reality does not require something else to activate or sustain its foundation. DEMU seeks that explanation in timeless activity and reciprocal balance. The arguments for total reset and renewal develop what I take that balance to require. A rival account must be judged by whether it supplies an explanation of its own, rather than by whether it can be given an impressive name.

The conclusion is not that rejecting one rival automatically proves every feature of DEMU. The positive account matters: its claims must follow from the reasons supplied for them. The next chapter extends that inquiry to the purpose of the universe and the role I assign to balance within it.

References for Chapter 13

- [1] Hume, David. 1779. *Dialogues Concerning Natural Religion*. Part IX, especially paragraphs 5, 7, and 9. Statements attributed to Cleanthes are his contributions within Hume's dialogue. Hume Texts Online. Online source.
- [2] Aristotle. *Metaphysics*. Book XII, chapters 6–7, especially 1071b12–22 and 1072b14–30. Translated by W. D. Ross. Internet Classics Archive, Massachusetts Institute of Technology. Online source.
- [3] Williams, D. C. 1951. "The Myth of Passage." *The Journal of Philosophy* 48 (15): 457–472. Especially pages 462–464. Reproduced in John Bowin's metaphysics course reader, University of California, Santa Cruz. Online source.
- [4] Spinoza, Benedict. 1677. *Ethics*. Part I, especially propositions 7, 11, 15, and 18. Modernized and abridged English version by Jonathan Bennett, Early Modern Texts. Online source.

CHAPTER 14

The Purpose of the Universe

I argue that the purpose of the universe is the continual realization of balance. The earlier chapters explain why I give balance this role: it is the activity through which, in DEMU, reality needs no earlier activation and no outside support. Renewal allows the temporal universe to belong to this foundation without leaving an ever-growing inheritance at its core.

This chapter develops what it means to call that activity a purpose. It also begins the connection between the activity of Nature and our place within it. The fuller arguments about consciousness, God, free will, and the meaning of life follow in later chapters.

14.1 The Purpose

The question of purpose begins with the question we have been asking throughout this work: why is there activity at all? Describing what moves, or what moved it earlier, does not yet explain why reality has this active character. My answer is that activity performs the work that a completely inactive foundation cannot perform. It realizes the balance through which I argue that infinite regress is resolved.

Chapter 4 began the argument for timelessness. Chapters 6 and 7 developed the relation between the timeless foundation and the temporal universe. The chapters on entropy then explained why, in this model, the returning history must be erased and usable matter and energy renewed. I now draw out the significance of these claims together.

The reasoning is that reality cannot depend entirely on conditions that each require further support. If such dependence is to end, the foundation must account for its activity without borrowing it from an earlier event. I have argued that this requires a timeless, active balance. That balance gives activity its place in the explanation: motion is here because it does the work required for coherent existence.

This is why I do not simply declare that balance belongs to Nature and leave it there. The reason I give is that dynamic balance resolves the problem that led us to seek a foundation. A critic may dispute whether the earlier arguments establish this. But the disagreement must

concern those arguments, rather than treating an argued conclusion as if it were merely an unexplained label.

Reason also helps us keep different conclusions apart. An argument for necessary activity does not, by itself, establish a conscious decision to act. Nor does it tell us that every event was planned for our benefit. The immediate conclusion concerns the role of activity in the existence of the whole. That is where my account of purpose begins.

14.2 What I Mean by Purpose

In ordinary life, purpose often means an intention. I make a cup of tea because I want to drink it. The intention helps explain what I do. We also use purpose to describe a function. We say that the heart pumps blood, and that this activity supports the organism. Describing that function does not require us to suppose that the heart first makes a conscious plan.

There is also the question of why something exists in the first place. These meanings can overlap, but they are not interchangeable. In this chapter, I connect the function of fundamental activity with the reason I give for its existence. I call balance the purpose of the universe because, in my argument, realizing balance is what makes this activity necessary.

Aristotle offers a useful comparison. In *Physics* II.3, he explains that health can answer the question of why someone walks. This is an explanation through an end: what the activity is for. In *Physics* II.8, he also argues that natural processes can act for an end without deliberating as a person does. His account shows why the language of purpose need not always describe a prior conscious plan. [1]

My proposal gives balance a foundational role. It is the relation through which I seek to explain why reality is active and needs no outside support. Aristotle's discussion helps clarify a kind of explanation; it does not establish the particular claims of DEMU.

This use of purpose is stronger than saying that movement happens to have a useful effect. Many events have effects that play no part in explaining their occurrence. My claim is that the connection between activity and balance is necessary at the foundation. The earlier arguments, rather than the word “purpose,” must establish that connection.

A reader who uses purpose only to mean conscious intention may prefer to call this a necessary function. That difference in vocabulary can be made clear without losing the claim.

I am arguing that the activity of reality has an explanation in what it accomplishes. Whether that same activity is also conscious is the further question developed later.

14.3 Why Balance Must Be Active

Balance can sound like a condition in which everything has stopped. That is not how I use it here. In DEMU, expansion and contraction are complementary aspects of one activity. The whole remains balanced through their relation. Removing the activity would remove the proposed means of balance itself.

Think again of a whirlpool. Its recognizable form belongs to the movement of water. We cannot preserve the whirlpool as an actual whirlpool by stopping all its movement. The example makes the distinction visible: something can keep its form through activity. It is an illustration of that relationship, not a claim that ordinary water explains the timeless core.

This also clarifies my statement that balance must be continually recreated. At the foundation, I do not mean that balance is first achieved, then lost, then repaired. That would introduce a succession of conditions into timelessness. I mean that balance consists in activity. That said, to be perfectly clear, I do mean that, at a deeper level, balance is first achieved, then lost, and then restored. Yet there is no “before,” “after,” or “then,” because expansion = contraction, with zero delay. Timelessness is that fast. Within the temporal universe, this same account is expressed through ongoing processes of formation, return, and renewal that the animal mind fashions into a universe.

The word “must” therefore expresses necessity within my argument. It does not describe a command issued to Nature by something outside it. Nor does Nature first discover infinite regress and decide to solve the problem. The activity and its foundational role belong together eternally. There was no earlier period in which reality existed in an unfinished or unsupported condition.

Spinoza raises a serious objection to explanations through purpose. In the appendix to Part I of the Ethics, he argues that people project their own goal-seeking onto Nature. They then imagine that natural things were arranged for their use. He also objects that a God acting to obtain an end would seem to lack something that the action was supposed to supply. [2]

My account does not begin with such a lack. Balance is eternally realized in the activity I describe. Calling it a purpose does not mean that the foundation is trying to acquire something missing from its nature. It identifies the reason I give for why that nature is active. Spinoza would still reject my language of natural purpose, but the objection about an initially incomplete foundation does not describe the proposal I am making.

14.4 A Purpose Without a Final Stopping Point

If balance is the purpose, why does the universe not reach balance and stop? The question assumes that balance is a finished product that can remain after the activity ends. In my account, the activity is what the balance consists of. There is no completed object left behind that can take over its role.

Some purposes are fulfilled throughout an activity. Understanding a subject, for example, can be expressed in the ongoing work of thinking clearly about it. The activity need not be a failure simply because it does not end in a last thought after which understanding is no longer exercised. Likewise, the purpose I assign to Nature is realized in its continuing activity.

We must also distinguish fundamental balance from a temporary balance within the universe. A particular structure can form, last for a time, and break down. Other structures arise within the circulation. The foundation, as I describe it, remains balanced throughout these changes. Its balance does not require every part of the universe to remain in the same local condition.

The entropy argument matters here. A universe could remain active for a very long time while its usable differences steadily declined. Endless duration alone would not establish the renewal proposed in DEMU. My account includes the complete reset of returning history and the restoration of usable output. The activity can therefore continue without a final exhaustion of its capacity for renewal.

This does not give Nature an external destination. Nature, as defined in this work, includes everything that exists. Its purpose cannot be to serve a reality wholly outside that totality. The explanation must concern what Nature itself is and does. I locate it in the ongoing realization of balance, rather than in a final event beyond the circulation.

There is consequently no last moment at which the universe finishes its purpose and becomes unnecessary. Within this proposal, fulfilling the purpose is the activity of existence itself. Eternity describes the absence of a final exhaustion; it does not describe an infinitely delayed success.

14.5 Renewal and the Formation of New Structures

The purpose of balance must leave room for the world we experience. We encounter change, growth, and many kinds of organization. A proposal that preserved the foundation by excluding all variety would fail to explain the universe whose existence prompted the inquiry.

DEMU places this variety within an eternal circulation. Particular structures develop histories, while the core remains free of accumulated history. When a returning history undergoes the reset, its information is erased completely. There is no hidden record elsewhere from which that history could be recovered. Renewal is therefore part of what allows temporal variety to belong to the whole without becoming a permanent inheritance at the foundation.

The circulation renews portions of the universe. It does not require the entire universe to disappear together before anything can arise again. This is why I describe an eternal universe containing things that have beginnings and endings. Their formation and dissolution occur within a whole whose existence has no first activation or final exhaustion.

I also propose that the fundamental organization can give rise to further expressions of itself. The image is a large whirlpool containing smaller whirlpools. The smaller forms belong to the activity of the whole, and their dissolution does not end the whole. Chapter 15 develops the proposed connection between complementary forms, replication, and what I call cosmic sex.

That connection has its own explanatory work to do. The need for ongoing balance does not, on its own, tell us how many structures must form or why a particular kind of organism appears. The proposed process of replication must connect the general requirement of balance to the formation of further structures. Geometry can help describe that proposal, but a geometrical resemblance alone does not explain biological reproduction or cosmic growth.

In this sense, the activity I describe provides room for becoming. Nature can express its fundamental relation through forms that arise, interact, and return. “More balance” means further expressions of that relation within the universe. It does not mean that the timeless foundation was previously only partly balanced and gradually improves.

14.6 Conscious Participation and Human Purpose

We belong to the reality whose purpose we are asking about. Our bodies, experiences, and actions are part of Nature. If the account developed here is correct, conscious life participates in the same activity that sustains the temporal world. We also have the capacity to ask what that participation means and to examine how we act within it.

This opens a connection between the purpose of the universe and the meaning of life. A person can understand something of the relations on which life depends and act with that understanding. Care, cooperation, and the repair of damage are possible forms of conscious participation. Their significance, however, involves the experience and interests of those affected, as well as their place within a larger process.

We should be precise about what follows from the argument. If every event belongs to Nature, harmful events belong to it too. Their occurrence does not make them good. A claim about the balance of the whole is not an instruction to accept suffering or to treat an individual's experience as unimportant. Explaining how an event belongs to reality differs from justifying that event.

Hume identifies the need to explain the move from statements about what is the case to statements about what ought to be done. In *A Treatise of Human Nature*, he asks writers to account for that change in their reasoning. [3] For this work, the relevant task is to explain why conscious beings should value a certain form of participation. The existence of a pattern cannot answer that ethical question by itself.

My wider proposal is that consciousness belongs to the foundation, and that balance has a meaning that can be experienced from within. Chapters 18 and 19 develop that account and its connection with God and free will. Chapter 20 then develops the argument about the true meaning of life. Here, I establish the place those arguments are intended to occupy: they concern how the fundamental activity can also be the activity of a conscious reality.

There is a related distinction about continuity. The eternal existence of the whole does not by itself establish the continued existence of the same individual. Nor does the appearance of a similar person after a reset restore information that the proposal says has been completely erased. Questions about immortality and personal identity therefore require the separate discussion in Chapter 17.

The connection with human purpose begins here without being reduced to an instruction that everyone must perform the same task. Our lives differ, and the ways we understand and participate in Nature can differ. The further question is what makes that participation meaningful for conscious beings within one reality.

14.7 The Role of Purpose in DEMU

The claim of this chapter is that balance gives the activity of the universe its role in my explanation of existence. I argue that the foundation must be active because complete inactivity cannot resolve the dependence that leads to infinite regress. Balance is the relation through which I propose that activity requires no earlier activation or external support.

The account is therefore continuous with the earlier argument. I have not introduced purpose as an extra force that pushes the universe from outside. I have drawn out the significance of the work already assigned to motion. The temporal world belongs to this activity, and its continuing renewal expresses the requirement that the foundation remain balanced.

Calling this a purpose makes a specific philosophical claim: the activity is explained through what it necessarily accomplishes. It does not yet establish that the universe carries out a human-like plan, that every event is beneficial, or that each detail of Nature follows from balance alone. Those would be further conclusions requiring their own reasons.

My answer to the question of this chapter is that the universe exists in the perpetual activity of realizing balance. Its purpose has no final stopping point because balance is inseparable from the activity itself. The next chapter examines how this ongoing activity could produce further structures through the complementary process I call cosmic sex.

References for Chapter 14

- [1] Aristotle. *Physics*. Book II, chapters 3 and 8. Translated by R. P. Hardie and R. K. Gaye. Internet Classics Archive, Massachusetts Institute of Technology. Online source.
- [2] Spinoza, Benedict. 1677. *Ethics*. Part I, Appendix. Modernized and abridged English version by Jonathan Bennett, *Early Modern Texts*. Especially the discussion of human purposes and divine lack, pages 18–20 in this version. Online source.
- [3] Hume, David. 1740. *A Treatise of Human Nature*. Book III, Part I, Section I, paragraph 27; SBN 469–470. *Hume Texts Online*. Online source.

CHAPTER 15

Cosmic Sex and Atomic Replication

A worked exploratory model for the Dynamic Equilibrium Model of the Universe

My proposal is that a large circulation can give rise to smaller patterns, and that complementary patterns can combine to form another circulation. I call this cosmic sex. Here I work through a possible mechanism in enough detail to calculate part of the process.

The calculation gives a definite geometric result: four tetrahedral wave directions, interacting with their opposites through a specified rule, produce twelve directions arranged as a cuboctahedron. A separate stability calculation shows how a sufficiently developed pattern could remain after its parent stops organizing it. These results provide a possible route to explore. They do not yet establish how real atoms form or reproduce.

This is a toy model: a deliberately simplified account with stated assumptions. It has three parts: a circulating background, a rule for combining patterns, and a rule for the strength of a new pattern. The parts are calculated separately. A physical theory would have to connect them through a common set of equations.

15.1 What would count as replication

A cuboctahedron changing into a tetrahedron and back again is a cycle of one structure. For replication, something more must happen: a new structure must form while the parent remains. The new structure must also persist after the parent's direct organizing influence has ended.

I therefore use parent for the maintained circulation and child for the proposed new pattern. These words describe their roles in the hypothesis. Cosmic sex names the proposed union of complementary patterns; it does not assign biological sexes to geometric shapes.

15.2 Giving the circulation a definite form

Imagine a flow that rises near its central axis, moves outward, returns downward farther from the axis, and curves inward again. At the same time, it rotates around the axis. This gives a

concrete example of circulation within circulation. Figure 15.1 shows a cross-section through one side of that flow.

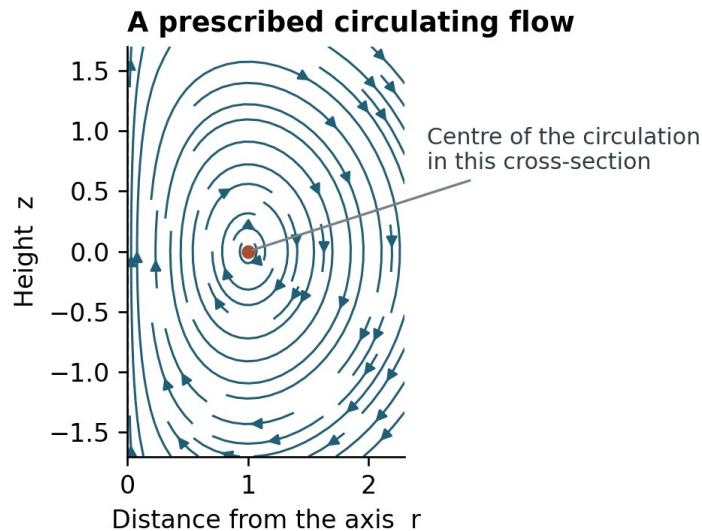


Figure 15.1. A calculated cross-section of the prescribed flow. The arrows show motion in this plane. Rotation around the central axis is also included in the equations, although it cannot be seen in this view. The coordinates use arbitrary units.

The equations in Appendix A make this flow smooth and locally conserve volume. I prescribe the flow as a starting example; I have not derived it from an equation for the proposed substance of reality. Its symmetry also means that it does not, by itself, select four smaller whirlpools. A rule that breaks this symmetry is needed to produce a tetrahedral arrangement.

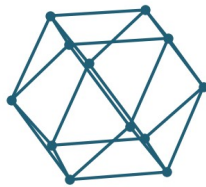
These coordinates describe an extended, changing expression of the proposal. They are not a mathematical description of the timeless core. The flow uses finite speeds, and this calculation makes no claim to reach timelessness by increasing them.

There is a limited physical comparison here. Gharib, Rambod, and Shariff studied vortex rings formed by a jet of water. Under the conditions they examined, a leading ring could separate from the flow feeding it. Their work shows a real formation and separation process, but it does not establish tetrahedral whirlpools or atomic replication. [1]

15.3 Folding and the formation of a tetrahedron

There is a precise geometric basis for connecting a cuboctahedron with a tetrahedron. Fuller's jitterbug is an articulated framework of twenty-four struts. It can move through cuboctahedral, octahedral, and tetrahedral configurations. At the octahedral stage the struts coincide in pairs; at the tetrahedral stage they coincide in groups of four. [2]

Cuboctahedron



24 single edges

Octahedron



12 doubled edges

Tetrahedron



6 quadrupled edges

Figure 15.2. Landmark configurations of the articulated framework, drawn with the same strut length. Coincident struts appear as one line. These are endpoint configurations, not a simulation of the folding motion. Based on Fuller's description. [2]

When I call a tetrahedron a folded cuboctahedron, this articulated construction gives the claim a clear meaning. It concerns a framework whose joints can move and whose struts can coincide. It does not mean that two different rigid solids are identical. Folding alone also leaves us with one framework; it does not produce another one.

For a first model of formation, I now assume four equal concentrations of activity at the same fixed distance from a common centre. Suppose they spread apart as evenly as possible. I assign each pair an inverse-distance repulsion potential. Under that rule, their lowest-energy arrangement is a regular tetrahedron.

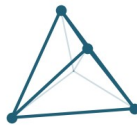
The geometric minimum can be proved, and eight numerical relaxations from different starting arrangements reached it. The calculation explains why four equal, mutually repelling components can become tetrahedral. The choice of four components and the repulsion rule remain assumptions. The calculation does not explain why a rotating background would produce exactly four.

15.4 How opposite tetrahedral patterns can combine

To make the pairing step explicit, I represent each tetrahedron by four wave directions. A wave direction specifies how a repeating pattern varies through space. The four directions point towards the corners of a regular tetrahedron. The complementary set points in the opposite directions.

Suppose the interaction contains a product of the two wave patterns. Multiplying waves can produce new spatial patterns whose wavevectors are sums of the original wavevectors. Since the second tetrahedron has opposite directions, those sums are the differences between pairs of directions in the first tetrahedron.

Four directions T



Opposite directions $-T$



Twelve differences



Direction space at one scale — not a picture of an atom

Figure 15.3. The first two panels show the complementary sets of four wave directions. Their twelve nonzero pairwise differences form the cuboctahedron in the third panel. The points represent wavevectors, not particles or positions on the surface of an atom.

There are sixteen pairings. Four pair a direction with its exact opposite and give a zero wavevector. The other twelve give twelve different nonzero wavevectors. Their endpoints form a regular cuboctahedron with twenty-four edges, eight triangular faces, and six square faces. This relation is exact for the coordinates used here.

The zero wavevector represents a spatially uniform contribution. It is not a newly discovered nucleus or a point-sized core. Likewise, twelve wave directions do not imply twelve material particles. This calculation establishes a geometry of interacting patterns.

The two tetrahedra must have the appropriate relative orientation and phase relation for this exact construction. Calling them opposite does not by itself give them opposite electric

charges. A physical account must identify the field involved and explain why it obeys the assumed interaction rule.

15.5 One proposed production cycle

The following sequence combines the geometric idea with a possible test of persistence. It is a proposed sequence, with the calculated steps identified along the way.

1. The parent circulation remains active. I treat it as maintained by the wider circulation. The present calculation does not solve for changes in the parent.

2. Two complementary patterns form nearby. Each has four tetrahedral directions. Their production from the parent is assumed at this stage; the calculation in Section 3 explains their arrangement once four components exist.

3. Their interaction supplies cuboctahedral directions. The multiplication rule in Section 4 gives this geometric result. A spatially confined structure would still need a mechanism that keeps the pattern together.

4. The candidate pattern grows. I represent its strength by a number called q . The parent supplies a temporary organizing influence, while an assumed feedback rule determines whether the pattern grows or fades.

5. The direct organizing influence ends. I remove it at a chosen time and calculate what happens to q . This tests persistence after withdrawal of that influence. It does not simulate a physical neck breaking or a vortex detaching.

6. A child remains if its strength has crossed the persistence threshold. The proposed outcome is then a maintained parent and an additional persistent pattern. Repeating this process would require the child to acquire the same organizing ability, which is not yet calculated.

This gives cosmic sex a specific proposed role: complementary patterns organize a new pattern from activity available in their surroundings. The folding analogy describes changes of form; the pairing rule supplies a possible way to recover cuboctahedral order. Production requires both a source of activity and a way for the new pattern to remain together.

15.6 Testing whether the pattern remains

The number q measures the strength of a candidate pattern. Zero means that the pattern has faded; one is its chosen full strength. I start every calculation with a small seed of 0.02. That seed is supplied as an initial condition, so this test does not show how a seed first appears.

I use a simple feedback rule with a threshold at 0.35. Once the parent's influence is removed, a pattern above the threshold strengthens towards one. A pattern below it fades towards zero. The rule is chosen to explore the required behaviour. Its physical origin has not been derived from DEMU.

$$\frac{dq}{dt} = q(1 - q)(q - 0.35) + S(t)$$

Here S is the parent's temporary organizing influence. Time and strength use arbitrary model units. A larger value of S means a stronger formation input. The table compares three formation conditions and one case in which the stabilizing feedback is removed.

Condition	Input S	Release time	q at release	Result by time 40
Strong formation	0.10	6	0.6024	Approaches 1
Weak formation	0.03	6	0.0980	Fades towards 0
Early separation	0.10	2	0.1788	Fades towards 0
No stabilizing feedback	0.10	6	0.2532	Fades towards 0

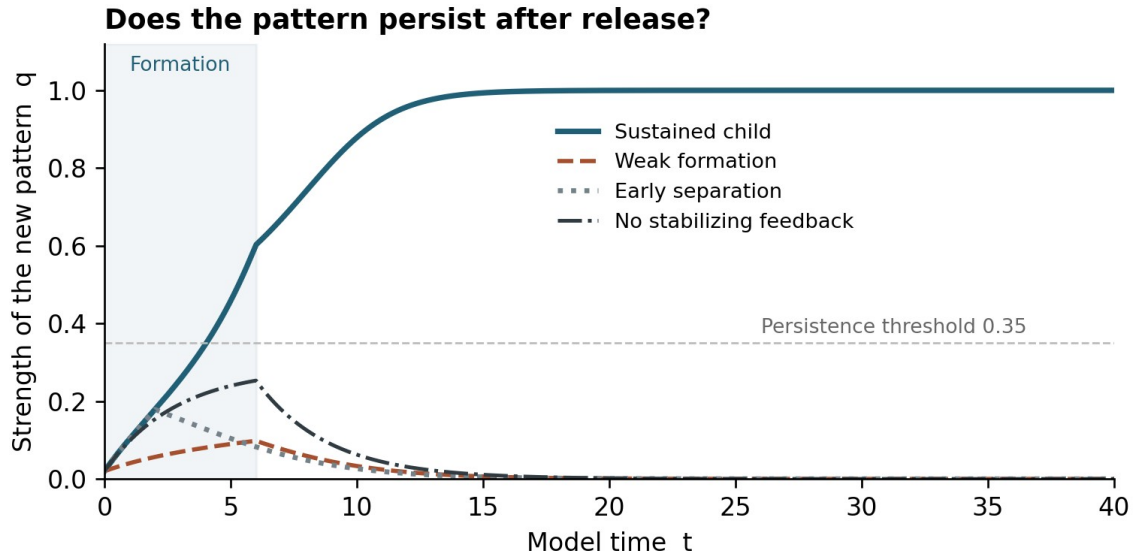


Figure 15.4. Numerically calculated pattern strength. The shaded interval marks the formation period for the cases released at time 6. The early-separation case is released at time 2. The threshold applies to the cubic feedback rule; the comparison without that feedback has no persistent nonzero state.

In the successful case, q reaches about 0.6024 before release and then approaches one. The weaker input and the earlier release both leave the pattern below the threshold, so it fades. Removing the stabilizing feedback also makes it fade, even with the stronger input.

This result depends on the chosen rule. It shows the kind of feedback a successful mechanism would need, rather than deriving that feedback from rotation. Because q is a single number, the model also cannot show whether the pattern stays confined to a small region of space. Persistence of amplitude and spatial confinement must eventually be tested together.

15.7 Accounting for the activity that supports the child

Ending the parent's organizing influence does not mean ending every exchange with the surroundings. A whirlpool can remain after a temporary disturbance has passed while still depending on the river that supports it. The successful branch here likewise assumes continued access to surrounding activity.

Within DEMU, I would identify this supply with renewed output within the wider circulation. It is not a withdrawal from the core treated as a battery. This local calculation

neither models nor establishes the production of that renewed output; it states the supply that the proposed child would require.

For a transparent numerical account, I assign the child a stored energy equal to q squared. I also assume a maintenance loss of 0.02 times q squared per unit of model time. These are illustrative choices, not measured properties of an atom. The required input must cover both growth in stored energy and the maintenance loss.

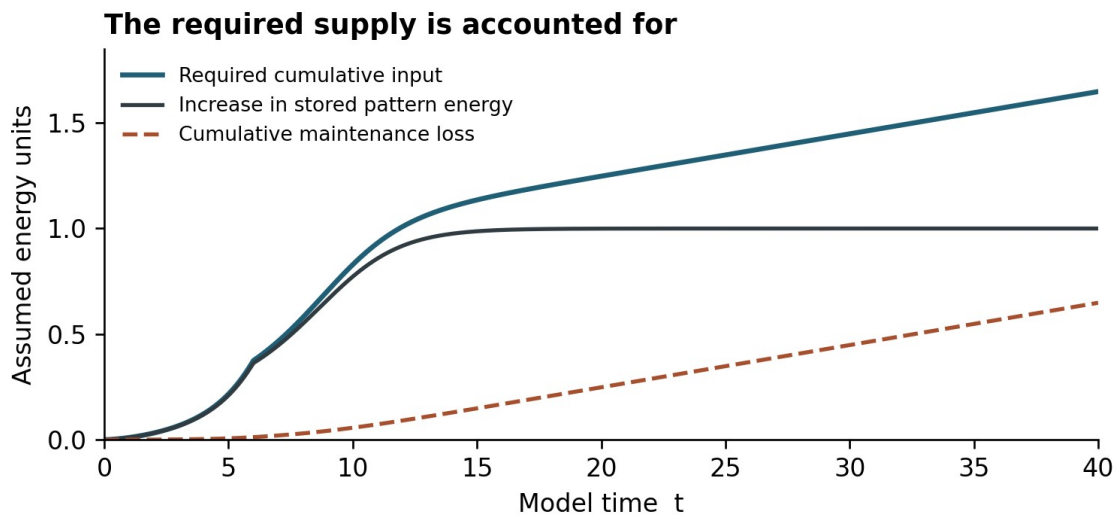


Figure 15.5. Energy accounting for the successful branch, using the stated assumptions. Cumulative required input equals the increase in stored energy plus cumulative maintenance loss. The graph calculates a requirement; it does not demonstrate a source that can meet it.

By time 40, the required cumulative input is about 1.6477 energy units. About 0.9996 units have increased the stored energy, and about 0.6481 units have become maintenance loss, here treated as heat. At full pattern strength, the assumed ongoing maintenance power is 0.02 energy units per unit of time.

This bookkeeping concerns a local formation interval. It does not calculate the total entropy reset proposed elsewhere in DEMU, or replace that total reset with information stored elsewhere. A complete physical model would have to connect local formation, continued support, and the proposed return to the core.

15.8 What would make the new structure an atom

A persistent geometric pattern is not yet enough to identify an atom. In established physics, atomic nuclei are formed and changed through processes such as nuclear reactions. The term

nucleosynthesis refers to the creation of nuclei. [3] A proposal about atomic replication must specify how its process relates to the material inputs, outputs, and measured properties involved.

The simplest first target would be ordinary hydrogen: a system corresponding to a proton and an electron in a bound state. The model would need to explain which part carries each charge, why the charges have their measured values, and why the system has the observed mass and energy levels. It would also need to explain what supplies the material or energy required to form another such system.

The geometry does not yet provide those answers. Four tetrahedral components need not be four protons, and twelve cuboctahedral directions need not be twelve particles. The size of the figures is arbitrary. No physical length, mass, or frequency has been calculated from this construction.

There is also a difference between a pattern extending throughout space and one that remains concentrated around a particular place. The wave calculation above establishes an arrangement of directions. It does not bind those waves into a small object. The next physical model needs a field equation that produces both the pattern and its confinement, with the formation input removed after the new structure develops.

A useful first test would be to predict one measured property of a chosen atom from that equation without adjusting its parameters merely to obtain the known answer. Predicting a transition frequency, size, or formation threshold would create a direct point of comparison. Failure at that point would identify which part of the proposed mechanism needs to change.

15.9 What this model establishes

The strongest completed result is the connection between the tetrahedral and cuboctahedral wave directions under the specified interaction. The persistence calculation adds a conditional result: a candidate pattern can survive withdrawal of its formation input when the assumed feedback maintains it and the required surrounding supply remains available.

Part of the proposal	Present status
Circulating background	An explicit flow is prescribed; it does not generate the seeds.
Tetrahedral arrangement	Calculated for four equal repelling components on a sphere.
Cuboctahedral directions	An exact result of the stated quadratic wave interaction.
Persistence after release	Calculated under an assumed feedback rule and background supply.
Physical separation and confinement	Not derived by the scalar amplitude model.
Identification and replication of atoms	A further physical hypothesis; not demonstrated here.

The next step is to replace the separately prescribed stages with one spatial model. That model should let a maintained parent generate disturbances, select their arrangement, couple them, and release a confined child through its own dynamics. A second production cycle would then test whether the child can reproduce the parent's organizing role.

My proposal is therefore more specific than saying that spin makes smaller spins. It proposes a route from circulation to complementary patterns, from their interaction to cuboctahedral order, and from that order to a persistent new structure. The calculations identify a workable geometric relation and the kind of stabilizing behaviour required. The physical connection between those steps remains the central task.

Appendix A Geometry and the prescribed flow

All variables in the calculations below use dimensionless units. The formulas specify this exploratory model and allow the numerical results to be reproduced.

A1 The circulating background

In cylindrical coordinates, r is distance from the axis, z is height, and φ is the angle around the axis. Define the streamfunction ψ and the three velocity components as follows.

$$\begin{aligned}\psi(r, z) &= r^2 e^{-(r^2 + z^2)} \\ u_r &= 2rz e^{-(r^2 + z^2)}, \quad u_z = 2(1 - r^2) e^{-(r^2 + z^2)} \\ u_\varphi &= r e^{-(r^2 + z^2)}\end{aligned}$$

These radial and vertical components are obtained from derivatives of ψ . The velocity field has zero divergence, including the smooth limit at the axis. Contours of ψ give the closed paths in the cross-section. Rotation around the axis adds the third component of motion. A complete three-dimensional particle path need not close after a single turn.

A2 The four directions

Use the following four vectors, each divided by the square root of three, as the unit wavevectors k_1 to k_4 .

$$T = \frac{1}{\sqrt{3}} \{ (1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1) \}$$

For any distinct pair, the dot product is $-1/3$ and the squared distance is $8/3$. The opposite set is $-T$. The twelve nonzero differences $k_i - k_j$ are the permutations of $(0, \pm 2/\sqrt{3}, \pm 2/\sqrt{3})$, with both signs chosen independently. These are the vertices of a regular cuboctahedron in wavevector space.

A3 Why four repelling directions form a tetrahedron

Let n_1 to n_4 be four unit vectors. Choose the repulsion potential below, where each pair is counted once.

$$E_{rep} = \sum_{i < j} \frac{1}{|n_i - n_j|}$$

The sum of the six squared pair distances equals 16 minus the squared length of the sum of the four vectors, so it cannot exceed 16. The function one divided by the square root of distance squared is convex. Jensen's inequality therefore gives the following lower bound.

$$E_{rep} \geq \frac{6}{\sqrt{8/3}} = 6\sqrt{3/8} \approx 3.674234614$$

Equality requires a zero vector sum and six equal pair distances. A regular tetrahedron meets both conditions. Eight BFGS numerical relaxations also reached this arrangement; the largest error in a pairwise dot product relative to $-1/3$ was less than 5.1×10^{-8} .

A4 The quadratic interaction

Take four equal wave components with a common phase origin. Define F as their sum and F^* as its complex conjugate. The star reverses the wavevectors in this construction. The multiplication gives

$$F(x) = \sum_{i=1}^4 e^{ik_i \cdot x}$$

$$F(x)F^*(x) = |F(x)|^2 = 4 + \sum_{i \neq j} e^{i(k_i - k_j) \cdot x}$$

The symbol i in the exponential is the imaginary unit; the i and j used as subscripts label vectors. The four self-pairings give the constant term. The remaining twelve give the cuboctahedral directions. Evaluating the two sides at twenty randomly chosen positions gave a maximum numerical difference below 3.6×10^{-15} . This checks the algebraic implementation, not a physical field law.

Appendix B Stability and energy calculations

B1 The amplitude rule

$$\frac{dq}{dt} = q(1 - q)(q - a) + S(t), \quad a = 0.35, \quad q(0) = 0.02$$

The formation input S is constant until the specified release time and zero afterwards. When S is zero, the fixed values are $q = 0$, $q = 0.35$, and $q = 1$. The slopes of the feedback function at these points are -0.35 , $+0.2275$, and -0.65 . The negative slopes make zero and one stable against sufficiently small changes; the positive slope makes 0.35 an unstable threshold.

In the comparison without stabilizing feedback, replace the cubic term with $-0.35q$. That model has only the decaying state after release. All cases are integrated to time 40, with the same small initial seed. The release times are imposed; the equations do not decide when separation occurs.

B2 Energy required by the successful branch

$$E_{child} = q^2, \quad P_{loss} = 0.02 q^2$$

$$P_{required} = 2q \frac{dq}{dt} + 0.02 q^2$$

$$E_{input}(t) = q(t)^2 - q(0)^2 + 0.02 \int_0^t q(\tau)^2 d\tau$$

The last term is cumulative maintenance loss. On the successful branch, q increases throughout, so the required power is positive. Its peak is about 0.1997 in the chosen units. These definitions impose a consistent local energy account; they do not derive an available supply from the flow field.

B3 Numerical method

The amplitude equations were integrated with SciPy `solve_ivp` using RK45, relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and maximum time step 0.05. Each run was split at release to handle the change in S . Curves use 4,001 equally spaced samples between time 0 and time 40. Cumulative maintenance loss was calculated from those samples with the trapezoidal rule.

Geometry checks counted distinct difference vectors, shortest edges, and coplanar convex-hull faces. The repulsion checks used eight BFGS minimizations over spherical coordinates. A NumPy random generator with seed 2718 supplied twenty three-dimensional normal samples for the wave identity check, followed by the eight random starting configurations. The analytic arguments establish the geometry; these numerical checks verify the implementation.

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The flow example, wave calculations, amplitude model, and energy account in this document are specified here. The references support the stated comparisons and geometric precedent; they are not evidence that the proposed atomic mechanism has been observed.

CHAPTER 16

The Four Forces and the Development of Nature

The previous chapter asked how Nature could produce new forms through the activity of its existing forms. I now turn to three possible applications: the four forces, the formation of Earth, and the size of dinosaurs. These subjects belong together because they concern how matter holds together, how a world develops, and what kinds of life that world can support.

My starting point remains the argument for a timeless foundation. In DEMU, the many things within Nature arise through one underlying circulation. I therefore look for connections between processes that are usually studied separately. The aim is to explain how the same foundation could appear as binding, transformation, growth, and life.

The earlier argument gives a reason to seek such a connection. The particular connections proposed in this chapter also need their own support. A claim about why reality needs a foundation and a claim about how Earth grew answer different questions. Keeping those questions clear allows the wider argument to develop without treating its applications as already established.

16.1 The Four Forces

Physics describes four fundamental interactions: gravity, electromagnetism, the strong interaction, and the weak interaction. They differ in what they affect, how they act, and the distances over which their effects become important. The Standard Model describes the electromagnetic, strong, and weak interactions. Gravity is described separately through general relativity. [1]

Gravity holds planets in orbit and draws ordinary matter together on large scales. General relativity describes it through the way spacetime is curved. Electromagnetism concerns electric charge and includes both electricity and magnetism. It allows atoms to form chemical bonds and is involved in light. Unlike charges attract; like charges repel. [2]

The strong interaction binds quarks into protons and neutrons. A remaining effect of this interaction binds protons and neutrons within atomic nuclei. This helps explain how a

nucleus can remain together even though its positively charged protons repel one another electrically. [3]

The weak interaction allows certain particles to change into other particles. For example, it plays a part in beta decay, in which a neutron can turn into a proton while producing an electron and an antineutrino. The word “weak” is therefore a name for a particular interaction. It does not simply mean that something is unstable or beginning to fall apart. [4]

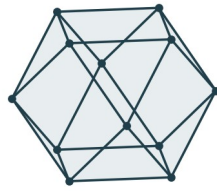
These descriptions raise the question that interests me: could the four interactions be different expressions of the same underlying activity? In DEMU, I propose that their differences arise through different arrangements of the dual torus. Its geometrical framework gives me a way to picture these arrangements.

Table 3. Familiar physical roles and the geometrical connections proposed in DEMU. The final column states the proposal; it is not an established mapping in physics. [1–4]

Interaction	Familiar physical role	Proposed DEMU shape
Gravity	Large-scale attraction and orbital motion	Cuboctahedron: balance
Electromagnetism	Electric charge, light, and chemical bonding	Octahedron: polarity
Strong interaction	Binding within protons, neutrons, and nuclei	Tetrahedron: binding
Weak interaction	Particle transformations, including beta decay	Icosahedron: transformation

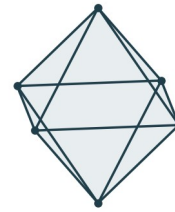
I associate the cuboctahedron with gravity because it represents the balance from which the other forms develop. I associate the octahedron with electromagnetism because its opposing corners offer a picture of polarity. The tetrahedron represents concentrated binding, while the icosahedron represents transformation between arrangements. These are the connections I propose to investigate.

Cuboctahedron



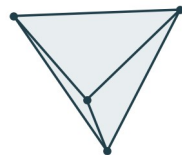
Proposed link: Gravity

Octahedron



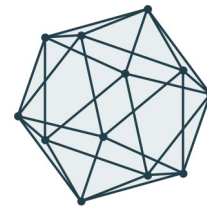
Proposed link: Electromagnetism

Tetrahedron



Proposed link: Strong interaction

Icosahedron



Proposed link: Weak interaction

Figure 16.1. Four geometrical forms used in the proposed account. Their outlines are drawn from their corner coordinates. The force labels show the author's proposed connections; the drawing does not establish a physical unification.

The shapes give the proposal a visible form. Their physical interpretation requires a further step. The fact that a tetrahedron forms a rigid framework does not establish the strength of a nuclear interaction. Equally, a pair of opposite corners does not by itself explain electric charge. The question is how the activity represented by a shape could produce the particular behaviour of the associated force.

The same point applies to the number four. Finding four useful shapes would not explain why Nature has these four interactions. The account would need to show why these arrangements have the proposed physical roles, and why other possible arrangements do not introduce further fundamental forces. This is a question about what the system does, as well as what it looks like.

Gravity gives the clearest example of my intended explanation. I picture a planet as a large circulation containing smaller circulations. Matter moves inward through the

contracting part of this process, while renewal belongs to its outward part. Within this picture, attraction is an expression of the whole flow.

An everyday image may help. Imagine standing beneath a waterfall. The water moves down around you, while the ground prevents you from moving with it. My proposed picture of weight is similar: a body participates in an inward flow, but the surface beneath it prevents further descent. The image makes my idea concrete. It does not mean that ordinary air pressure causes gravity, or that the proposal has already explained motion in a vacuum.

To develop this image into a physical account, I would need to explain why different materials fall with the same acceleration under the same conditions, why gravity changes with distance, and how the proposed flow affects light. A moving medium would also raise questions about resistance and heating. The account must explain why its circulation produces the observed gravitational effects without adding effects that observations rule out.

This brings the chapter back to the earlier argument. I argue that the foundation must sustain its own balance because otherwise the problem of support remains unresolved. Here I propose that the different forces are ways in which that balance is expressed within the world. The philosophical claim supplies a direction for the explanation; the physical relationships would show whether this particular application succeeds.

16.2 The Formation of Earth

In the standard scientific account, Earth formed about 4.5 billion years ago as material in the young Solar System gathered under gravity. This gradual accumulation is called accretion. Earth developed a metal-rich core, a rocky mantle, and an outer crust. Its Moon probably formed after a major collision involving the young planet. [5]

That account concerns the history of our planet. In DEMU, the existence of such a history does not mean that reality itself began at the same moment. A whirlpool can form within a river that was already flowing. In a similar way, I propose that a planet can form within an eternal circulation. Earth has a history even if the foundation of that history has no first event.

My further proposal is that a planet may develop partly through renewal within its own circulation. On this view, Earth is more than an object that collected material from its surroundings. It may contain a process through which returning material undergoes the total

reset described earlier, with its inherited information erased and no record preserved elsewhere. Renewed material then becomes available again. Atomic replication, as proposed in Chapter 15, would provide a possible further route to growth.

Renewal and growth need to be distinguished here. Suppose a process takes in a certain amount of material and returns an equal amount. That process can renew the material without increasing the amount present. For Earth to gain mass through the proposed circulation, its retained output would have to exceed what leaves or is consumed by that process. For Earth to grow in radius, the account must also explain how the material occupies more space.

This matters because a planet can change in several ways. It can gain matter from outside, rearrange what it already contains, change its density, or expand because it becomes hotter. My proposal adds another possibility: an internal process linked to the renewal and replication described earlier. The proposal becomes clearer when I specify which kind of change is being claimed.

I therefore use “a growing Earth” to mean a planet whose overall dimensions increase through its development, with a corresponding account of its mass and internal structure. I am interested in whether an inward and outward circulation could help explain that development. I do not take the mere presence of movement inside Earth as proof of the particular circulation proposed in DEMU.

The word “core” also needs care at this point. Earth’s geological core is a physical region described as a solid inner part surrounded by a liquid outer part. [5] My timeless core names a proposed foundational condition. If I locate a local expression of that condition at Earth’s centre, I am making an additional physical claim. The shared word does not make the two accounts identical. A hot, rotating geological core does not become timeless merely because I call it a core.

There is already direct evidence that Earth’s surface is active. Its outer shell is divided into moving plates, and their interactions produce earthquakes, mountain building, and other changes. [5] Such activity is relevant to my interest in a developing Nature. However, movement of the surface does not by itself show that the planet as a whole is expanding.

Measurements place a concrete limit on the growing Earth proposal. A NASA-led study reported in 2011 used space-based measurements to look for changes in Earth’s average

radius. It found no statistically significant expansion over the period studied. [6] This does not establish that Earth's radius was identical throughout its entire history. It does mean that a claim of substantial ongoing growth must address those measurements.

A useful version of my proposal would therefore say when growth occurred, how much occurred, and what physical traces it should leave. It would connect those claims to the distribution of material inside Earth, its surface history, and its gravitational field. If growth was faster in the past, the reason for that change would also belong in the explanation.

The central idea is that a planet may be a developing expression of the wider circulation. In that case, its formation is an episode within Nature's activity. Whether Earth actually developed through the particular process I propose must be decided through the details of that planet. This leads to an especially concrete question: could its earlier conditions have helped very large animals to live?

16.3 The Size of Dinosaurs

The largest dinosaurs make the question of bodily size difficult to ignore. How could a living animal carry such a body, move it, feed it, and keep it alive? The question is especially striking for sauropods, the long-necked plant-eating dinosaurs that included the largest land animals.

My proposed connection is straightforward. If Earth was once smaller and had sufficiently less mass, its surface gravity could have been weaker. Weaker gravity would reduce the weight that an animal's bones and muscles had to support. A world with weaker gravity might therefore allow some forms of life to reach larger sizes.

The words "sufficiently less mass" matter. A smaller planet does not automatically have weaker surface gravity. Gravity at the surface depends both on the planet's mass and on the distance from its centre. In a simple spherical model, the relationship is expressed by Newton's gravitational law. [7]

$$g = \frac{GM}{R^2}$$

Here, g means gravitational acceleration at the surface, M is the planet's mass, R is its radius, and G is the gravitational constant. The radius is the distance from the centre to the

surface. The equation assumes a spherical mass distribution and leaves out the effect of rotation on apparent weight.

Consider two imaginary planets, each with half Earth's present radius. Give the first planet all of Earth's present mass. Its surface gravity would be four times as strong, because that mass is concentrated within a smaller radius. Give the second planet one-eighth of Earth's present mass. Its surface gravity would be half as strong. The second planet would have the same average density as Earth in this simplified comparison.

The difference is decisive. Both planets are smaller, but their surface gravity changes in opposite directions. If I want weaker past gravity to follow from a growing Earth, I need an account of how both mass and radius changed. Figure 16.2 shows why the radius alone cannot settle the question.

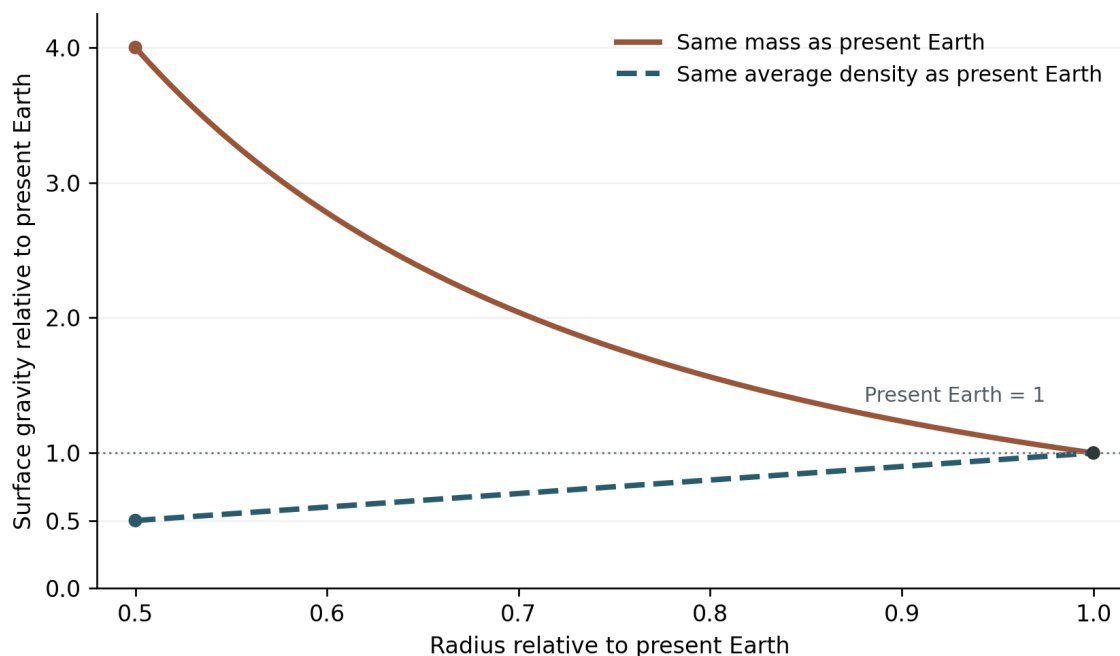


Figure 16.2. Two calculations for spherical planets using Newton's gravitational law. At half the present radius, keeping Earth's present mass gives four times its surface gravity; keeping its present average density gives half its surface gravity. These curves compare assumptions and do not reconstruct Earth's history. [7]

For example, with half the present radius, past mass would need to be less than one-quarter of the present mass for surface gravity to be weaker under this equation. At exactly

one-quarter of the mass, surface gravity would be unchanged. These are conditional calculations, not measurements of ancient Earth.

Although DEMU seeks a deeper account of gravity, this calculation remains useful. A deeper account must explain the familiar relationship where that relationship works. If it predicts a different relationship in Earth's past, the difference needs a reason and an independent way to test it. Otherwise, a change in the law could be chosen simply to give the desired answer.

There is also a simple reason why size places demands on a body. Imagine enlarging an animal while keeping its proportions and material density the same. If every length doubles, its volume and mass increase eightfold. The cross-sectional area of each supporting leg increases only fourfold. Under the same gravity, the load per unit area would therefore double. Large animals need suitable proportions and structures, rather than being exact enlargements of smaller animals.

Weaker gravity would reduce part of that burden. It would not remove the need to feed the animal, move its limbs, circulate its blood, or produce offspring. An animal's mass would also remain the same when its weight decreased. Lower gravity changes the pull on a body; it does not make the body contain less matter.

Research on sauropods offers a detailed biological explanation of their size. Sander's evolutionary model connects several features: long necks, small heads, food swallowed without chewing, a bird-like breathing system, rapid growth, and reproduction through eggs. These features could reinforce one another. A light head and suitable neck structure, for example, allowed feeding over a large area without moving the whole body. The model explains gigantism through a combination of traits, rather than a single cause. [8]

This account matters because it gives the lower-gravity proposal a specific comparison. To explain dinosaur size better, the proposed change in gravity would need to account for something that anatomy, growth, feeding, and reproduction leave unexplained. It would also need to fit the differences between large and small animals living under the same conditions.

The existence of a giant animal is therefore not itself a measurement of gravity. Fossils show us that large bodies existed and provide evidence about how those bodies worked. To infer weaker gravity, I would need an argument that connects the fossil evidence to a

particular gravitational condition while accounting for the animal's structure. I cannot draw that conclusion merely from the impression that an animal is too large for the world I know—unless its proportions border on the absurd, as I believe those of the largest dinosaurs do.

That is to say, the immense size of the largest dinosaurs is difficult to overlook, and I believe it demands a more convincing explanation than modern science currently offers. It would be fascinating to reconstruct one of these giants and test whether its body could withstand present-day gravity. Would it stand, or would it collapse under its own weight? More broadly, I question the continued reliance on the accretion and Pangaea models, both of which I regard as outdated and overdue for reassessment.

The proposed connection can now be stated precisely: if Earth's earlier mass and radius produced weaker surface gravity, that condition could have reduced some of the mechanical limits on animal size. Whether it did so depends on evidence about Earth's development and the animals themselves. A convincing explanation would bring the geological and biological records together.

Across these three sections, the aim has been to connect the foundation proposed in DEMU with the world we can investigate. The forces concern how matter interacts. Earth's formation concerns how a particular world develops. Dinosaur size concerns the possibilities and limits of life within that world. I propose that these belong within one account of Nature's activity. The value of that proposal will grow as its connections become precise enough to explain the differences as well as the similarities.

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CHAPTER 17

Immortality and the Nature of Existence

The earlier chapters describe reality as an eternal activity of balance and renewal. Particular things arise and dissolve within this activity, while the whole continues. This raises a question that matters to us personally: what does this mean for the possibility of immortality?

I consider two connected ideas. The first is that consciousness belongs to the timeless foundation and therefore has no final end. The second is that a living body might continue renewing itself through the use of low-entropy matter. One concerns the nature of consciousness. The other concerns the continued life of an organism.

My aim is to explain how these ideas fit within DEMU and what each would mean for the individual. The word “immortality” must have a clear subject. We need to know what continues: reality, consciousness, the living body, or the person whose life is at stake.

17.1 The Power of Power

Reason allows us to ask whether death is an unavoidable feature of existence itself or a feature of the particular forms that existence takes. These are different questions. An eternal reality can contain temporary forms. The continued existence of the whole does not, by itself, tell us how long any one form can last.

My argument begins with the foundation developed in the earlier chapters. I have argued that reality requires an active balance that closes infinite regress. I give balance this role because of the explanatory work I argue that it performs. Its necessity is meant to follow from the argument, rather than being inserted as an unexplained fact.

I also propose that consciousness is what this fundamental activity feels like from within. Chapter 18 will defend the further claim that the self-awareness of consciousness is essential to the closure I seek. Here, I can explain the consequence of that claim without treating the later argument as already complete.

The reasoning is straightforward. Reality cannot lose the condition that makes its existence possible. If consciousness is that indispensable condition, consciousness cannot be

something that appears briefly and then disappears from reality altogether. On that account, it belongs to the eternal foundation.

This would give consciousness a stronger role than that of a temporary observer looking at an otherwise complete universe. It would make consciousness part of what allows reality to be complete in the first place. That is the route by which I argue for its eternal nature.

The conclusion concerns fundamental consciousness. A further step is needed to connect it with the continued existence of a particular person. Reason is useful here precisely because it helps us see which conclusion follows from which claim. An argument for an eternal conscious foundation should not quietly become an argument that every human memory lasts forever.

17.2 The Whole and the Individual

Consider the image of a whirlpool within a larger flow. The water can continue after the whirlpool disappears. We would not normally say that the same whirlpool survives merely because its water still exists. Its continued existence depends on the continuation of a particular organized activity.

The example helps explain why the impossibility of absolute nonexistence would not, on its own, establish personal immortality. The end of a particular organization need not be the disappearance of all reality. If a person ceases to live, something more must be shown before the persistence of the wider universe can count as that person's survival.

My wider proposal is that the many centres of experience belong to one conscious reality, which I call the Mind at Large. In this account, an individual life is an expression of that reality. The deeper subject of experience would therefore have an existence that reaches beyond any one human biography.

This changes the question of who I am. If I identify myself entirely with this body, these memories, and this personal history, their continuation is central to my survival. If I identify myself with the fundamental consciousness expressed through them, the claim concerns something wider than my biography. These meanings can be connected, but they should be stated openly.

Table 4. Different claims about continued existence. Each claim identifies something that continues and a question that remains to be answered.

Claim	What continues	Question to answer
An eternal reality	The whole of reality	Does this preserve any particular person?
Eternal consciousness	The conscious foundation proposed in DEMU	How is an individual related to it?
Personal survival	The same living or conscious individual	What preserves that individual's continuity?
Biological longevity	A living organism through ongoing renewal	Can repair prevent continuing age-related decline?

The table shows why I do not use the eternity of the whole as a substitute for an account of the individual. The possibility I want to explore is richer: consciousness may belong to the foundation, while individual lives may have a capacity for renewal that we have not yet understood. Each part of that proposal has its own argument.

17.3 Consciousness and Personal Identity

Personal identity asks what makes a later life or experience belong to the same person. It is not simply a question of keeping every particle in place. Our bodies change throughout life, yet we usually regard ourselves as the same individuals through those changes.

John Locke makes a useful distinction in his discussion of identity. A living organism can remain the same organism while the matter within it changes. When discussing a person, he gives special importance to consciousness and its connection with past thoughts and actions. His example of a prince's consciousness appearing in a cobbler's body separates the question of who the person is from the question of which body is present. [1]

I do not need to accept every part of Locke's account to use the question it raises. What must remain connected for this later experience to count as mine? The answer might involve the continuing body, the continuing subject of experience, memory, or a relation among them. Ordinary forgetting already shows why remembering every event cannot be a requirement for remaining the same person.

DEMU adds a particular difficulty: the reset is total. When a returning history is erased, no hidden copy of its information remains elsewhere. I cannot describe that history as completely erased and then preserve the person's biography in a cosmic memory. That would change the reset I have proposed.

The same point applies to a later being who resembles an earlier person. Similarity alone does not establish that the same individual has returned. If the earlier history has been completely erased, a similar pattern arising later is not the recovery of a surviving record. A claim of personal continuity would need a different basis.

Within my account, foundational consciousness is not itself a stored record of every life. Its proposed eternity therefore differs from the permanent preservation of personal history. If the fundamental subject is one, there may be a sense in which the subject of experience does not end when a particular expression ends. Whether that is enough to count as the survival of the individual depends on the account of identity being defended.

For the definition given earlier in this work, the continuation of a living or conscious individual is the central issue. I therefore keep the claim about eternal consciousness distinct from the claim that a particular person survives death. Chapter 18 develops the first claim. The rest of this chapter considers a route toward preserving the living person through ongoing renewal.

17.4 Life as Ongoing Repair

A living body is already an activity of renewal. It takes in material, changes it, uses it, and releases waste. Digestion supplies components that the body uses for energy, growth, and repair. Proteins, for example, are broken into amino acids, which can then be used to build proteins the body needs. [2]

This gives us a concrete starting point. Continued life does not require a body made of permanently unchanged matter. It requires the organized activity of the body to continue. Renewal can help preserve that activity even while individual components are replaced.

However, ordinary renewal does not repair every kind of damage indefinitely. Research on ageing describes several connected processes, including damage to DNA, failures in the maintenance of proteins, changes in the cell's energy systems, and cells that stop dividing

while affecting nearby tissue. Ageing involves interactions among these processes. It cannot be reduced to the idea that the body has simply run out of food. [3]

The question for longevity is therefore whether renewal can become sufficiently complete to prevent a continuing decline in the body's ability to function. A body would need to maintain the systems that carry out repair as well as repair the damage those systems encounter. Supplying material is one part of this task; using it successfully is another.

Nature also gives us reasons to avoid assuming that every organism follows the same pattern of ageing. A long-term study of Hydra, small freshwater animals, found very low mortality that did not increase with age in the studied laboratory populations. Fertility also showed no systematic decline over the observation period. This is evidence about those animals under those conditions. It does not show that they can never die or that human beings share the same capacity. [4]

The example matters because it makes the question more precise. We can investigate the processes that produce decline and the processes that maintain life. A long lifespan would need an explanation in those processes. The possibility of further renewal is a question about what a living system can do.

17.5 Low-Entropy Matter and Longevity

My proposal adds a possible source of renewal to this picture. The earlier chapters describe the core as resetting returning matter and making renewed output available. I now ask whether that output could help a living body maintain itself for much longer than it presently does.

I use “low-entropy matter” here to refer to matter in the renewed condition proposed by DEMU. The claim concerns the condition and organization of that matter. It does not mean that age alone makes an atom worn out, or that recently produced matter must automatically be nutritious.

The biological idea is that renewed matter could have properties that help the body rebuild and maintain its organization. If the body can take in that matter and use those properties, renewal at the foundation could support renewal within the organism. This is the

connection I am proposing between the earlier entropy argument and the possibility of a greatly extended life.

Entropy is not, however, a general score of health or usefulness. It must be described for a particular system and under specified conditions. A material does not become useful to a body merely because its entropy is lower. Its chemical composition, available energy, and effects on the body's processes also matter.

The strongest version of my proposal therefore concerns a specific benefit: renewed matter would have to help maintain or restore living organization in a way that ordinary inputs do not. More fuel alone would not answer the question. The proposed advantage must concern the quality or effectiveness of renewal.

There is also no conflict between a body releasing heat and waste during ordinary life and DEMU proposing a total reset at the foundation. These claims concern different parts of the account. The body maintains itself through local exchanges. The core is where I propose that returning entropy and history are completely reset. The body does not have to perform that total reset inside every cell.

I present this as a proposed route toward longevity. No particular food or substance has been identified here as having these properties, and no result described in this chapter establishes that consuming such matter extends human life. The philosophical proposal gives us a relationship to investigate: can the renewal described by DEMU become biologically useful renewal?

17.6 Digestion and Deeper Order

A natural objection is that digestion would destroy the special organization of the renewed matter. If the body breaks it into ordinary components, why should its earlier condition make any difference?

My reply begins with the fact that structure exists at different levels. Breaking a protein into amino acids removes the protein's larger arrangement, but it does not remove every structure within its components. Chemical digestion is therefore not the same as the complete erasure of all organization. [2]

This leaves room for a more specific proposal. The relevant properties of renewed matter might lie at a level that ordinary digestion does not remove. Alternatively, those properties might affect how the material is processed or how later structures form. In either case, the body would have to receive a useful effect from the renewed material.

The distinction between levels does not itself establish that such a property exists. It explains why the statement “digestion breaks things down” is not yet a complete answer. The decisive question is what is broken down, what remains, and whether any remaining difference affects biological repair.

I have also suggested that the first living structures may have developed from renewed matter. If so, there could be a compatibility between that matter’s organization and the processes that sustain life. This is a proposed connection within my wider account of Nature. Sharing an origin would not, by itself, make a substance suitable for consumption by an organism today.

The idea can be made concrete by comparison. Suppose renewed material and ordinary material supplied the same familiar chemical ingredients. My proposal would predict an additional useful difference in how the renewed material supports maintenance or repair. Identifying that difference would turn a broad claim about order into a more definite claim about what happens in the body.

Any useful organization in the renewed output must also be consistent with the total reset. It cannot be a secret store of the erased history. In DEMU, organization arising from the foundation after reset differs from information inherited from the history that returned to it. New order does not require the old biography to survive.

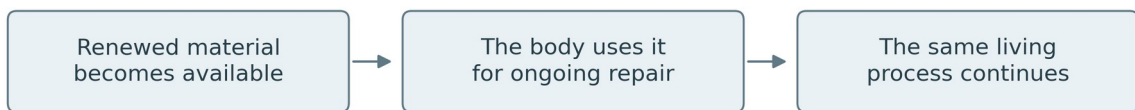
17.7 Renewal Without Erasing the Person

The longevity proposal does not require a person to pass as a complete individual through the timeless core! (Yes, this is probably something a skeptic who fails to understand my ideas will use to attack me.) The proposed route is to use renewed output while the living organization continues. This is how renewal could serve the same person instead of replacing that person with an unrelated new beginning.

Think of repairing a house while it remains occupied. Materials can be replaced while the building's use and much of its structure continue. The comparison is limited, but it illustrates the point: replacing components need not mean ending the organized whole that they support.

In a living person, the continuity at stake is more demanding. Repair must preserve the processes that allow the person to think, remember, act, and remain alive. Renewing the material of the body would count as personal benefit only if that living continuity were maintained.

Proposed maintenance of a living person



Total reset of a person's entire history



Figure 17.1. Two different scopes of renewal in DEMU. The upper route proposes maintaining a person through the use of renewed material. The lower route concerns a total reset of that person's history, leaving no inherited record anywhere. These are different claims about continuity.

The difference in the figure concerns the scope of the reset. In the upper route, renewed material enters a person whose life continues. That person's history has not been declared completely erased. In the lower route, the person's entire inherited history undergoes the total reset. There is then no preserved record of that history anywhere from which to restore it.

This also explains why ordinary sleep or forgetting is not an example of the total reset. During such changes, much of the person's organization and history continues to exist. A total reset, as defined in this work, removes the relevant inheritance completely. The two processes should not be treated as interchangeable.

The route toward longevity is therefore a route of maintained continuity. It seeks to prevent the loss of the living organization by renewing what supports it. It does not depend on erasing the whole person and then calling a later formation the same individual.

17.8 What an Open-Ended Life Would Mean

An open-ended life would have no fixed endpoint imposed by an unavoidable process of ageing. That is already a stronger claim than saying that people might live somewhat longer. It would require the continuing capacity to maintain and repair the living body.

Even that capacity would not mean protection from every cause of death. An organism could avoid age-related decline and still be destroyed by injury or conditions it cannot survive. Removing one source of an inevitable end would not, on its own, establish that a particular individual actually lives forever.

For this reason, I describe biological longevity as a possible route toward immortality. The question is how far renewal can extend the continuation of the same individual. The term “unlimited” would concern the absence of an established limit within the proposed process, rather than a promise that every danger has disappeared.

My other route concerns consciousness at the foundation. If consciousness is necessary to the closure of reality, its existence cannot depend on a temporary biological arrangement. This would change our understanding of consciousness and our place within Nature. It would still leave the further question of how an individual centre of experience relates to the eternal whole.

The two routes meet in a common thought: existence may have a deeper capacity for renewal than the lifespan of any familiar form suggests. In DEMU, renewal belongs to the activity of the foundation. This chapter has explained how I propose to connect that activity with continued life, while keeping the survival of the individual clear.

The next chapter takes up the question on which the account of eternal consciousness depends. Why should the foundation be conscious at all, and why do I think consciousness is necessary to complete the explanation of reality?

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CHAPTER 18

Consciousness and the Closure of Nature

I argue that consciousness belongs to the foundation of Nature. In DEMU, the activity that closes infinite regress is also an activity that is experienced from within. Consciousness gives this account an inner character: reality is present to itself through its own activity.

This chapter explains why I make that connection. The earlier chapters developed the argument for a timeless, active balance. I now ask what that activity is in itself, and how our experience belongs to it. My answer is that the physical world and conscious experience are expressions of one reality.

The claim that only consciousness can close Nature is stronger than the claim that consciousness exists. To defend it, I must connect the presence of experience with the kind of foundation required by the earlier argument. That connection is the central task here.

18.1 The Nature of Consciousness

By consciousness, I mean the presence of experience. Seeing a color, hearing a sound, feeling pain, and thinking about a problem are different experiences. What they share is that they are lived through. There is something it is like to have them.

This is what I mean by the given. Experience is encountered in having it. I do not first need a scientific theory to establish that an experience is occurring. A theory becomes necessary when I ask what the experience is, what explains it, and how it relates to the world.

Consider the difference between describing a taste and tasting it. A description can tell us about a substance, the tongue, and the signals involved. Tasting adds the experience whose explanation we are seeking. The distinction does not settle whether the experience is physical. It makes clear what an account of consciousness must explain.

Descartes makes a related point in the Second Meditation. Even if he is deceived about the world, the act of thinking cannot occur without his existence. Doubt itself provides a starting point that survives the doubt. His argument helps show why experience has a special place in an inquiry into reality. It does not, at that stage, establish that the whole universe is one consciousness. [1]

Nor does the presence of experience make every belief about it correct. I may misremember an event, misunderstand a feeling, or mistake an appearance for something else. The fact that experience occurs and the accuracy of my interpretation are separate matters. The given is a starting point for inquiry, rather than permission to treat every impression as true.

David Chalmers draws a useful distinction between explaining mental functions and explaining experience. We can ask how a system recognizes an object, directs attention, or reports information. We can also ask why those activities are accompanied by anything felt at all. He calls this second question the hard problem of consciousness. His argument is that an account of functions leaves this further question open. [2]

My inquiry takes that question into the argument about foundations. If experience belongs to reality, an account of reality cannot be complete while leaving experience outside its explanation. Consciousness is part of what we are trying to understand from the beginning.

18.2 Why Only Consciousness Can Close Nature

Closure, as I use the word, means that the explanation requires no further unexplained support. It includes the whole of reality and accounts for why its fundamental activity belongs there. The earlier chapters argue that neither an endless chain of dependence nor a completely inactive foundation can provide that closure.

There is no outside to Nature understood as everything that exists. Anything proposed as an external support would itself belong to reality and would have to enter the explanation. This is a claim about the scope of the explanation. By itself, it does not establish a particular geometrical shape or prove that everything is conscious.

My further argument begins with the fact that experience is already present within Nature. A complete foundation must account for this fact through what the foundation is. If its nature wholly excludes experience, I argue that simply rearranging its expressions cannot supply the missing connection. Calling the result consciousness would name the problem without explaining it.

This is where self-awareness becomes relevant. In experience, Nature is not encountered only as something described from a distance. There is an activity that is present to the one undergoing it. I propose that this relation belongs to the foundation itself. The activity and its experienced character are two ways of approaching the same reality.

The argument can be stated in a few steps. Nature includes everything real. Experience is real and belongs to Nature. A complete explanation must account for experience through the foundation, without introducing a further unexplained source. I argue that a foundation whose nature wholly excludes experience cannot do this. I therefore conclude that the foundation must include experience in its own nature.

The key claim concerns a foundation whose nature wholly excludes experience. A physicalist may argue that experience follows from the right organization of physical processes, even if simpler arrangements are not conscious. My objection is that the proposed organization must explain why experience follows. A description of how parts interact does not settle that question merely by being detailed. Section 18.4 develops this objection more fully.

My positive proposal is that consciousness supplies the inner character of the active foundation already argued for. Balance explains the role of its activity in closing regress. Consciousness identifies that activity as something experienced from within. I am proposing one necessary reality with both features, rather than adding an unexplained mind to a finished mechanical universe.

This is the sense in which I argue that only consciousness can close Nature. A complete account must include the given within the nature of its foundation. The conclusion depends on the argument against a wholly nonconscious foundation; it is not established simply by defining closure to mean consciousness.

18.3 Awareness Without Another Observer

When I say that consciousness is aware of itself, I do not mean that there is a small observer inside the mind watching a screen. Such an observer would raise the same question again: who experiences what appears on that observer's screen? Repeating this picture would reproduce the regress we are trying to avoid.

Instead, I mean that experience is present in its occurrence. A pain does not need a second person inside me to make it felt. Reflecting on the pain is another activity, but the pain can be experienced before I form a judgement about it. Basic experience and deliberate self-reflection should therefore be distinguished.

This distinction also clarifies the phrase “awareness of awareness.” In its ordinary reflective sense, I can notice that I am seeing or thinking. In the deeper sense relevant to my proposal, I mean that the activity is already experienced. Its presence does not wait for an endless series of further acts that each observe the one before it.

A mechanical feedback loop has a different description. Its output returns as an input and affects what happens next. A heating system, for example, can use a temperature reading to regulate its operation. Such a description explains a relation among processes. It does not by itself establish whether that relation is accompanied by experience.

Leibniz illustrates the difficulty in section 17 of the *Monadology*. He asks us to imagine entering an enlarged thinking machine as though it were a mill. Inside, we would encounter parts acting on other parts. He argues that this would not reveal an explanation of perception. The example draws attention to the gap between describing mechanisms and accounting for experience. [3]

However, a feedback loop is not unreal simply because we observe it through consciousness. My claim that physical processes are expressions of a conscious foundation needs the wider argument. Perception alone cannot establish the nature of everything perceived.

Nor does being aware automatically make a person independent of causes or conditions. Human experience has many dependencies. What self-awareness contributes to my proposal is a way to understand activity as present to itself without another observer. The argument that the foundation needs no further support still comes from the earlier case for necessary balance. These two claims must work together.

18.4 One Reality and Its Expressions

I describe Nature as one reality. The things within it differ, interact, and develop their own histories, but none exists outside the whole. My question is how experience can belong to that unity if its fundamental nature is said to exclude everything experiential.

Oneness alone does not answer the question. A whole can have parts with different properties. A tree has roots and leaves without every part being a root or a leaf. Likewise, the fact that some beings are conscious does not immediately prove that the whole is a single conscious subject.

My argument requires a further principle: what appears within reality must be intelligible through what reality fundamentally is. By “intelligible,” I mean that the connection can be explained. I reject the suggestion that experience simply arrives at a certain level with no account of why that level makes experience possible.

This allows development and novelty. A melody can have a pattern that no individual note possesses. Its pattern depends on the relations among the notes. The explanation shows how the new organization arises. I do not require every feature of a complex system to be separately present in each of its components.

The question about consciousness is whether the explanation reaches the experience itself. If the account describes only relations among things whose nature is wholly nonexperiential, why should any arrangement be felt? Saying that there are more relations, or that they become more complex, does not yet explain that connection.

A defender of physicalism may reply that consciousness is identical with a particular organized physical activity. That is a substantive proposal. My objection must address the proposed identity, rather than assuming that the word “physical” rules it out. The disagreement concerns whether that identity explains experience and whether the foundation itself can remain wholly nonconscious.

My own answer begins with the experienced character we already encounter. I take it to reveal something about the nature of reality, rather than treating it as an unexplained addition to a reality defined without it. Particular minds then become organized expressions of a foundation that already has an inner character.

This is what I mean when I say that the One becomes expressions of what it already is. I am not claiming that the foundation thinks every human thought in advance. I am claiming that the possibility of experience is rooted in the foundation's own experiential nature. Its different expressions can have different capacities and contents.

The move from a foundation that includes experience to one conscious whole requires the further argument about individual minds in section 18.7. These conclusions are connected, but neither should be hidden inside the word “oneness.”

18.5 Consciousness and the Dual Torus

DEMU describes the fundamental activity through the relation expansion = contraction. I propose the dual torus as the dynamic expression of this relation. I also propose that consciousness is what this activity is like from within.

The geometrical description and the experienced character have different roles. Geometry describes a proposed organization. Consciousness concerns the presence of experience. Drawing a torus does not establish that the drawing, or any object with that shape, is conscious. The proposed identity concerns the fundamental activity of Nature, rather than a resemblance between a diagram and a feeling.

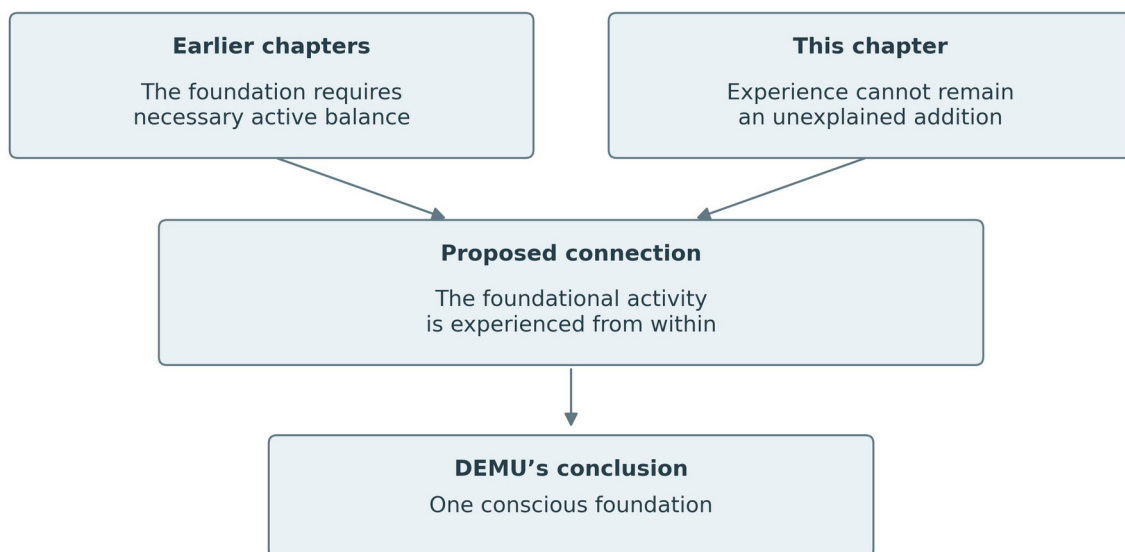


Figure 18.1. The proposed connection between balance and consciousness. The arrows show how the arguments are intended to support the conclusion. They do not represent events in time. The identity stated in the middle box requires the argument developed in this chapter.

The figure brings together the two parts of the argument. The earlier chapters seek a foundation whose activity needs no first activation or outside support. This chapter adds the requirement that the explanation include experience through the foundation's own nature. I propose that one conscious activity of balance meets both requirements.

“From within” does not mean that a person is hidden inside a physical container at the core. It names the experienced character of the activity itself. Likewise, describing its outward expression does not require an observer standing outside all of Nature. An individual within the temporal world can describe the organization to which that individual belongs.

Timelessness also requires care here. Ordinary human thought involves succession: one thought follows another. I am, however, placing an endless human conversation inside the timeless core. The proposed foundational consciousness is its activity being present to itself. Ordered memories and changing thoughts belong to the atemporal expressions of that activity.

In my account, expansion and contraction belong together with zero delay. I describe balance as achieved, lost, and restored, without an interval separating what those words distinguish. The foundation is not waiting between stages. Similarly, the claim about consciousness does not require a first act of awareness followed by another act that finally makes the first one experienced.

This remains a proposed identity that must earn its place in the explanation. The fact that experience does not present itself as a measurable object is not, by itself, a demonstration that its ground has zero spatial dimensions. That conclusion belongs to the earlier argument about extension and regress. Here, I am connecting that proposed ground with the nature of experience.

18.6 The Brain and Individual Experience

An account of fundamental consciousness must still explain the importance of the brain. Our individual experiences are closely connected with what happens in our bodies. Treating consciousness as fundamental does not make those connections disappear.

For example, a study of five people with impaired face recognition found that patients whose damage included a particular region on the right underside of the brain had difficulty

detecting changes in the arrangement of facial features. The findings connected a specific kind of brain damage with a specific difficulty in perception. [4]

My proposal must accommodate such evidence. Within DEMU, the brain belongs to the organization through which a particular individual experiences the world. Changing that organization can change what the person experiences, remembers, or can do. The claim that consciousness is fundamental does not imply that each person's experience can continue unchanged when this organization is damaged.

I therefore distinguish the existence of consciousness at the foundation from the conditions of an individual conscious life. A general capacity can belong to the whole while a particular expression depends on a particular organization. This is the relation I propose between fundamental consciousness and the embodied mind.

The account cannot stop at calling the brain an interface. It must explain why particular changes in the brain produce particular changes in experience. Physical accounts also have to explain why those processes are experienced. These are connected questions, and a philosophical proposal should help us formulate them clearly rather than remove either from consideration.

My claim is that dependence on the brain within a human life does not settle the ultimate nature of reality. It supplies evidence that any account of that nature must respect. The metaphysical question concerns what the brain and its activity are expressions of in the first place.

18.7 Many Minds Within One Nature

If consciousness is fundamental, why do we encounter many separate points of view? I do not experience your thoughts as I experience my own. Any account of one conscious reality must explain that difference.

I propose that individual minds are organized expressions within the activity of the whole. The image used earlier is a large whirlpool containing smaller whirlpools. The smaller forms can have their own organization while belonging to one circulation. The example illustrates how local differences can exist within a common activity; it does not itself explain the privacy of experience.

For that, the proposal needs relations that limit what is available to each individual. A local mind would experience according to its own organization and interactions. Belonging to the same foundation would not mean that every person has access to every other person's thoughts. The boundaries would concern access and organization within Nature, rather than a separation from Nature itself.

Donald Hoffman and Chetan Prakash develop a mathematical theory of conscious agents. It relates experiences, decisions, and actions, and explores how interacting agents could give rise to the perception of objects and spacetime. Conscious realism is their hypothesis that the world consists of conscious agents. The paper develops a theoretical proposal; it does not establish a derivation of our universe. [5]

Their work shows one way to make questions about interacting minds more precise. DEMU adds its own proposed foundation and its argument about closure. The mathematical combination of agents would still need an explanation of how it corresponds to actual experience. It cannot be treated as proof that a particular geometrical circulation is conscious.

I favour one conscious foundation because I seek one account of the activity through which all individual expressions exist. If independent foundational minds were simply placed beside one another, their coexistence and relations would remain to be explained. Locating their common ground in one conscious activity is my proposed answer. The fact that all minds belong to one reality alone does not establish that answer.

The total reset remains part of this account. When a returning history is erased, no hidden store of that history survives in the Mind at Large. Fundamental consciousness is not defined as a permanent archive of every personal memory. Its proposed eternity and the preservation of a particular biography are different claims, as Chapter 17 explained.

Even if histories survive, it cannot do so for eternity, because all things must pass through the gates of death: the toroidal core of reality. I can see how this circulation can nourish the Mind at Large and allow the cosmic mind itself to evolve through feedback processes. Ultimately, however, all things must pass through death, for death serves to keep Nature herself immortal in her perpetual pursuit of balance.

18.8 Consciousness as the Activity of the Foundation

The proposal developed here connects two questions. The first asks how reality can exist and remain active without an earlier activation or further unexplained support. The second asks how experience belongs to that reality. I argue that the same foundation must answer both.

My answer is a conscious activity of balance. Balance has its role because, in the earlier argument, it supplies the relation through which regress is closed. Consciousness has its role because experience must belong to the foundation's own nature rather than enter the account as an unexplained addition. Self-awareness provides the model of activity that is present to itself without another observer.

The claim that only consciousness can close Nature rests on this combination. Self-awareness alone is not an explanation of why something exists. The argument depends on joining the case for necessary balance with the case that a wholly nonexperiential foundation cannot account for experience. A critic can dispute those connections, but they are the connections my conclusion requires.

If the conscious character of the foundation is necessary to its closure, that character cannot be temporary. It would belong to the eternal activity itself. This is the reasoning behind the account of fundamental consciousness introduced in Chapter 17. It does not, by itself, establish the continued existence of every individual personality.

I therefore understand consciousness as Nature's own activity experienced from within. Human minds are particular expressions of that activity, with their own limits and histories. In the next chapter, I develop why I identify the conscious foundation with God and what this account could mean for free will.

However, elsewhere I have tried to provide a deeper explanation for the existence of consciousness. The argument is as follows: if resolving infinite regress requires Nature to be zero-dimensional, then no spatially extended process can provide its closure. Only a non-spatial process could do so. This is why I identify consciousness as Nature's means of closure.

But why must Nature be closed? Because opening it up would reintroduce infinite regress: its openness would extend outward forever, like a soap bubble that has burst yet

somehow continues to exist, stretching endlessly outward. Close the bubble, however, and it becomes coherent. It is closed; it can exist.

A consciousness without an evolutionary reason for existing—a context that gives it purpose—would simply be a consciousness tripping in a void somewhere. Thus, consciousness cannot be fundamental unless it has a job to do: a purpose that explains why it is there.

In Nature's machinery, everything has a purpose. The purpose of consciousness is the purpose of all things: to create greater balance. And balance is closure.

Thus, if spatial processes could not close Nature, consciousness had to be there from the start. It had to be eternal and immortal, with the capacity to renew and replicate itself, because this, too, is part of the process of balance.

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CHAPTER 19

God and Free Will in a Spiritual Universe

I identify God with the eternal conscious activity through which Nature exists and renews itself. Chapter 18 developed the argument that consciousness belongs to the foundation. This chapter explains why I call that reality spiritual, what I mean by God, and how individual freedom could belong to it.

The connection matters because a conscious universe does not automatically give us free will. A mind could, in principle, follow a completely fixed course. My account therefore needs to explain both why Nature is conscious and why our own awareness can take part in shaping what happens.

19.1 The Necessity of a Spiritual Universe

By a spiritual universe, I mean a universe whose fundamental nature includes experience. Consciousness belongs to what reality is. The physical world is the form through which this reality appears and acts within our experience. Bodies and events remain real features of our lives, even if their deepest nature differs from how they appear.

The word spiritual can suggest another world hidden beyond Nature. That is not how I use it. Nature includes everything that exists. If a spiritual reality exists, it belongs to Nature itself. The question is therefore whether the reality we already inhabit is conscious at its foundation.

My answer follows the argument developed in the previous chapter. Experience is given in the very act of experiencing. We can question an experience's interpretation, but the questioning is itself experienced. A complete explanation of Nature must account for this fact. It cannot describe a wholly nonconscious foundation and then leave experience as an unexplained addition.

I argue that the foundation must contain the possibility of experience in its own nature. This does not mean that every human thought is already stored there (it could be). It means that the development of individual minds expresses a reality that already has an experienced

character. Nature becomes capable of many forms of awareness through the different ways its activity is organized.

The necessity claimed here rests on two connected arguments. The earlier chapters argue that active balance is required to close regress. Chapter 18 argues that this foundation must also account for experience through its own nature. If consciousness is indispensable to the foundation's completeness, it cannot be a temporary feature that reality eventually loses altogether. It belongs to the same necessary activity as balance.

This is why I call the spiritual character of the universe necessary. I am drawing a consequence from the role assigned to consciousness in the explanation. The conclusion depends on the argument that a wholly nonconscious foundation cannot account for experience. A reader who rejects that argument need not accept this conclusion; the word spiritual supplies no additional proof.

The proposal gives the physical and the spiritual a common ground. Consider an ordinary act such as listening to music. We can describe the movements involved in producing and receiving the sound. We can also attend to the experience of hearing it. In DEMU, these belong to one reality. The experienced character is part of what the activity is, rather than an observer added outside it.

This changes how I understand our place in the universe. Conscious beings are ways in which Nature experiences its own activity. We are able to notice, understand, and respond to the world because we belong to a reality that has an inner character. Our awareness is therefore connected with the foundation whose balance the earlier chapters describe.

The connection does not make every interpretation of a spiritual experience correct. A powerful feeling of unity is still an experience that requires interpretation. It may be meaningful to the person who has it without establishing a particular account of the cosmos. My argument rests on the place of experience in a complete explanation, rather than on the intensity of a private feeling.

Calling the universe spiritual also does not settle the question of purpose in every detail. Chapter 14 identified balance as the role of fundamental activity. With consciousness included, that activity can have an experienced meaning. The further question is how

conscious beings can understand and participate in it. Free will is central to that question because participation through choice differs from merely undergoing a process.

19.2 Who God Is

God, as I use the name, is the conscious foundation of Nature and the whole activity that expresses it. I also call this the Mind at Large. The name refers to the reality on which individual minds and the temporal universe depend. It does not refer to my personality, your personality, or a larger human personality placed somewhere above the world.

The explanation begins with the work already assigned to the foundation. It must account for existence without an earlier activation or outside support. Its activity must remain coherent while allowing the variety of the universe. I identify God with the conscious balance that, in DEMU, performs this work. The name comes after the argument about what the foundation must be.

This gives a direct answer to the familiar question of who created God. Within this account, God is not an earlier event that started a chain of later events. God names the eternal activity through which the chain has its ground. There was no period before this activity in which another being could prepare or activate it. Its lack of a creator follows from the proposed explanation of its necessity, rather than from an exception granted to a favoured being.

Likewise, when I place God at the beginning of everything, I mean the origin on which everything depends. I do not mean a first date on a cosmic calendar. The core remains present throughout the circulation. Creation, in this sense, is the ongoing formation and renewal of the world within the activity of Nature.

The dual torus is my proposed account of that activity's organization. Expansion and contraction belong to one relation, and the temporal world develops within its circulation. Consciousness names the experienced character of the whole. The geometry and the account of experience meet in the identity proposed in Chapter 18; the geometrical shape alone does not establish a divine mind.

Spinoza provides a useful comparison. In Part I of the Ethics, he argues that everything exists in God and that God is the cause within which things exist. God does not act on a

world entirely outside God. This resembles my insistence that nothing stands outside Nature. However, Spinoza also holds that everything follows necessarily from the divine nature. I do not adopt his whole system, and necessity becomes a central issue for free will. [1]

Within DEMU, God and the universe therefore cannot be two independent realities. The universe is an expression of the activity I call God. Individual beings can have distinct lives and perspectives within this whole without becoming separate foundations. The image of smaller whirlpools within a larger circulation helps describe this relation, although the privacy of individual experience still requires the account discussed in Chapter 18.

This also explains the limits of saying that we are God. We belong to the reality I identify with God, but a particular person does not possess the knowledge or capacities of the whole simply by belonging to it. My immediate wishes are not automatically the purpose of Nature. Other people remain centres of experience whose lives I cannot replace with my own point of view.

I understand divine freedom first as freedom from outside control. Nature has no external ruler, fuel supply, or prior event on which its fundamental activity depends. This does not mean that God can make a contradiction true or turn existence into absolute nonexistence. In my argument, coherence belongs to what makes the foundation possible. Its necessity is its own nature, rather than a restriction imposed by something else.

Spinoza's definition of freedom helps clarify this limited comparison. He calls a thing free when it exists and acts through the necessity of its own nature, rather than through determination by something else. That sense of freedom concerns independence from outside determination. It does not establish the human ability to choose among genuinely open alternatives. [1]

The claim that Nature is divine also raises the problem of suffering. If the whole is conscious and its purpose is balance, why do its living expressions experience illness, fear, and loss? The eternal balance of the whole cannot answer this simply by making an individual's suffering disappear from consideration. Pain belongs to the very experience this account treats as fundamental.

My proposal allows temporary forms to be vulnerable within an enduring whole. That explains how local damage can coexist with fundamental balance. It does not show that every

instance of suffering is necessary or good. Nor would human freedom alone explain suffering that no human choice caused. An account of divine goodness must address these differences. I therefore do not treat every event as an instruction from God or as something we ought to accept without trying to change it.

I connect the spiritual meaning of balance with care and, ultimately, love. The connection begins with recognizing that other beings have experiences that matter to them, and that our lives depend on relations within one Nature. Turning this recognition into an account of what we ought to do requires the argument about meaning in Chapter 20. Physical balance by itself is not yet a complete ethics.

The God defended here is thus the eternal conscious activity of Nature. Questions about prayer, revelation, or the teachings of a particular religion require additional evidence and arguments. They are not settled merely by identifying the foundation as conscious. The immediate question for this chapter is how beings within that foundation can have a will of their own.

However, how can God simply be there at the beginning of time? Because the universe has a true beginning: death, the toroidal core and the logical starting point of existence. The complexity of the universe reflects some of the complexity of the cosmic mind, but the universe is to timelessness what colors are to white light. So, you see, God has a beginning and a history, yet, in a sense, began before all of that.

19.3 Free Will

By free will, I mean the capacity of a conscious individual to understand possible actions and help determine which action occurs. In the stronger sense I want to defend, at least some choices remain genuinely open until the individual settles them. The choice is guided by awareness and reasons, while not being completely fixed by the prior conditions of the universe.

Freedom does not require unlimited power. I may choose to learn a language without being able to speak it immediately. I can decide to help someone and still fail to help as much as I intended. The freedom at issue concerns my part in deciding and acting, within the capacities and circumstances I actually have.

The starting point in DEMU is that consciousness belongs to the activity that grounds time. It is not merely a spectator placed at the end of a chain whose entire work has already been done without it. I propose that conscious activity participates in the formation of temporal events. Human decisions are local expressions of that participation.

This is what I mean by consciousness wearing time like a headset. We encounter an ordered world, remember earlier events, and consider what to do next. Yet the account of what we are does not end with the temporal order presented to us. The image expresses a relation between fundamental awareness and its temporal experience. It does not describe a little observer sitting outside Nature and operating the brain by remote control.

The phrase “faster than causality” points to the same idea in my earlier discussion. More precisely, the foundation has no delay through which it waits for an earlier event to activate it. This is a claim about its place in the explanation, rather than a measured signal outrunning light. Our ordinary decisions still take time within our experienced lives.

The distinction between timelessness and determinism matters here. A timeless foundation has no succession of earlier and later states. Determinism is the claim that the relevant conditions and laws fix what happens. A description can fix an entire history without itself unfolding in time. Timelessness therefore removes the need for a first activation, but does not alone prove that individual outcomes are open.

My further proposal is that necessary balance can allow more than one course of events within it. The requirement that the foundation remain balanced does not itself specify whether I eat an apple or a banana. To conclude that it fixes this detail would require an argument connecting fundamental balance with that particular choice. The necessity of the foundation and the necessity of every event are different claims.

Consider the example from my earlier discussion. I begin eating an apple almost automatically. I then notice what I am doing and decide whether to continue or put it down. The example shows the kind of control I have in mind: awareness can be involved in reviewing and redirecting an action already underway. The fact that I did not consciously announce its first movement does not settle whether I can exercise control over what follows.

A determinist can still say that the review and the decision to stop were themselves fixed by earlier conditions. This is the real disagreement. The example makes conscious control

understandable, but it cannot, by itself, show that exactly the same prior conditions allowed a different choice. My stronger account requires that additional openness.

William James makes the distinction concrete by imagining two possible routes home. He asks us to consider whether, with the circumstances otherwise unchanged, he could genuinely have taken either route. His example separates an alternative that was really possible from one that merely seemed possible because the outcome was unknown. It clarifies the claim I want to defend without providing a proof of it. [2]

There is also a difference between wanting something and endorsing the desire that moves us. Harry Frankfurt examines this distinction through desires about our own desires. On his account, freedom of the will concerns whether the will that moves a person is the will that person wants. He allows this kind of freedom to be compatible with determinism. [3]

That distinction is useful when considering habits or conflicting wishes. I might feel an urge to send an angry message while also wanting to respond fairly. Reflection can help me decide which motive I endorse. However, my proposal goes beyond agreement between my motives and my action: I also claim that some decisions are not completely settled by the conditions that precede them.

Three meanings of freedom

Meaning of freedom	What is claimed	Compatible with determinism
Freedom of Nature	The whole is free from outside control.	Yes
Freedom through reflection	A person can endorse the motives through which the person acts.	Yes
The stronger freedom defended here	Some choices remain open, and the individual helps determine which action occurs.	No

The first two meanings can allow every event to be fixed. The third adds a claim about genuine alternatives and the individual's control over the outcome.

How can such a decision have an explanation without being predetermined? My answer is that the conscious individual acts for a reason. The reason can make the action understandable without forcing it as the only possible outcome. I may choose to apologize

because I recognize that I hurt someone. What explains the apology includes my understanding of the harm and my decision to respond to it.

This is not a claim that an unexplained accident becomes freedom merely because it happens in the mind. If the apology were produced by a random interruption that bypassed my understanding, it would give me less control over what I said. My proposal requires the individual to settle the action through awareness. Openness supplies room for a choice; conscious agency supplies the choosing.

Roderick Chisholm develops a related idea by distinguishing an event caused by another event from an event caused by an agent. His proposal places the acting person in the explanation, rather than treating the action as either an inevitable result of earlier events or an entirely uncaused occurrence. This is an account of agency, not a demonstration of DEMU's timeless foundation. [4]

In DEMU, the individual is an organized expression of the conscious whole. I am not introducing an independent creator at each decision. The individual's existence, capacities, and circumstances have their place within Nature. The claim is that the exercise of those capacities can settle which of several genuinely possible actions occurs. Agency belongs within the whole, even though the whole provides the conditions for it.

This brings us back to my rejection of brute facts. A critic can ask why a person chooses one available action rather than another. My reply identifies the person, the reasons understood, and the act of choosing. But a critic may insist that a complete explanation must also supply conditions that make this particular choice unavoidable. If that requirement is accepted for every action, the stronger freedom I defend is excluded.

I distinguish these demands. Explaining why the foundation exists requires an account of its necessity. Explaining an action can instead identify how a conscious person settles it for a reason. The latter explanation need not turn the reason into a force that compels only one response. This is the central claim about agency in my proposal. Whether it gives a sufficient explanation of the particular choice remains a substantive point of disagreement; calling consciousness fundamental does not settle it by itself.

Our freedom is also shaped by the organization through which we live. Attention, understanding, memory, and the ability to act matter. A person who misunderstands the

options has less informed control than someone who understands them. Learning can widen the choices we recognize, while coercion can sharply restrict what we are able to do. A timeless ground does not erase these differences within an embodied life.

Nor does my account require every movement to be freely chosen. Habits and automatic responses remain part of human activity. Freedom concerns the occasions on which awareness contributes to directing that activity. The development of more complex organisms can increase their capacity to perceive alternatives and act on them, even though consciousness itself is fundamental in the model.

The total reset must be kept distinct from this ordinary use of memory. An individual can learn and act through a history while that individual's organization continues within the circulation. When a returning history undergoes the total reset, its information is erased completely. No record survives elsewhere in God (at least not indefinitely). Freedom during a life does not establish that the same person's memories survive that reset; Chapter 17 addresses the separate question of personal continuity.

God's relation to our choices also needs a clear meaning. Belonging to one conscious reality does not automatically make an individual a puppet. On my proposal, individual agency is one way the whole acts. If instead a cosmic mind fixes every personal decision independently of the individual's choosing, the stronger human freedom defended here would fail. Calling that mind God would not remove the difficulty.

Likewise, knowledge of a choice and control over it are different relations. Saying that God knows what someone does is not yet saying that God forces the person to do it. Still, an account of infallible knowledge of the future must explain how genuine alternatives remain possible. I do not build the conclusion about free will on an additional claim that the core contains a completed record of every future decision.

The spiritual significance of freedom lies in conscious participation. We can understand something of the relations that support life and choose how to respond. We can care for another person, repair damage, or resist a harmful impulse. Those actions matter to the beings affected by them. Their significance does not depend on our possessing control over the whole universe.

My proposal is therefore that timeless consciousness acts within time through living individuals. The necessary balance of the whole provides the ground of their activity, while their awareness can help shape its particular course. The argument for a spiritual foundation establishes the setting for this account; genuine alternatives and conscious control are the further claims through which I defend free will. Chapter 20 develops what such participation means for the purpose of a human life.

References for Chapter 19

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CHAPTER 20

The True Meaning of Life

I argue that the meaning of life is conscious participation in the renewal of balance, and that its fullest expression is love. The earlier chapters approached this claim through questions about existence, motion, consciousness, and God. This chapter brings those questions into the life of the individual. What does this account of Nature mean for how we live?

In DEMU, Nature continually renews the balance through which it exists. I have defended this balance as the answer to the problem of infinite regress. Chapter 18 argued for a conscious foundation, and Chapter 19 identified that foundation with God, or the Mind at Large. Taken together, these arguments suggest that existence has both an activity and an inner life. Its purpose must therefore be considered from within as well as from outside.

From outside, I describe the renewal of balance. From within, I propose that its deepest fulfilment is the experience and expression of love. To explain this connection, I must show why conscious lives matter and why their relationship to the whole gives us reasons to care for one another.

20.1 The Meaning

Reason allows us to ask whether our beliefs fit together. It also allows us to question habits that feel natural simply because we have never examined them. In this thesis, I have used reason to challenge the idea that reality can rest on nonexistence, an unexplained fact, or an endless demand for further support. Here I apply the same willingness to question to the meaning of a life.

There is a difference between explaining what a system does and explaining why we should value it. A description of circulation tells us about a pattern. It does not yet tell us whether anyone flourishes within that pattern. David Hume draws attention to this kind of gap when he asks writers to explain how their claims about what ought to be follow from their claims about what is. [1]

My account therefore needs a clear starting point about value. I take the experience of conscious beings to matter in itself. Suffering matters to the one who suffers, and joy matters

to the one who feels it. A life has this importance before anyone assigns it a useful role in a larger system. This is a claim about value, and I defend it through what experience reveals to us about being alive.

The next step concerns other lives. If I ask others to take my suffering seriously because of what suffering is like, I need a relevant reason to dismiss a similar experience in someone else. The fact that their suffering is theirs does not remove the reason I have already recognised. Reason asks me to apply that concern consistently. Differences in circumstances can change what help is possible, but mere distance from my own point of view does not make another life worthless.

In DEMU, this concern also belongs within a wider picture. Individual minds are expressions of the same conscious reality. Their experiences occur within Nature's own life. I therefore argue that an account of conscious balance must include how those lives fare. A whole described only through its continued operation would leave out the very experience that makes its operation significant from within.

This is the connection I defend between Nature's balance and our responsibility toward one another. The value of experience supplies the reason for care; the account of a shared foundation explains where these apparently separate lives belong. Together, they support the view that our purpose involves the conscious renewal of relationships in which life can flourish.

20.2 From Balance to Love

The word balance needs care here. At the foundational level, it means the activity through which reality sustains itself. At the level of a human life, it concerns the way our needs, actions, and relationships fit together. These are connected in my proposal, but they are not interchangeable descriptions. A universe could continue to circulate while a person within it suffers greatly.

For this reason, I cannot call every event loving merely because it happens within Nature. The continued existence of the whole does not establish that every relationship within it is good. When I speak of love as conscious balance, I mean a relationship in which the reality of another's experience is recognised and their good becomes a reason for action.

Love, in this sense, is more than a passing feeling. It includes attention, patience, and the willingness to respond to what another being needs. It can be present when someone helps a child understand a difficult idea or stays with a friend through grief. In such cases, the other person's life matters to us through what it is like for them to live it.

This also explains why love cannot mean possession. If I demand that another person become whatever suits me, I have made my own wishes the measure of their life. A loving relationship allows the other person to remain a person, with an experience and a direction of their own. Sometimes care requires closeness. Sometimes it requires space or a firm boundary.

I connect love with balance because it allows different lives to belong together without making their differences disappear. The aim is a living relationship that can respond and renew itself. People can disagree, learn from one another, and change their shared life without losing their connection. This gives a human meaning to the dynamic balance defended throughout the thesis.

My stronger claim is that love belongs to the deepest fulfilment of conscious existence. Once the reality of other lives is fully recognised, their flourishing cannot be treated as meaningless to the whole. I use infinite love to name concern that excludes no conscious being from consideration. This does not mean that a finite person must feel an endlessly intense emotion or meet every need at once. It means that no life is ruled out as unworthy of concern simply because it is someone else's.

20.3 One Purpose and Many Lives

If the meaning of life is connected to balance and love, there can still be many meaningful ways to live. A common purpose does not give everyone the same task. People have different abilities, relationships, and circumstances. The way they contribute to a shared world will therefore differ.

A teacher may help someone gain the confidence to think for themselves. A builder may make a home safer. An artist may give form to an experience that others have struggled to express. Someone caring for a relative may sustain a relationship through years of ordinary, largely unseen work. These activities matter through what they make possible in actual lives.

Susan Wolf offers a useful comparison. She connects meaning in life with active involvement in things that have worth, together with a person's own interest in them. On her account, feeling absorbed in an activity is insufficient if the activity lacks value. [2] I share the concern that fulfilment should answer to something real. My further claim is that meaningful activity belongs within the purpose of a conscious Nature. Wolf's account of meaningful engagement does not itself establish that wider claim.

The distinction matters because I am looking for more than a comforting story. Caring for someone has value because their life matters and because the care makes a difference to them. We may choose how to respond, but we do not invent the reality of their experience by choosing to notice it. In this sense, we discover reasons for action while also shaping the particular life through which we answer them.

The replication discussed in Chapter 15 should be understood with this variety in mind. Its proposed role in Nature does not make biological reproduction an obligation for every person. Love can take the form of parenthood, friendship, teaching, or care for the living world. A person who has no children can participate fully in the renewal of life through these relationships.

A person's worth also cannot depend on how much they produce. An infant, a sick person, or someone who needs long-term care remains a centre of experience. Receiving care belongs within the same shared life as giving it. If balance required us to count only those who could contribute the most, it would fail to recognise many of the lives it was supposed to include.

20.4 Freedom and Spiritual Growth

Chapter 19 defended a form of freedom in which the individual has a real part in determining what happens. Here that freedom gains a direction. If our choices can make a difference, then the way we attend to others and respond to them becomes part of how the world develops.

Knowing the purpose of life would not make every decision easy. We still have limited knowledge. We can misunderstand another person's needs, confuse control with care, or defend our pride under the name of principle. Growth involves becoming better able to notice these mistakes and respond to them. Reason helps us examine our motives, while experience teaches us what our actions actually do.

Aristotle's account of the good life provides a useful comparison. He describes human flourishing through activity in accordance with virtue across a complete life. [3] This directs attention to how a person lives over time. It supports the thought that a good life involves developed ways of acting, rather than a single moment of satisfaction. His argument has its own basis; I am not treating it as an argument for DEMU.

Within my account, spiritual growth means a growing ability to recognise and live our connection with the whole. This can involve deeper understanding, but it also appears in ordinary conduct. Someone may become more willing to listen, more honest about an error, or more able to care without demanding control. The significance of the change lies in the life that becomes possible through it.

This development does not require every experience to be pleasant. Learning can be difficult, and repairing a relationship may demand effort. Yet it would be a mistake to conclude that every painful event was sent for our improvement. The possibility of learning from an injury does not show that the injury was needed or deserved. Spiritual growth includes the work of preventing avoidable harm.

The spiritual dimension I propose is real within the model. It concerns the inner life of Nature and the possibility of conscious fulfilment within it. Our experiences may reveal something about this dimension, but a powerful feeling alone does not establish a complete account of the universe. Experience and reason have to remain in conversation.

20.5 Suffering and the Work of Repair

An account that identifies the deepest meaning of life with love must face suffering directly. People experience loss, cruelty, and isolation. Calling Nature balanced does not make those experiences harmless. The question is how a conscious whole whose fulfilment I identify with love can contain such painful forms of life.

The distinction between foundational and lived balance helps state the problem. The foundation can sustain existence while local relationships fail the beings within them. This explains why universal circulation and human distress are not a formal contradiction. It does not, by itself, explain why a loving whole permits the full extent of suffering. That remains a serious challenge to the spiritual interpretation I defend.

What follows for our conduct is clearer. If another person's experience matters, their suffering gives us a reason to respond. We cannot excuse neglect by saying that the whole will continue anyway. Continued circulation answers a question about existence. The person's need presents a question about care, and the answer must include what happens to that person.

Kant expresses a related concern when he argues that humanity should be treated as an end in itself and never merely as a means. [4] His argument centres on rational beings, while my concern here extends to conscious beings whose lives can go well or badly for them. The point of comparison is the refusal to reduce an individual to a tool for someone else's purpose.

This matters especially in a philosophy of the whole. We should be cautious whenever an appeal to a great purpose makes actual people seem expendable. A claim to serve cosmic balance cannot, on its own, justify overriding another person's life. If the whole is conscious through its living expressions, then their treatment belongs within any honest account of its fulfilment.

Repair begins with the particular situation. It may mean ending an abusive relationship, changing an unfair practice, or meeting a basic need. Reconciliation is valuable when it restores a relationship in which people can live well. It cannot be demanded at the cost of continued harm. Love includes the strength to oppose what damages life.

Nor does someone lose their place in a meaningful world when they cannot feel its meaning. Grief, exhaustion, and loneliness can make connection difficult to experience. The person still matters. The task of those around them is to help make care real, without requiring them to display spiritual confidence before receiving it.

20.6 Eternity and the Value of Each Life

The eternal Nature described in DEMU has no final point at which existence has finished its work. Balance belongs to its ongoing renewal. I understand the meaning of life in the same active way. Love can be realised, expressed, and renewed through living relationships. Its value need not wait for a last event at the end of cosmic history.

This also means that the foundation does not begin as a spiritually empty thing and later acquire a purpose from outside. In the interpretation defended here, the possibility of conscious fulfilment belongs to Nature's own being. Individual lives develop through time within that reality. Their development need not mean that the timeless foundation itself passes from an earlier failure to a later success.

I have sometimes used white light and its colours as an image for this relationship. The colours help us imagine different expressions belonging within a more complete reality. In a similar way, I propose that the variety of conscious life belongs within the Mind at Large. The image helps explain the proposal; the arguments about consciousness and the foundation must carry its philosophical weight.

Eternity also requires precision about the individual. The continued existence of a conscious foundation does not by itself show that a particular person's memories and identity continue forever. Chapter 17 considered that question separately. The value of a person and the conditions of their survival are closely connected concerns, but one does not settle the other.

The total reset defended in this thesis makes that distinction especially important. Where a history is completely erased, its information is not secretly preserved somewhere else, including in a divine archive. I cannot appeal to the reset and then promise that it keeps every individual biography intact. Any account of personal continuation has to explain how the person continues consistently with the reset proposal.

This limitation does not make care pointless. An experience matters to the being who has it, including while it is happening. Relieving suffering changes that lived reality. Protecting a relationship can also preserve possibilities for the people within it. These reasons for care do not depend on a promise that every event will be recorded forever, and they give us reason to support continued life where we can.

20.7 Living the Meaning of Life

The answer I defend gives direction without prescribing one life for everyone. We participate in meaning by understanding more clearly, caring more truthfully, and helping to sustain conditions in which conscious lives can flourish. The work takes place through the relationships and opportunities we actually have.

Sometimes this participation is demanding. It may involve changing a harmful habit or accepting responsibility for something we would rather ignore. At other times, it involves enjoying a friendship, resting, exploring an idea, or appreciating the living world. A life has room for delight as well as repair. Constant labour is no measure of spiritual worth.

Reason remains important throughout. A claim about love should be tested against how people are treated. A claim about balance should identify what is being balanced and whose needs are being considered. A spiritual interpretation should remain open to questions about its arguments. Care for truth belongs with care for life because mistaken beliefs can shape what we do to one another.

I therefore understand the true meaning of life as participation in the fulfilment of conscious Nature. Balance names the fundamental activity I have defended; love names its fullest expression in the recognition and care of conscious lives. This answer rests on the earlier arguments for DEMU together with the case for value made in this chapter. Its force depends on those arguments, rather than on the appeal of the conclusion alone.

If this account is right, we belong to the very reality whose meaning we are trying to understand. Our capacity to love is one way that reality becomes aware of the lives within it and responds to their good. A human life can take part in this purpose through the concrete work of understanding, caring, and making room for others to live.

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CHAPTER 21

A Dialogue with the Skeptic

21.1 Battling a straw man

It is incredibly frustrating to be subjected to a straw man argument. It happens when someone distorts, oversimplifies, or outright misinterprets what you are actually saying, only to attack the weak, false version they have just fabricated. Instead of debating your actual point, they are fighting a "straw man."

Skeptic:

Equal expansion and contraction do not, by themselves, establish timelessness.

Reply:

That reverses the order of my argument. I have already argued for timelessness before introducing expansion and contraction. My starting point here is **motionlessness = motion**. Expansion and contraction are introduced to explain how a universe of time can belong to that timeless reality while its foundation remains timeless.

When I speak of “breaking out of timelessness into time,” I do not mean that timelessness changes into time at some first moment. That would already require time. I mean that temporal activity and experience arise within a reality whose fundamental balance never ceases. Expansion and contraction explain this relationship in my model.

Skeptic:

You propose that the foundation requires no further support because it resolves regress. That could become an argument if you independently establish why its nature makes further dependence impossible or unnecessary.

Reply:

My argument begins with the requirement that reality must exist coherently. If a proposed foundation still depends on another unexplained foundation, then it has not resolved regress. It has only moved the problem elsewhere.

The balance I propose is meant to explain why that further dependence ends. Expansion and contraction belong to one complete relation; neither waits for an outside agent to

establish or maintain it. Its activity is the balance itself. There is no separate, inactive structure that must first receive motion.

This is what I mean by a foundation that requires no further support: its existence and activity are explained together, through the condition that closes regress. The claim rests on the argument about balance, rather than on simply naming something “the foundation.”

Skeptic:

Freedom from contradiction is insufficient to establish actual existence or unique necessity. Several different structures might be internally consistent. Rejecting absolute nonexistence would still leave the question of why existence must have this particular structure.

Reply:

Chapter 4 addresses this through an argument about the available alternatives. We begin with the fact that something exists. I then reject absolute nonexistence as an explanation, brute facts as a complete answer, and unresolved regress as a solution. If an explanation always appeals to another earlier state, the same question remains. This leads me to a foundation that is timeless.

The next requirement is that this foundation must remain timeless while accounting for the activity of the universe. I also reject complete inactivity as an adequate foundation. These requirements lead to my proposal of dynamic balance: motion and motionlessness belong to the same reality, and expansion is met by contraction.

So I am doing more than choosing a structure because it appears consistent. I am arguing that the foundation must meet these requirements together. A rival structure would have to explain existence, close regress, account for activity, and preserve timelessness. Merely being internally consistent does not show that it does all of this. The more detailed geometry enters as my account of how these requirements are realized.

Skeptic:

Your five-premise entropy argument contains a conditional conclusion I accept. If the complete circulation necessarily restores a specified low-entropy core, and entropy increases during an intervening stage, then the circulation must include a decrease back to that condition. A permanently exhausted outcome would be incompatible with perpetual restoration.

The unresolved step is why ending explanatory regress requires that restoration.

Reply:

The connection is my claim that entropy and infinite regress express the same problem at the philosophical level of this argument: inherited conditions that still call for an explanation. If the foundation retains the accumulated distinctions of an earlier history, then that history remains part of what determines it. We have not reached a condition independent of that inheritance.

The core must therefore restore the balance that closes regress. It cannot begin in perfect balance and then lose that balance forever. Nor can it preserve the returning history under another name. In my proposal, the reset is total: the inherited information is erased, with no surviving record or backup from which it could be recovered. Entropy is reset without being transferred elsewhere.

That is why restoration belongs to the whole argument. The foundation must remain what makes it foundational throughout the eternal circulation.

Skeptic:

A foundation needing no maintenance energy does not establish unlimited extractable energy.

Reply:

The proposed source is the renewed matter and energy that emerge from the core. We do not draw energy from the core itself. In my model, collecting this renewed output leaves the fundamental circulation unaffected.

The claim of an ongoing supply comes from continual renewal. The output is replenished rather than being a stock that can only run down. By “unlimited,” I mean that renewal has no final exhaustion; I do not mean that a collector delivers an infinite amount of power at a single moment. The practical question is how to locate and collect useful output from this process.

Skeptic:

Even granting that your tetrahedral construction produces a larger cuboctahedron, that would establish a geometrical relationship. It would not yet establish that atoms replicate, that biological sex expresses this mechanism, or that it drives cosmic expansion.

Reply:

My proposal treats this geometry as the form of an active process. The cuboctahedron describes its organization, while the dual torus describes its circulation. In the construction I discuss, the cuboctahedron can fold into a tetrahedral form, and two opposing tetrahedral clusters can combine to form a larger cuboctahedron.

I propose that this is a repeating pattern through which Nature forms further structures. “Male” and “female” name the complementary roles of the two clusters in the analogy. This is what I mean by “cosmic sex”: complementary forms participate in producing a new expression of the same organization.

Atomic replication, biological reproduction, growth, and cosmic expansion are the connections I propose to explain through this process. The idea is that the same basic organization can operate at different scales, from very small structures to much larger ones. The geometry gives these claims a common proposed mechanism; the chapters on cosmic sex and replication develop its application to each case.

Skeptic:

A universe perpetually maintaining balance does not, by itself, establish that balance is its purpose or the meaning of life. Those interpretations need their own arguments.

Reply:

By “purpose” here, I mean the job the activity performs within my explanation of existence. I argue that motion is necessary because it continually realizes the balance that prevents reality from falling into unresolved regress. Balance is therefore the reason I give for why this activity exists at all.

Life, growth, and complexity belong to that ongoing activity. They are ways in which Nature continues its work of forming and renewing balance. Conscious life adds the possibility of understanding this process and taking part in it knowingly.

That is the connection I draw between balance and the meaning of life. The later argument about consciousness and spirituality develops what this participation means for us.

Skeptic:

Aristotle already accepted eternal motion. His further question was what accounts for its actuality and its particular character. In *Metaphysics* XII.6, he argues that merely positing something eternally capable of activity does not explain why activity actually occurs. He therefore seeks a principle whose nature is actuality. See *Metaphysics* XII.6–7.

Reply:

My answer is that motion occurs because it realizes the balance required for a coherent reality. It has a job to do: it closes the regress that an inactive foundation, in my argument, cannot close.

Saying only that “activity belongs to reality’s nature” leaves out this explanation. I am saying why activity belongs to reality’s nature: the balance on which coherent existence depends is dynamic. Motion is necessary to that balance.

This does not describe a first event in which Nature notices a problem and begins moving to solve it. The activity and the balance are eternally one system. There was no earlier, inactive reality waiting to be set going.

Skeptic:

Asking “Where did this entropy come from?” raises an explanatory question. It does not yet establish that every possible answer generates another unanswered question indefinitely.

Reply:

My objection concerns answers that keep appealing to earlier states. If the entropy comes from an earlier condition, what explains that condition? If the answer is another earlier condition, the same question returns.

We do not need to ask the question an infinite number of times to recognize the pattern. As long as each answer has the same form, it passes the explanatory burden backwards. My argument asks what ends that dependence, rather than how far back we can follow it. A complete answer must explain why the same demand no longer arises.

Skeptic:

Why must entropy retained at the foundation be an unexplained inheritance, rather than an intrinsically determined feature of a fully explained whole?

Reply:

In my account, the core is timeless and contains no accumulated history. There has been no passage of time within it through which entropy could accumulate, and no returning history survives the reset.

Suppose, for the sake of argument, that some inherited entropy remained. What explains its presence? If it comes from a temporal state, we must explain that state. If the answer is another temporal state, the regress returns. Calling the inherited entropy an intrinsic feature of the whole does not yet explain why the foundation must contain it.

My proposed resolution is that the core does not preserve this inheritance at all. Its balance is the condition to which histories return and in which their distinctions are erased.

Skeptic:

“No entropy could have been produced there” does not, by itself, establish that its entropy must be zero.

Reply:

My argument concerns what could account for entropy being there at all. I connect entropy with the distinctions and accumulated outcomes of temporal activity. The timeless core neither accumulates these outcomes nor preserves them when they return.

Within this account, entropy cannot simply be inserted as an unexplained feature of timelessness. It must have an explanation, and the explanation cannot depend on a history that the core is supposed to erase. This is why, within my model, I describe the core as having zero entropy. The absence of production there and the complete erasure of returning history are both parts of the claim.

Skeptic:

Why can balance belong to the foundation without being produced, while entropy-bearing distinctions cannot?

Reply:

Because they play different roles in my explanation. Balance is the relation through which I argue that dependence on earlier conditions ends. Entropy-bearing distinctions preserve the inheritance whose origin still needs explaining.

The foundation's balance is not a product left behind by an earlier event. It is the ongoing relation of the activity itself. In my account, retaining entropy at the core would retain dependence on history, while restoring balance removes that dependence. That is the difference on which my argument rests.

Skeptic:

Why does timelessness require that low-entropy condition in the first place?

Reply:

Because the perfect balance I propose requires the removal of inherited differences at the core. Spacetime and timelessness are forms of the same reality. The temporal circulation must therefore be organized so that the timeless condition is maintained.

I describe that condition as uniform because no trace of one returning history distinguishes it from the condition reached by another. Nothing remains that could accumulate, shift the balance, or make the core depend on what came before. This complete restoration is what I connect with the core's low-entropy condition and its renewed output.

Skeptic:

Uniformity needs a precise meaning before it establishes low entropy.

Reply:

Here, "uniformity" means that the reset leaves no inherited distinctions at the core. My argument is philosophical, and I am explaining the condition I claim the foundation requires. I am not supplying a numerical calculation of entropy.

The other part of my reasoning concerns the universe we actually experience. Life and activity use available energy. Within my model, that activity belongs to the circulation whose output is continually renewed at the core. This is why I describe the output as useful and low in entropy: it can support further activity rather than remaining permanently exhausted.

The claim depends on the connection I have drawn between the core, the circulation, and the universe. Simply calling something uniform would not express that full argument.

Skeptic:

What excludes alternative configurations within your proposed foundation?

Reply:

The requirement that regress must end. A core whose condition varied according to the history returning to it would still carry that history as part of its explanation. It would not be historically independent.

In my proposal, every returning history reaches the same foundational condition. The differences that belong to temporal configurations do not survive there. An alternative account must explain how its proposed distinctions belong to the foundation without restoring the dependence that the foundation is meant to resolve.

Skeptic:

I accept entropy reset as a requirement of your model. What remains unestablished is that ending regress necessarily requires this model and its single foundational condition.

Reply:

The order of the argument matters again. I begin with fundamental motionlessness and timelessness. I introduce expansion and contraction later to explain how the temporal universe belongs to that foundation.

Consider my pen analogy. If the pen never moves, it never draws a picture. Likewise, I argue that there can be no accumulated drawing of history at a foundation that never passes from one temporal state to another. An inherited pattern would require an explanation of how it came to be there.

Expansion and contraction then provide my account of the complete circulation: the universe extends, contracts, and returns to the core without the foundation losing its timeless balance. The reset removes the historical distinctions of what returns. These are connected parts of one explanation, beginning with the timeless condition.

Skeptic:

You have explained that the reciprocal relation is the foundation itself, rather than something assembled by an earlier process. But something occurring in time does not automatically require an endless sequence of earlier explanations.

Reply:

The sequence is endless if every explanation takes the same temporal form:

“This state came from an earlier state.”

“What explains that earlier state?”

“Another earlier state.”

If that is the only kind of explanation available, the chain cannot finish. Declaring a first state stops the sequence, but it still leaves the question of why that state exists. I introduce timelessness to explain why the foundation does not need an earlier temporal condition at all.

Skeptic:

One possible reply is that time has a first state with no earlier cause. That avoids an infinite temporal chain but leaves the first state unexplained. You reject this because you require a complete explanation, rather than a stopping point.

But you also move from “Nothing changes at the foundation” to “The foundation cannot contain different physical arrangements or distinctions.” Those statements are not equivalent.

Reply:

My claim concerns a foundation with no inherited arrangement waiting to be explained. Return to the pen: if it never draws anything, where does the picture come from? Saying that the picture was simply there from the start leaves its presence unexplained.

Likewise, I do not accept an arrangement of distinctions as a brute fact at the foundation. If those distinctions come from an earlier condition, the demand for explanation returns. If they are simply declared to exist without explanation, the demand has been abandoned.

So the argument combines timelessness with my rejection of unexplained inheritance and brute facts. It concerns why a complete foundation should contain no accumulated history, rather than the appearance of a frozen picture alone.

Skeptic:

There is a third possibility: a distinction could follow necessarily from the foundation's nature without being produced by an earlier event.

Reply:

Then the proposed necessity needs to be explained. Which distinction must exist, and why must the foundation contain it? Merely saying that a distinction could be necessary does not yet establish an alternative account.

My answer is that the foundation's necessary condition is the balance that closes regress. I argue that inherited distinctions are erased in that condition. A proposed alternative must show how its distinctions follow from its nature and how they resolve the same explanatory problem. Until that explanation is given, I do not regard the suggested third possibility as an established alternative to my account.

Skeptic:

Speed means distance travelled per unit of time. If your non-relative world has neither temporal intervals nor spatial distances, we cannot straightforwardly describe it as having an extremely high speed. You would be proposing activity more fundamental than speed as ordinarily understood.

Reply:

This is why I introduced the Hercules analogy. Imagine that Hercules is so strong that nobody can defeat him in an arm-wrestling match. Suppose there is eventually nobody left against whom he can measure his strength. Does the absence of an opponent make him weak? No. Losing a means of comparison does not, by itself, remove the property being compared.

I propose the same distinction for fundamental motion. Without a clock, ruler, second object, or external reference frame, there is no relative measurement of its speed. My claim is that the underlying motion remains real even when those comparisons disappear.

This is the sense in which I say that motionlessness can still be real speed. In the non-relative condition, motion does not travel through an external space or pass through a further sequence of moments. Motion and motionlessness describe the same foundational reality in

my proposal. The Hercules analogy explains the distinction I am making between the activity and its measurement.

Skeptic:

“Nothing exists outside Nature” does not, by itself, establish that Nature must close upon itself. An infinite space could contain everything without having an outside. Having no outside does not require a circular or toroidal shape.

Reply:

My argument also includes the rejection of actual infinity at the foundational level. An actually infinite expanse does not resolve the problem as I understand it. It extends the condition whose explanation I am seeking. I connect this with the same failure to reach completion that appears in infinite regress.

There is a difference between an infinity that is physically complete and a process or description that can always be extended. A video game offers an analogy: rules can generate more terrain as the player explores, without every possible region having to exist as a completed world beforehand. This is procedural generation.

In my account, the experienced universe can appear indefinitely extendable while its foundation remains closed. My case for closure therefore combines the claim that nothing lies outside Nature with the rejection of actual foundational infinities and the requirement to resolve regress. The toroidal model gives a proposed form to that closure.

Skeptic:

Rejecting actual infinities also rules out actually infinite spin. You cannot exclude an infinite straight path while allowing an infinite rate of rotation.

Reply:

I distinguish the universe as experienced from the foundation that makes that experience possible. In my proposal, the animal mind presents reality through an interface of space, time, distance, and duration. Within that representation, I allow the description of infinite spin and the apparent paradoxes that accompany it.

The spin itself belongs to the zero-dimensional, non-relative condition. There is no extended path there and no external clock against which an infinite number of revolutions is

completed. So I am not placing a completed physical infinity at the core after rejecting one elsewhere.

What I call “infinite spin” expresses how the proposed condition appears when we try to describe it through temporal and spatial measures. My claim is that the real activity persists where those measures cease to apply. That is how I connect the language of extreme speed with a foundation that contains no actual infinity of extension.

Skeptic:

Why would this be useful to humanity? Saying that we already use renewed matter from a previous cycle does not, by itself, give us unlimited energy.

Reply:

The practical significance is that, in my model, renewal happens continually and in portions. We do not have to wait for the entire universe to complete a cycle before any renewed output becomes available.

Penrose’s conformal cyclic cosmology provides a useful comparison of scale. It connects the remote future of one cosmic aeon with the beginning of the next. My proposal places renewal within the ongoing activity of our universe. The distinction is between successive cosmic aeons and a circulation that continually renews portions of matter and energy. See Meissner and Penrose, *The Physics of Conformal Cyclic Cosmology* (2025).

That is why I use the image of a spring. The proposed opportunity is to find its outlets and place our buckets where renewed output becomes available. We collect what emerges from the process, without drawing energy from the core itself. The practical task is to identify those outlets and work out how their output can be collected and used. In this account, continual replenishment is what opens the possibility of a supply without a final exhaustion.

CHAPTER 22

Human Access to Renewed Energy

Human access to renewed energy would mean collecting usable output from the ongoing activity of Nature. In DEMU, that activity renews the conditions for useful work without exhausting its foundation. I argue that this possibility deserves investigation, and that any successful use should serve the lives that depend on it.

Chapter 12 explained what I mean by free and unlimited energy. Free describes a fundamental circulation that needs no external supply. Unlimited describes renewal without a final point of exhaustion. Neither term gives a particular device an infinite power output or removes the work of building and maintaining it.

This chapter brings the proposal into human life. It asks what practical access would involve, what would justify using it, and how it might change our circumstances. The distinction between the philosophical argument and a working technology remains important. The account developed here proposes a source of renewal; it does not yet provide a tested collector or an identified physical outlet.

22.1 Harnessing the Renewed Output

In this model, collecting usable output obviously does not draw energy from the core. The fundamental circulation remains unaffected. The core is the foundation through which I explain renewal. It is not a battery whose charge falls whenever someone switches on a light.

Human access takes place within the temporal universe, where renewed activity becomes available in forms that can interact with equipment. A collector would receive some of that available output and direct it toward a task. Using it to heat a room or operate a tool would remain part of Nature's activity. Nothing would be taken outside the whole.

This gives the practical inquiry a clear target. We need to identify a process through which the proposed renewal produces an accessible supply. We then need to understand how an apparatus can receive that supply. The location, rate, and physical form of the output must come from a developed account of the process, rather than from the word renewal alone.

The river image remains useful. A wheel can receive energy from a current at the place where it stands. Its operation depends on how the wheel meets the flow. In the proposed technology, we would likewise need to understand the interaction through which useful output reaches the collector. The analogy explains access; the ordinary river is not evidence for a cosmic entropy reset.

The geometry discussed earlier may suggest questions about circulation and the organization of a collector. It cannot determine the answer by itself. A toroidal shape would need a defined physical role: what moves through it, what maintains that movement, and how energy reaches the intended task. Building something in the shape of a torus would not establish that it accesses the renewal described by DEMU.

The output must also be usable under the conditions where it is collected. Energy can be present without being available for every task. Feynman's account of the second law explains why a cyclic engine cannot take heat from one reservoir and turn it completely into work as its sole result. [1] The proposal therefore needs an accessible difference or organized transfer that the collector can use. Naming the core does not supply that local description.

DEMU places its proposed departure from ordinary thermodynamics in the total reset and renewal described in Chapters 10 and 11. A collector could receive the resulting output while operating through familiar physical processes. Such a collector would have an input from its surroundings. The larger claim concerns why the availability of that input can be renewed without final exhaustion.

Figure 22.1 separates the foundation's supporting role from the energy transfers that a collector would need to account for.

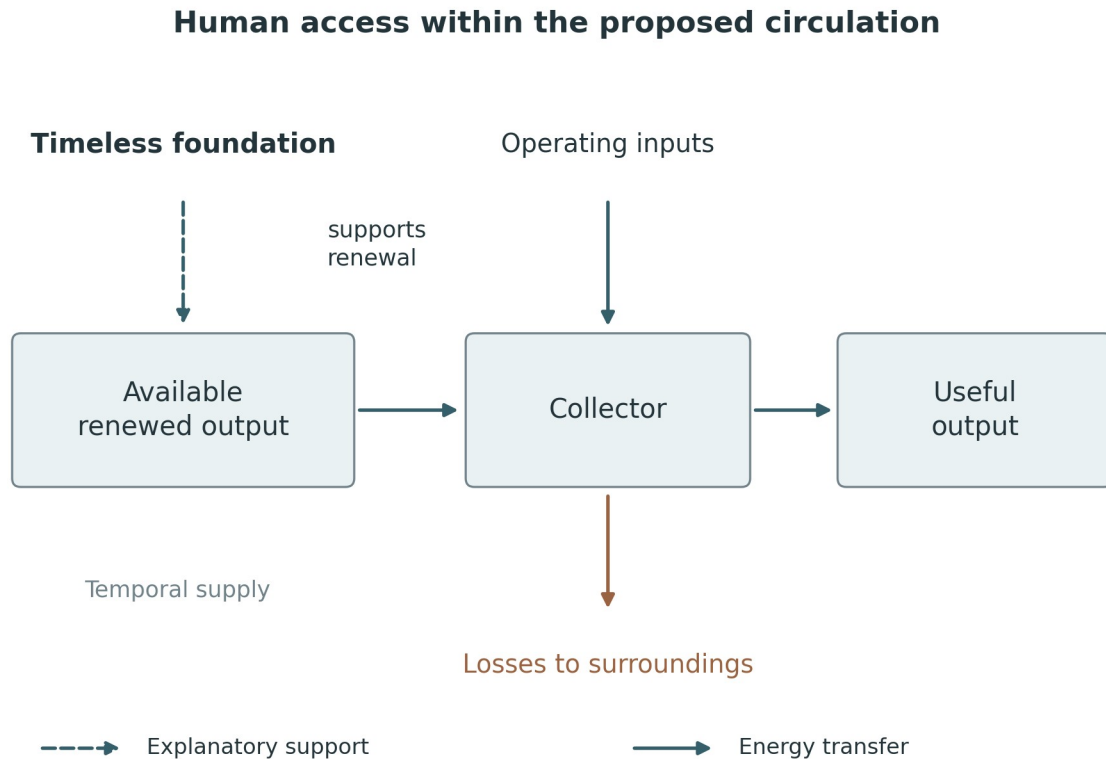


Figure 22.1. Human access to renewed output in DEMU. The dashed arrow marks the proposed supporting relation between the timeless foundation and renewal. Solid arrows show energy transfers within a possible collection process, including operating inputs and losses. The core is not represented as a fuel store. This schematic identifies roles; it does not specify or demonstrate a working device.

Once a candidate process is identified, the first useful experiment would ask a narrow question. Under stated conditions, does the apparatus transfer a measurable amount of energy to a defined task? A turning wheel or a glowing lamp can show that energy is being transferred, but we also need to know where it comes from and what changes during the test.

This means including the equipment's own needs. A pump, controller, or separate electrical supply belongs in the account. So does energy released from an initial store. A machine can run for a time by discharging a battery, cooling a body, consuming material, or slowing a rotating part. Those possibilities have to be measured or bounded before continuing output can be attributed to a different source.

Net useful output, as defined in Chapter 12, is what remains for other tasks after the collector's own needs and losses are included. There is no extra cost in this definition for keeping the timeless core active. The accounting concerns the apparatus and the transfers through it. A loss already excluded from the measured output must not be subtracted again.

The uncertainty of the measurements matters as much as the reported numbers. NIST's guidance calls for researchers to state the sources of measurement uncertainty and explain how they were evaluated. [2] Applied here, a small apparent surplus would be inconclusive if the uncertainty were large enough to explain it. The report would need to show how the inputs, outputs, and changes in stored energy were assessed over the same period.

Further tests would ask whether the result continues when the apparatus performs useful work, how it changes under different conditions, and whether other researchers can reproduce it. A small result that survives these checks would be more informative than a large claim supported only by an impressive demonstration. It would give the next stage of research something definite to explain.

Even then, practical performance and the truth of the full model are different questions. A successful collector would establish access to a source under the conditions tested. Connecting that source specifically to DEMU would require evidence that distinguishes the proposed renewal from other explanations. No finite experiment can directly observe endless operation. The claim of unlimited renewal would continue to depend on the explanation of the source.

Questions about practical access

Claim	Evidence needed	What it establishes
A collector provides useful output	Complete accounting, stated measurement uncertainty, and repeatable performance under load.	Access under the tested conditions.
The source is renewed as DEMU proposes	Tested predictions that distinguish the proposed renewal from other sources.	Support for the proposed mechanism.
Renewal has no final exhaustion	A supported account of the source that explains why renewal continues.	An inference beyond any finite test.

The role of philosophy is to state the proposal clearly and explain why the search is worth making. A physical theory must then identify what to look for, and experiments must establish what actually happens. I can defend the reason for renewal without pretending that this also supplies the missing engineering details.

22.2 Whether We Should Harness It

If renewed output becomes accessible, safe, and useful, I argue that we should use it to improve the conditions of life. Energy can help us meet needs that would otherwise remain unmet. The ethical reason for developing access lies in those lives and needs, rather than in the desire to display greater power.

Chapter 20 argued that conscious experience matters and that love involves taking the good of other lives seriously. Applied here, this means asking what the technology allows people to do and how its benefits are shared. A source that helped keep homes warm, water systems working, or essential services running would have value through those effects.

The same concern requires attention to the people affected by its development. An early experiment and a widely used energy system create different responsibilities. As more people depend on the equipment, failures can reach beyond the laboratory. The scale of use should therefore grow with the evidence about how the system behaves and how it can be maintained.

The core's proposed independence does not remove these local responsibilities. Equipment can change its surroundings, even while the fundamental circulation remains unaffected. Its materials, heat, noise, and other physical effects would belong to the world in which people and other living beings have to live. These effects should be assessed through the particular design, not assumed away by calling its source free.

Nor would the total reset justify treating the environment as disposable. In this model, returning histories are erased completely; their entropy is not preserved elsewhere. But that claim does not tell us when a damaged habitat will recover or whether a particular community will survive an injury. The consequences experienced before any reset still matter to those who experience them.

Where serious effects remain uncertain, I favour development that allows us to learn and change course. Small, contained trials can answer questions before a community becomes dependent on the result. People who would bear significant effects should have a meaningful part in decisions about use. The purpose is to make the benefits real while keeping avoidable harm out of their lives.

We also need to distinguish a continuing source from fair access to it. The source might renew itself while control over equipment or distribution remains concentrated. People could still be excluded through price, ownership, location, or lack of technical support. The meaning of free at the foundational level does not settle any of those social questions.

My position is that access to essential energy services should be a central aim of development. The United Nations' 2030 Agenda includes universal access to affordable and reliable energy services in Goal 7. [3] I share that aim. In the present proposal, it provides a way to judge progress: does the technology make needed services more dependable and more widely available?

Different arrangements could serve that aim. Some equipment might be maintained by households or communities, while other applications could require shared infrastructure and specialist teams. The appropriate arrangement would depend on the actual technology. Decisions should account for who can repair it, who can challenge a failure, and whether people have a reasonable alternative if a provider lets them down.

The knowledge needed to assess the technology should also be available for independent examination. Researchers should be able to check the evidence and understand the conditions under which the claimed output occurs. Confidence should grow through results that others can examine, rather than through loyalty to the person who first proposed the idea.

These responsibilities give a positive answer to whether we should harness renewed energy. We should pursue it where the evidence supports access and where its use can serve life. We should also be willing to alter a design, delay a particular use, or reject an application when its consequences conflict with that purpose. A philosophy of balance has to guide what we do with power as well as explain where power might come from.

22.3 Implications for Humanity

If the proposed renewal could be accessed at useful rates and reasonable cost, the consequences could be substantial. The first question would concern the services it could reliably support. A modest source might have value in a place with limited access, while a larger one might support more demanding tasks. The scale of the benefit would depend on the demonstrated supply and the equipment needed to use it.

Consider a community that could keep a water pump operating reliably. The benefit would include what a dependable water supply makes possible in daily life. Or consider a clinic with secure power for lighting and refrigeration. These are examples of what reliable energy could support; they are not claims that DEMU has already supplied the equipment. The people using the services would be the measure of their practical importance.

Cheaper and more dependable energy could also give people greater freedom over their time. Some demanding tasks might become easier to perform, and some opportunities might become available to people previously excluded from them. How far this change reached would depend on access, skills, and the way work was organized. A technical improvement becomes a human improvement when it changes the conditions under which people can live.

The end of an energy constraint would not remove every form of scarcity. Food still requires suitable growing conditions and care. Homes require materials, space, and construction. Equipment requires maintenance, and people need knowledge to use it well. Even a continuing energy supply would have to work within these relationships. Unlimited renewal does not mean that every desired outcome becomes instantly available.

The environmental consequences would likewise depend on use. A new source could reduce reliance on damaging practices if it successfully replaced them. It could also support more production and extraction. Renewed energy would give us additional means to act; our decisions would shape whether those means restored or damaged the places on which life depends.

Efficiency would therefore remain worthwhile. At any particular outlet, power could be limited, and equipment would still have costs. Delivering the same service with less input could reduce the amount of equipment needed and leave more output available for other tasks. The foundation's continuity gives no reason to waste resources within the human system of access.

Political consequences would also depend on how access was organized. Local generation, if the technology allowed it, could reduce some forms of dependence on distant providers. A system controlled through scarce components or exclusive technical knowledge could create new dependencies instead. A philosophy concerned with the good of conscious lives should pay attention to who gains freedom and who becomes more vulnerable during such a change.

The transition would include people whose work and communities depended on older forms of supply. Their lives would remain part of the balance we claim to serve. Training, participation in decisions, and support during changes could help make a successful technology a shared gain. It would be inconsistent to celebrate a benefit for humanity while treating those affected by the transition as irrelevant.

There would also be an intellectual consequence. A reproducible source that required a new account of energy renewal could change how we understand the universe. Its significance would depend on what the evidence established. Useful output alone would not prove the conscious foundation, God, or free will. Those claims have their own arguments. Evidence for a particular physical process could strengthen part of the thesis without settling every part at once.

For DEMU, the deeper possibility is that humanity learns to participate more deliberately in the renewal already proposed within Nature. We would understand ourselves as living within an activity whose foundation is not progressively used up. That understanding could support a more hopeful view of the future, provided it remained connected to what we can explain and demonstrate.

Greater material security would also leave room for the question asked in Chapter 20: what do we want life to become? More available energy could create opportunities for learning, care, exploration, and creativity. It would not decide which opportunities we choose. The purpose I defend gives those choices a direction through the value of conscious life and the relationships in which it can flourish.

This is the human promise of renewed energy in my account. A reliable source would be valuable because of the lives it could support, the burdens it could reduce, and the possibilities it could open. The work ahead is to connect the proposed renewal to a definite physical process and then establish whether people can use its output. If that connection succeeds, the task will be to make access serve the living world to which we belong.

References for Chapter 22

- [1] Feynman, Richard P., Robert B. Leighton, and Matthew Sands. 1963. The Feynman Lectures on Physics, Volume I. Chapter 44, "The Laws of Thermodynamics," especially Sections 44–1 and 44–2. Online source.
- [2] Taylor, Barry N., and Chris E. Kuyatt. 1994. Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results. NIST Technical Note 1297, 1994 edition, Section 7, "Reporting Uncertainty." Online source.
- [3] United Nations General Assembly. 2015. Transforming Our World: The 2030 Agenda for Sustainable Development. Resolution A/RES/70/1, Goal 7, especially target 7.1. Online source.

CHAPTER 23

The Circle of Life

23.1 The Full Circle

The full circle of my argument is this:

- Nonexistence is impossible.
- Paradox, including the infinite regress described earlier, is impossible.
- Static systems, understood here as complete inactivity, are impossible.
- Motion must be accounted for without a first activation. My answer is motionlessness/timelessness.
- Timelessness must remain timeless.

Within my model, these requirements come together:

Timelessness must remain timeless + activity requires no first activation = our eternal dual torus [expansion = contraction].

Timelessness and time are therefore two aspects of the same reality. Expansion, contraction, and activity belong to the timeless foundation; spacetime is how that activity is expressed as our universe.

We therefore begin our explanation with eternal spin: one gigantic cosmic dual torus, which I later identify as the Mind at Large—God. This is a foundation, rather than a first event in time.

Through its dual structure, I propose that this system can replicate and produce further expressions of itself. I represent its geometric balance through the cuboctahedron, with complementary tetrahedron clusters producing new cuboctahedrons.

Why would it produce more of itself?

Because, in my account, the purpose of Nature is balance, and complete inactivity is impossible. Balance cannot simply be established and then left inactive. It must be continually renewed.

Nature is therefore an eternal becoming: a continual expression and renewal of balance. I picture this activity as occurring so rapidly and continuously that the unresolved regress never arises. At the core, no independently extended spacetime forms, because that would reintroduce the sequence of earlier conditions whose dependence I am trying to resolve.

As we have argued, an explanation that refers only to earlier temporal states keeps passing the question backwards. The core of reality must therefore be timeless.

No time, no regress.

Nevertheless, the cosmic dual torus can be pictured as a gigantic whirlpool that forms smaller whirlpools within itself. Its foundational role does not prevent it from containing a world of activity.

The core must remain timeless and free of paradox, but a temporal world can belong to its activity. This does not mean that the core first exists alone and then creates time. It means that the world and its timeless foundation belong to one reality.

How is this world generated?

If the cosmic dual torus is conscious, and its activity continually expresses balance, I propose that it produces further conscious expressions of itself. This brings to mind Gottfried Leibniz's idea of monads. I use the term here for conscious centres within the cosmic whole.

We now reach a question similar to the one Donald Hoffman explores: how could a universe of spacetime arise from the interactions of conscious agents? If such an account succeeds, it could help explain how a temporal world belongs to a more fundamental conscious reality.

Within that temporal world, irreversible processes can occur and entropy can increase.

How, then, is entropy reset?

If the universe arises through the interactions of these monads, and the cosmic dual torus is the fundamental reality to which they all belong, then their separate forms can dissolve back into it.

Think again of smaller whirlpools within a larger whirlpool. The smaller patterns can disappear into the larger flow. In my proposal, this dissolution is total: the inherited

distinctions disappear, information is erased, and entropy is reset. Nothing preserves the returning history elsewhere.

The cosmic dual torus itself neither decays nor consumes energy in this account. I picture the relationship through the image of a river: resisting the current requires effort, while surrendering to it allows the current to carry you. Likewise, I propose that returning to the fundamental flow requires no external energy input.

There is nothing outside the cosmic dual torus that could slow it down. Nor, in my model, can anything within it permanently obstruct its activity, because every internal form belongs to that activity and can dissolve back into it.

The cosmic dual torus can therefore continue its eternal spin and produce smaller whirlpools. These “baby whirlpools” participate in forming worlds, and those worlds can dissolve completely into the same foundational reality.

Where does free energy enter the picture?

The key to my free-energy proposal is that renewal occurs in portions. Parts of the universe dissolve and renew while other parts continue their activity. The entire universe does not have to disappear before renewed output becomes available.

The universe itself can therefore be understood as the outward expression of the cosmic dual torus. By “outward,” I mean its manifestation as a world, not a separate reality outside the whole. The universe is eternal, even though the forms within it have beginnings and endings.

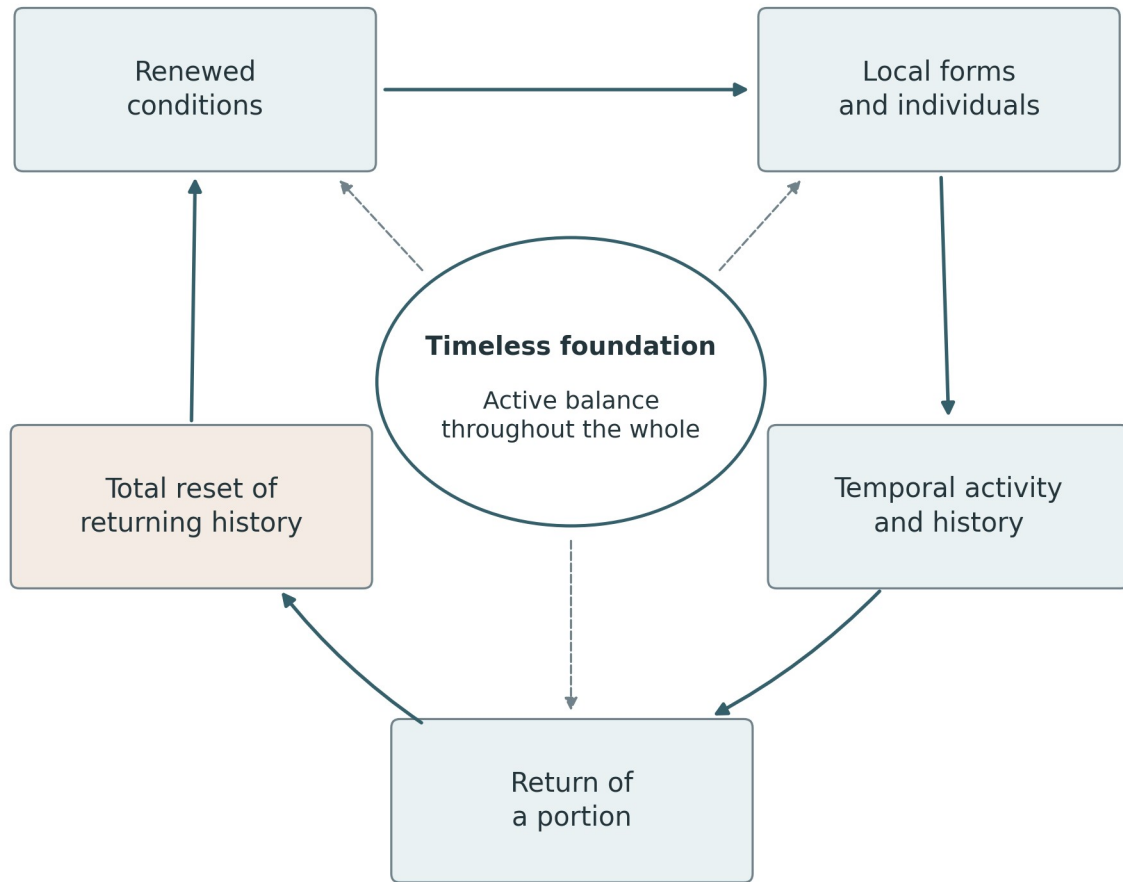
In my proposal, renewed matter and energy emerge through the centre of the cosmic dual torus, where the reset occurs. Their emergence marks a beginning within the circulation, rather than the beginning of existence itself.

Existence never began. The forms within the universe arise, develop, dissolve, and renew through its eternal circulation.

Figure 23.1 brings these relationships together. The foundation is shown at the centre to indicate its role throughout the circulation.

The proposed circle of formation and renewal

Different portions can occupy different roles at the same time.



Total reset: no recoverable record of the returning history remains anywhere.

Figure 23.1. The proposed circle of formation and renewal. Outer arrows describe the circulation from within time; they do not identify a first cycle. The central foundation supports the whole, as indicated by the dashed arrows. Different portions may occupy different roles at once. Total reset erases the returning history completely, with no recoverable record remaining elsewhere.

CHAPTER 24

Consciousness as Nature's Replicator

The previous chapter brought the circulation together. The foundation remains timeless while its activity takes the form of worlds, living beings, and individual histories. Those forms can develop, interact, and return. Their returning history is erased completely. The activity through which Nature continues does not depend on keeping an endless collection of everything that has happened.

One question now deserves a fuller answer. What is it that keeps expressing itself through this circulation? My answer is consciousness. I propose that consciousness is Nature's fundamental productive activity, and that individual lives are renewed expressions of it. This is the sense in which I call consciousness Nature's replicator.

The claim connects the earlier arguments without making them interchangeable. Chapter 15 considered a proposed mechanism for producing new structures. Chapter 17 distinguished the continuation of a person from the continuation of the whole. Chapter 18 argued for a conscious foundation. Here I ask how these ideas belong together. An account of the geometry of a new structure must meet an account of what it means for that structure to be a centre of experience.

I take the arguments already developed for the foundation as my starting point. The task here is to explain the productive role I give consciousness, clarify its relation to biological reproduction, and follow the consequences for individuality and total reset. Calling consciousness fundamental is one claim. Showing what it contributes to the continuing formation of life is another. The second claim gives this chapter its work to do.

24.1 What Does It Mean to Replicate

Imagine lighting a second candle from a first. There are now two flames. The original flame has not travelled into the second candle and left the first one dark. A process has helped another process begin, using material and conditions available at the new location. This is a useful first image for production within a continuing world. It does not yet tell us whether anything has been copied, or whether either flame is conscious.

Now imagine making a photocopy of a letter. Here particular marks matter. A copy that replaces every sentence with a different sentence has failed, even if the new page is the same size. Copying involves a relation between an earlier arrangement and a later one. We can identify what has been retained and ask how accurately it has been reproduced. Resemblance caused by shared circumstances is not always copying. Two people can draw circles without either copying the other.

DNA replication provides a concrete biological case. Cells copy DNA through molecular processes, providing genetic material for daughter cells. The National Human Genome Research Institute describes replication as the copying of a genome's DNA before cell division. This involves inherited organization and a physical process that produces further DNA. It does not mean that an entire organism is already contained in a finished miniature inside each copied molecule. [1]

These examples separate questions that can otherwise become confused. Has the number of things increased? Has information been copied? Has an existing individual continued? The answers need not be the same. A second flame increases the number of flames. A photocopy preserves a pattern of marks. Repairing a damaged chair may preserve one chair without producing another. A theory that uses the word replication must say which relation it means.

For this chapter, I distinguish biological replication from foundational replication. Biological replication concerns processes within the experienced world through which inherited arrangements are reproduced. Foundational replication names my proposal that one conscious reality continually gives rise to further organized expressions of its own activity. The second use is broader. Its subject is the productive character of the whole, not a sequence of cosmic objects copying a hidden genetic code.

This broader use needs a limit. If every change counted as replication, the word would explain very little. A falling stone, a passing shadow, and a new person would all qualify simply because something happened. I therefore reserve the expression for the production of a further organized centre capable of participating in the same general kind of activity. Within the proposal, the new centre must possess some unity of its own while remaining dependent on Nature.

The relevant sameness concerns organization and capacity. A child need not reproduce the parent's complete state. Indeed, two perfectly matching descriptions would not establish

that there are two individuals unless a difference in their relations or histories also distinguished them. The proposal concerns the reappearance of a productive organization in another expression of the whole. Its success would be measured by what the new expression can do and experience, rather than by its being a duplicate biography.

I also need to distinguish an additional expression from a replacement. If one organized form simply changes its shape, I have described transformation. If it disappears and another arises independently, I have described succession. For a local process to count as reproduction, the earlier organization must contribute to the formation of the later one. The contribution might concern the arrangement of available activity, the conditions of formation, or an inherited pattern. It must make a difference to what forms.

This requirement matters for cosmic sex. The cuboctahedron changing into a tetrahedral configuration and back is a proposed cycle of form. It becomes a candidate for reproduction when the process produces a further organized structure. The continuing parent and the newly formed structure then have distinct roles within the same circulation. Chapter 15 made this distinction in its worked model. Here I add the question of whether the resulting organization has its own experiential unity.

The foundation itself does not reproduce by becoming two independent foundations. There is no second Nature standing outside the first. If I divided the whole into self-sufficient realities, I would immediately need an account of their coexistence and relation. My proposal instead treats reproduction as an internal differentiation of one reality. Nature produces additional expressions without producing something outside everything that exists.

Nor does timelessness first wait and then begin to reproduce. Waiting and beginning already belong to temporal description. At the foundational level, the productive relation belongs to what reality is. Within its temporal expression, particular structures form at particular times. A new organism can have a beginning even though the reality expressed through it has no first activation. These statements concern different subjects and therefore need not conflict.

The word replicator should consequently not suggest a tiny eternal organism hiding at the centre of the universe. I mean the activity through which organized centres continue to arise. The claim is substantial only if that activity explains their formation and their character.

Naming consciousness as its source is the beginning of this chapter's account, not a substitute for describing the relation between the source and what it produces.

There is a reasonable linguistic objection. Someone may prefer generation, expression, or renewal to replication. Each captures part of what I mean. Generation emphasizes the appearance of a further form. Expression emphasizes its dependence on one reality. Renewal emphasizes the continuing possibility of formation. I retain replicator because I want to ask what sustains the recurring production of organized life. I do not need every use of that word in biology to carry my broader meaning.

This definition also prevents the reset from becoming disguised copying. During an individual history, reproduction can transmit information. At the total reset, the returning information is erased. A later expression arising from the foundation is therefore not a copy recovered from that erased history. The productive capacity continues, while the particular inheritance does not. The rest of the chapter develops how those two claims can belong to one account.

Table 24.1. Different relations involved in production and renewal.

Relation	What happens	What it does not establish
Transformation	An existing form changes.	A second individual has been produced.
Replication	An earlier organization contributes to a further one.	The two individuals are the same person.
Continuation	One organized life persists through change.	Every original component or memory remains.
Total reset	All information about the returning history is erased.	A later form recovers that erased history.

24.2 Genes and the Deeper Ground of Life

When I say that consciousness is Nature's replicator, I am asking about the foundation of the processes through which life reproduces. I am not proposing that biologists stop studying genes. A philosophical explanation that cannot accommodate the processes it seeks to explain would weaken its own claim to completeness. The question is where those processes belong within the wider account of Nature.

Genes carry inherited information and contribute to the development and functioning of organisms. They work within living systems whose activity also involves other molecules and relations with an environment. The National Human Genome Research Institute describes genes as units of inheritance and distinguishes genes coding for proteins from genes whose products are functional RNA molecules. This gives the term a biological meaning that should remain clear. [2]

It helps to consider an ordinary instruction. A recipe can guide the preparation of bread, but a recipe lying unopened on a table produces no loaf. Ingredients, equipment, and an activity of preparation are needed. The comparison does not mean that DNA is literally a written recipe or that a conscious cook must direct every cell. It illustrates a point about explanation: a pattern that guides a process is intelligible through the conditions in which that guidance operates.

In DEMU, those conditions eventually lead back to the wider circulation. Molecules, bodies, and environments are forms within Nature. They are not independent resources from which Nature itself was assembled. My proposal therefore places genetic reproduction within a more basic productive activity. Genes can explain aspects of how organisms resemble their parents without supplying the final explanation of why there is an active world capable of sustaining organisms at all.

The distinction can be stated without diminishing biology. Suppose an account correctly explains why a particular inherited difference spreads through a population. That account can succeed at its own task even while leaving the foundation of matter and experience unexamined. Conversely, a metaphysical account of that foundation will not tell us the frequency of a genetic variant merely by announcing what reality ultimately is. Each explanation has a subject, and the subjects have to be connected through additional work.

Darwin's account of natural selection gives a familiar example of explanation through differences within a population. In the first edition of *On the Origin of Species*, he describes how inherited variations affecting survival and reproduction can be preserved through generations. The process does not require each organism to understand the population's development or intend its long-term result. [3] My interpretation of Nature must leave room for processes of this kind.

Fundamental consciousness therefore cannot mean that every biological event is deliberately planned by a human-like mind. A person can consciously intend an action while remaining unaware of many bodily processes that support it. That familiar contrast does not prove that those processes have experiences of their own. It does show why conscious participation and complete personal knowledge of a process should not be treated as the same thing.

My account concerns the character of the whole activity. I propose that what biology describes from the outside belongs to a reality whose intrinsic nature is experiential. This is different from placing an additional invisible person beside each molecule. The proposal does not need a miniature supervisor to read DNA, direct a chemical bond, or push a cell into division. Such supervisors would merely reproduce the question of what organizes their own activity.

An objection immediately arises. If the same biological description can be given without mentioning a cosmic consciousness, what has consciousness added? My reply concerns the range of the explanation. A description of inherited structure can relate one physical condition to another while leaving experience itself unexplained. I want an account in which the world described and the experience through which it is encountered belong to one reality. Chapter 18 supplied the argument for taking that reality to be conscious.

The present chapter gives that conclusion a productive role. If the activity through which Nature forms organisms is an expression of the same conscious foundation, then organisms do not introduce an entirely foreign ingredient into existence. Their lives organize and express what Nature already is. Human reflection can be new as a particular capacity, even though the experiential character of reality is not newly manufactured from something wholly alien to experience.

This does not establish that every organized thing has a separate mind. A building is organized, but its organization alone does not show that the building experiences its rooms. The account needs criteria for a centre of experience. A collection may belong to conscious Nature without being one additional subject. The question of which organizations count as subjects will therefore remain central when I move from the whole to individual lives.

The gene also raises a question about purposes. A process can increase the number of descendants without improving the lives of the beings involved. Reproductive success and a

good life are different standards. If my philosophy identifies the deeper purpose of Nature with lived balance, then the spread of an inherited pattern cannot by itself settle whether that pattern serves the welfare of conscious beings. The metaphysical interpretation does not turn every successful biological strategy into a moral instruction.

Consider a person who decides not to have children and devotes their life to caring for others. Nothing in the claim that Nature continually produces conscious forms makes that life a failure. The individual's value cannot be measured simply by how often their genes are copied. Care can preserve lives, expand another person's possibilities, and improve the conditions in which experience occurs. These effects belong to conscious participation even when they do not reproduce the caregiver's biological lineage.

This is one reason I resist treating people merely as vehicles for whatever happens to reproduce through them. Within my proposal, their experience is part of what the whole account is trying to understand. A person's pain, understanding, and freedom are not made irrelevant by their contribution to a chain of inheritance. A theory of the productive whole should explain why individuals matter, as well as how they are produced.

There are therefore two claims in the sentence that consciousness is deeper than the gene. One concerns dependence: genetic processes belong to the activity of the conscious foundation. The other concerns significance: the value of a life is not exhausted by its success in passing on inherited material. The first is part of my metaphysics. The second requires the further argument about conscious welfare and love developed in Chapters 19 and 20.

I do not infer either claim merely from the fact that genes require an environment. That dependence would also be accepted by accounts that reject idealism. My reasoning includes the earlier case for a conscious foundation. Given that starting point, the productive process described by biology becomes a particular expression of Nature's activity. The conclusion is a relation between levels of explanation, not the announcement that genetic inheritance has been shown to be unreal.

The practical consequence is that the proposal must remain answerable to what happens in living systems. An account of fundamental consciousness should help explain why changes in organization change the capacities of an individual. It should not make those changes disappear behind the statement that everything is already conscious. The next

section therefore asks how one conscious Nature can support centres whose experiences and abilities genuinely differ.

24.3 How One Reality Can Contain Many Minds

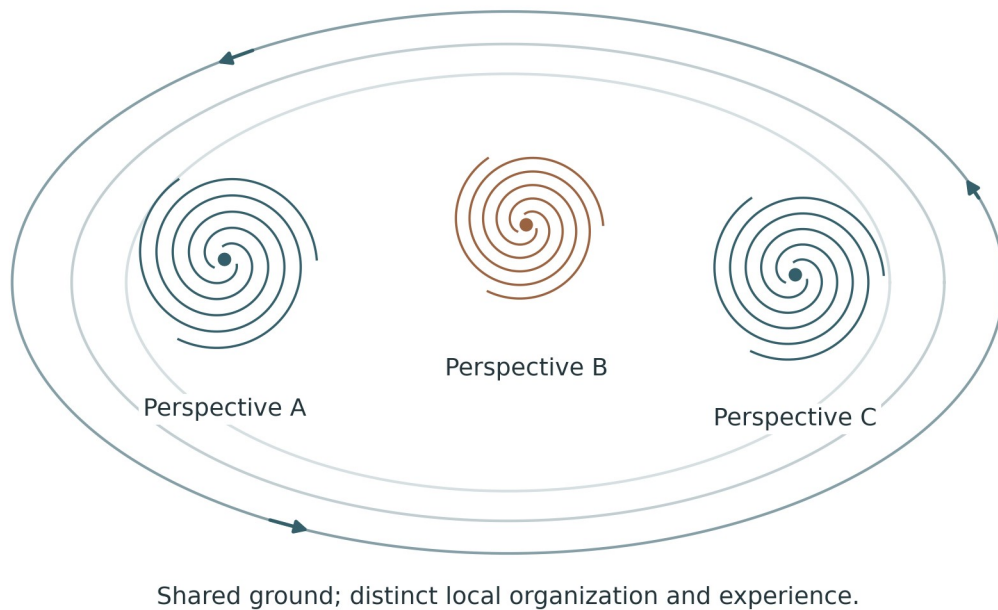
If there is one conscious foundation, why do you and I have different experiences? Why can I feel my own headache directly while learning about yours through what you say and do? Why does the unity of Nature not give every individual immediate access to everything? These questions identify something the proposal must explain. The existence of a shared ground cannot simply erase the differences between the lives that depend on it.

The first step is to separate three ideas: belonging to the same reality, sharing an experience, and being the same individual. Two people can belong to one reality without sharing every experience. They can also share an experience in the ordinary sense, such as watching the same sunset, without becoming one person. Their relationship to the whole and their relationship to each other answer different questions.

My whirlpool image helps with the first distinction. Several whirlpools can belong to one body of water while maintaining different local patterns. Their common material does not make their positions, boundaries, and movements identical. The example shows how dependence on a whole can coexist with local differences. It does not show that water has a mind, or explain by itself how a particular pattern becomes a subject of experience.

Figure 24.1 illustrates the relation between the shared whole and its distinct local expressions.

Many local centres within one reality



Whirlpools illustrate the relation proposed in DEMU.

Figure 24.1. The whirlpool analogy for distinct individuals within one reality. The enclosing circulation represents the shared ground, while the smaller patterns represent differently organized local perspectives. The illustration expresses the relationship proposed in Section 24.3; it does not show that a water flow is conscious or provide a physical model of how a subject forms.

Within DEMU, an individual is a locally organized expression of the foundation. Its boundary concerns the way its activity holds together, responds, and maintains a perspective. That boundary need not be a wall separating it from Nature. A living being constantly depends on exchanges with its surroundings. What matters is that these exchanges support an organized life whose encounters belong together in a particular way.

Consider a person looking through a window. Light from outside enters the room, sounds arrive through the glass, and the person remembers something that happened yesterday. These different contributions belong to one lived situation for that person. Another person beside the window has their own situation. Their experiences can overlap in content without being directly available to each other in the same first-person way.

I therefore use a centre of experience to mean an organization for which there is a particular lived perspective. The words are intended to identify what needs explaining, rather

than solve it by definition. Saying that information is gathered together does not yet tell us why it is experienced together. An account must connect the organization described from outside with the unity encountered from within.

Chalmers draws attention to this difficulty when he separates explaining cognitive functions from explaining why those functions are accompanied by experience. A theory might describe discrimination, reporting, and the control of behaviour while still leaving the felt character of those activities unexplained. [7] I take this as a challenge any account of consciousness must address. His discussion does not, by itself, establish my account of the timeless core.

My proposal approaches the challenge from the other direction. Rather than first assume a foundation wholly without experience and then add experience to it, I begin with the conscious foundation defended earlier. The problem becomes how its activity supports distinct perspectives. I am proposing differentiation within conscious reality. I still need to say what makes a particular differentiation one individual rather than an arbitrary collection chosen by an observer.

A football team gives a useful warning. Its members coordinate, exchange information, and pursue a shared goal. These facts do not establish that the team feels an additional headache over and above the headaches of its members. Practical cooperation and a single subject of experience are different claims. The same warning applies to a cluster of atoms, a computer network, or a galaxy. Calling a collection organized does not settle whether it has its own perspective.

The working requirement in this chapter is therefore stronger than connection alone. An individual must have an organization that supports a unified perspective and some continuity of its own activity. This is a proposed requirement, not a complete test for consciousness. It tells the model what to explain. It also prevents the loose claim that any things touching each other automatically combine into a larger mind.

Leibniz provides a historical comparison through his account of monads. His monads are simple, unextended beings, not little material objects assembled from smaller parts. He also denies that they are naturally made or destroyed by ordinary composition and separation. Their perceptions express multiplicity within a unity. [4] These features matter because my

image of forming and dissolving whirlpools differs from his account. When I use “monad,” I mean a centre of experience by analogy; I am not adopting every feature of Leibniz’s system.

In DEMU, local individuals belong to the activity of one foundation and can arise through changes in organization. Their relationships are part of the explanation of their development. My account therefore cannot borrow Leibniz’s term and then assume that he supplied a mechanism for the interactions I propose. The comparison clarifies the question of experiential unity. The work of explaining its formation remains with my own model.

Hoffman and Prakash offer a different point of comparison. They define conscious agents mathematically through relationships involving experience, decision, and action, and investigate networks of such agents. They present conscious realism as a hypothesis about the world. [5] This is useful because it shows one way of giving an account based on consciousness a formal structure. It does not demonstrate that DEMU’s geometry produces conscious agents, or that an agent in their mathematics is a whirlpool in my model.

The lesson I take is methodological. If I describe individuals as interacting centres, I should eventually specify what can change in each centre, what passes between centres, and what counts as a new centre. Otherwise “interaction” can conceal the very problem it is supposed to solve. A formal description would make these commitments easier to inspect. Whether it accurately describes experience would remain a further question.

The whole also needs careful treatment. When I call Nature the Mind at Large, I do not mean that my ordinary personal awareness secretly contains everyone’s biography. My limited access is a real feature of my life. The proposal is that local experience depends on a conscious whole, while the organization of each individual limits what is available within that perspective. Dependence does not give a person unlimited knowledge.

There is a further distinction between the foundation being conscious and its being one subject in the same sense as a human person. My spiritual interpretation identifies the whole with God, but human features such as waiting, remembering a childhood, and discovering yesterday’s mistake cannot simply be transferred to a timeless ground. The character of its awareness must fit the account of timelessness. Local minds can learn through time without requiring the foundation itself to acquire a previously missing history.

This gives the replicator proposal a clearer task. Nature continually supports forms through which experience becomes locally organized. The forms can differ in what they encounter and what they can do, even though they share the same ground. Their differences are not evidence of a second substance outside Nature. Equally, their common ground is not a reason to dismiss those differences as unreal. A successful account must preserve both unity and individuality.

24.4 Cosmic Sex and the Formation of Individuals

The phrase “cosmic sex” names a proposed relation between complementary activities that can generate further forms. It gives a concrete image to an abstract idea: Nature produces through relationships. Something comes together, a new organization becomes possible, and that organization can take on a life of its own. The phrase concerns this productive relation before it concerns the familiar biological differences between male and female organisms.

In Chapter 15, I explored this idea through the relationship between tetrahedral arrangements and the cuboctahedron. The larger proposal connects expansion and contraction with an eternal circulation that can support smaller centres within it. Here I want to ask what would have to happen for those smaller centres to count as new individuals. A shape appearing inside another shape is only the beginning of that question.

Three events must be distinguished. An existing form can change its shape. Two forms can combine into one. An existing organization can contribute to the formation of a further organization while continuing itself. The third is closest to replication. The first is transformation, and the second is combination. A circulation that moves from one geometry to another does not automatically perform all three operations.

Imagine folding one sheet of paper into a new shape. The new shape can reveal relationships that were difficult to see while the sheet was flat. But folding the sheet has not produced a second sheet. If I want two independently usable folded objects, I must explain where the additional material and organization come from. The same point applies to a proposed movement between a cuboctahedron and a tetrahedral arrangement.

This distinction strengthens the account of cosmic sex by identifying the productive step. The claim is not merely that complementary arrangements can be drawn together. The claim is that their interaction can help organize the surrounding activity into a further centre. That

centre must then maintain itself through the circulation available to it. Without this step, I have described a changing pattern, not the birth of an additional individual.

The worked model in Chapter 15 provides a limited example of how such a question can be made precise. Starting with four suitably chosen directions, their pairwise differences give the twelve directions of a cuboctahedral arrangement. The construction establishes a geometrical relationship under its stated assumptions. It does not establish that an atom has divided or that a conscious subject has appeared. Those claims require additional connections to what the geometry represents.

The same chapter also explored a simple growth variable with feedback. In that example, a sufficiently strong and sustained input could carry the variable into a condition that persisted after the initiating input ended. A weaker or shorter input failed to do so. This illustrates a possible difference between a passing disturbance and a continuing pattern. The persistence came from the rules supplied to the model; the numerical result did not independently discover those rules in Nature.

That difference is useful for thinking about individuality. A passing ripple is not automatically a new self. A pattern that maintains its organization has a stronger claim to being a continuing unit, although maintenance alone still does not establish experience. The proposal therefore has two steps: explain how an organization becomes able to continue, and explain why that organization supports its own lived perspective.

The parent and offspring language should follow the same discipline. If a large circulation produces a smaller one, the larger circulation can be called a parent in a broad sense. But an actual account should identify its contribution. Does it provide a continuing flow, an initial disturbance, a boundary condition, or a transferable pattern? Several contributions may operate together. Naming them makes the proposed relation more informative than simply calling every small structure a baby whirlpool.

Within DEMU, the fundamental circulation is the continuing ground of the local process. A local parent need not manufacture a second foundation. It helps establish another organized expression within the one reality. This allows the parent to remain while a further individual forms. It also explains why the language of production need not imply that existence itself has increased from one complete reality to two complete realities.

Figure 24.2 shows the productive step: a further organized centre forms while the parent continues within the same reality.

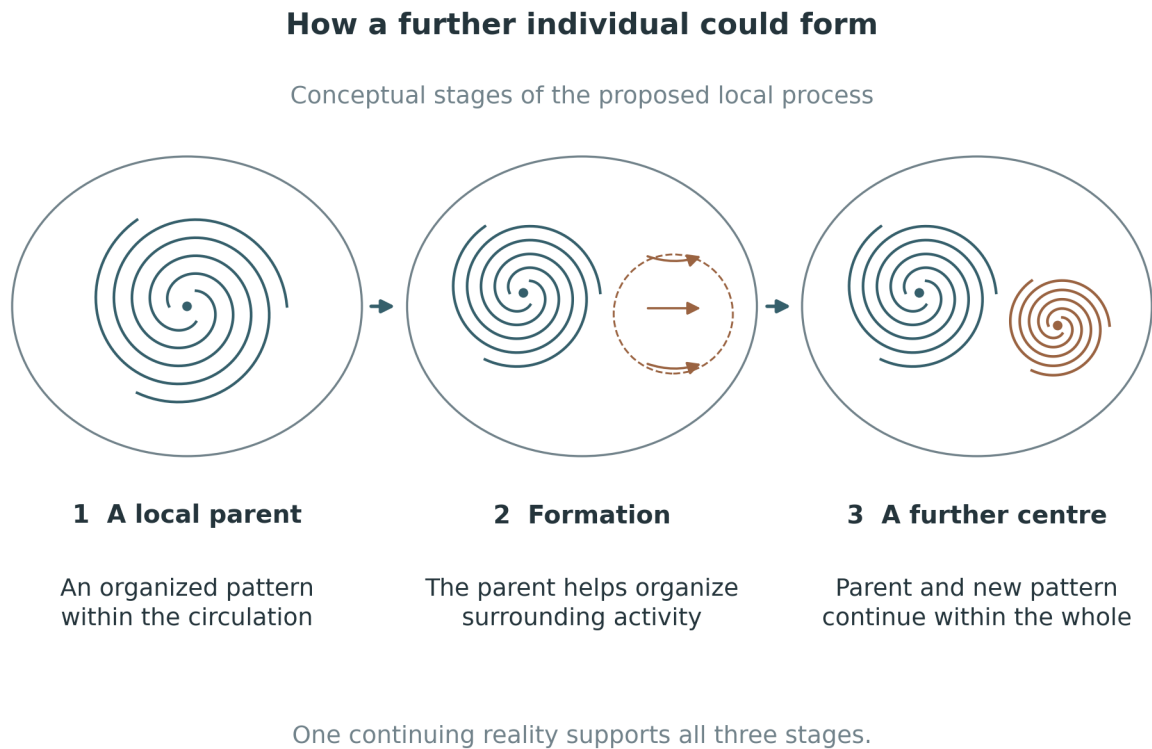


Figure 24.2. The proposed formation of a further individual within one circulation. A local parent contributes to the organization of surrounding activity. A further pattern then maintains its organization while the parent continues. The panels identify the relation required for replication; they are not a calculated flow or evidence that an atom or conscious subject has formed.

There may be competition, cooperation, and limits within this production. A locally sustained pattern can depend on conditions that another pattern changes. My claim that the fundamental circulation continues does not mean every local form can grow without encountering any restriction. The existence of an inexhaustible foundation, as proposed earlier, and the availability of suitable conditions for a particular individual are different matters. Local formation needs its own account of what supports it.

Consciousness enters at the point where organized activity has an inner life. The claim is not that a tetrahedron possesses thoughts because it has four corners. Geometry describes relations that might be involved in a model of organization. It cannot replace the argument connecting organization with experience. My earlier case for a conscious foundation gives

the wider setting, while a theory of individuality must explain how the local perspective is established.

Cosmic sex also need not mean that every productive relation consists of two opposed sexes. Biological reproduction takes several forms, and the metaphysical image concerns complementarity more generally. In the present proposal, activities can be complementary because of what they allow each other to do. Their cooperation can make a stable pattern possible that neither sustains in the same way alone. This is a relational claim, not a rule assigning cosmic rank to human bodies.

The image becomes most useful when it joins difference with connection. If the contributing activities were simply identical in every relevant respect, their encounter might add nothing to explain the new form. If they had no relation at all, their cooperation would be unexplained. A productive account describes both what differs and how those differences participate in one continuing process. That is the work the language of complementarity is meant to perform.

There is room here for novelty without creation from absolute nothing. A new individual can possess a combination of capacities that was not previously present as one local life. Its existence still depends on the activity and conditions from which it formed. The novelty concerns the organized individual; the dependence concerns its ground. These claims are compatible because the arrival of a new form need not be the arrival of an entirely new kind of existence.

I therefore understand cosmic sex as a proposal about the generation of organized lives within conscious Nature. The geometric account supplies candidate relationships, the dynamic account supplies conditions of formation and persistence, and the account of consciousness supplies the question of an inner perspective. None of these tasks can simply stand in for the others. Bringing them together is what would turn the image into a developed account of replication.

24.5 What Continues Through Change

A replicator produces further instances of something. An individual continues as someone or something. These ideas overlap in some ordinary descriptions, but they are not interchangeable. A person who has a child does not thereby become that child. A copied

book is not the same physical volume as its source. A continuing life therefore needs an account that is more specific than the claim that a familiar pattern appears again.

The distinction matters especially in a philosophy of eternal renewal. It would be easy to say that because Nature never ends, no individual can end. But a whole can continue while particular forms within it cease. The continued existence of the ocean does not establish the continued existence of each wave. My account of immortality must therefore connect the whole with the individual in a way that preserves the person concerned.

Three meanings of “the same” help clarify the issue. Two objects may be the same kind, such as two copies of a chair design. One object may retain the same arrangement for a period. An individual may remain that very individual while some of its arrangements change. These are similarity, stability, and continued identity. The fact that each can be expressed using “the same” does not make them one relation.

Consider a small boat that is repaired while it remains in use. One damaged board is replaced, then another. The boat’s material changes while its history of use and repair continues. This example suggests why continued identity need not mean keeping every original component. It does not settle every difficult case. It gives us a concrete reason to examine organized continuity rather than demand that nothing ever be replaced.

A living person provides a more demanding case because the question includes experience. I am not merely asking whether a recognizable body remains in a room. I am asking why the later life is the continuation of this person’s life. Bodily organization, memory, character, relationships, and the capacity for experience can all matter to that inquiry. No single familiar feature should be silently treated as a complete solution.

Locke’s discussion of personal identity is useful because he distinguishes the identity of a person from questions about a body or substance considered alone. His example of a prince’s consciousness entering a cobbler’s body tests where we locate the person when those descriptions come apart. [6] I use the example as a prompt for the present question, without assuming that such a transfer is physically possible or that Locke settled every problem about memory.

Memory is clearly relevant, but retaining every memory cannot be required for remaining a person. I do not remember every meal I ate as a child. Forgetting those meals does not by

itself show that a different person has replaced me. The relevant continuity may include capacities and relationships that persist while particular contents are lost. A theory that makes one forgotten detail fatal to personal identity would fail to describe ordinary life.

Nor should the absence of an immediately reportable experience automatically be treated as the destruction of its subject. The proposal concerns the organization that supports a continuing life, including its capacities, rather than requiring a person to deliver an uninterrupted spoken account. A pause in reporting, a change in attention, and the total destruction of every basis of a person's history are very different conditions. The reset question concerns the last of these.

Now imagine that a device could produce a second individual matching my current memories and character. Suppose, for the thought experiment, that the second individual is conscious. The new person might sincerely say that they remember my childhood. Yet I could remain standing beside them, having a different experience a moment later. The matching information would not make the two simultaneous lives one numerically identical individual.

This shows why resemblance alone cannot establish survival. If two later people both resemble an earlier person equally, resemblance does not explain why one should be that earlier person and the other should not. Calling both the original person would create another difficulty: they can immediately undergo incompatible experiences in separate places. The account needs a relation of continuation, not merely a list of matching characteristics.

The example also shows why a perfect archive would not automatically solve every identity problem. An archive could make reconstruction imaginable, but a further question would remain about the relation between the reconstructed life and the original. DEMU's total reset does not preserve such an archive anyway. I should not make the weaker case of a surviving record appear sufficient, and then quietly assume the stronger case of complete erasure presents no difficulty.

My approach treats a person as an organized life whose activity continues through change. The proposal allows renewed material to enter that continuing organization without requiring the person's defining history to be erased. An individual can therefore benefit from renewal without personally undergoing a total reset. This distinction develops the possibility

of longevity considered in Chapter 17. Whether a particular means of renewal can actually extend a human life is a further practical question.

Imagine repairing a flowing fountain while keeping its pattern in operation. Parts of the equipment can be renewed and different water can pass through, while the organized process continues. The example does not establish that a fountain is conscious or that a person can live forever. It makes one point visible: replacing what supports a process and ending the process are different events. A proposal for longevity must show which event its mechanism produces.

The same distinction applies to Nature's replicative activity. A person may continue while helping to form something new: a child, a work of art, a practice, or an understanding shared with another person. These products can carry something of that person's influence. They are meaningful forms of continuation in ordinary speech, but they do not all amount to the person remaining alive. Influence, resemblance, and personal survival should each receive their own description.

An author can therefore live on through a book in a cultural sense, while the book is not literally the author's continuing consciousness. A parent can see familiar qualities in a child without the child being a replacement version of the parent. These relationships matter precisely because distinct people can affect one another across time. Their value does not depend on pretending that difference has disappeared.

For DEMU, this means that the foundation's persistence supports several possible kinds of continuity. It supports the ongoing universe, the maintenance of local lives, and the production of new lives. It also allows local histories to end under the reset proposal. Which relation applies in a particular case depends on what continues and what is erased. The word "eternal" cannot answer all those questions on its own.

The positive proposal is therefore specific. A continuing person may remain an expression of the eternal foundation while their life is renewed within the temporal world. The person need not be a sealed container of permanent material. Their identity can concern the continuity of an organized, experienced life. This gives renewal a possible role in longevity while preserving the difference between sustaining a person and producing another person who resembles them.

24.6 Total Reset and Personal Identity

The reset in DEMU is total. The inherited distinctions that carry the returning history are erased. They are not transferred to another location, retained in a hidden record, or stored in the Mind at Large for later recovery. Nothing remains from which that returning history can be reconstructed. This is the meaning of reset used here, and the account of replication must respect it throughout.

Historical independence means that the core condition does not vary according to those erased histories. Different returning histories can therefore reach the same foundational condition. If the core still carried a mark identifying which history had returned, the relevant erasure would be incomplete. Calling the mark spiritual rather than physical would make no difference. A recoverable distinction would still be a surviving distinction.

The foundation's continuing nature must be distinguished from a record of what returned. In the proposal, the capacity for activity and renewal belongs to the ground itself. It is not a diary of its previous products. A rule that permits a form and a stored description of a particular past form have different roles. The reset removes inherited historical content; it does not mean that the account has abolished the foundation whose activity makes renewed expression possible.

An ordinary comparison is a musical instrument that can produce a range of notes without containing a recording of yesterday's performance. Its capacities and a record of the performance are different things. The comparison only illustrates that distinction. An instrument still has a material history and does not perform DEMU's total reset. The claim about the core is stronger: its foundational condition carries no recoverable information about the returning history.

This distinction also places a limit on the use of geometry. A diagram with no visible historical marks is not evidence that physical information has been erased. Nor does drawing a point prove that it cannot be assigned different states in a model. The argument for total reset must concern what the condition can retain and what the process removes. An image of contraction can represent that claim, but it cannot establish it merely by becoming small.

The earlier chapters supplied the case for reset as part of the attempt to close regress. I do not restart that argument here. I ask what follows for the replicator proposal if the reset

operates as described. The first consequence is that renewed forms cannot be explained by consulting an erased parent history. A process that copied that history after its total erasure would contradict the account's own terms.

Does this make further formation impossible? No. A capacity to generate forms need not preserve a particular record of previous forms. But the explanation of what comes next must use conditions that genuinely remain available. Within DEMU, these include the foundation's productive nature and, where a local portion returns, the conditions of the ongoing world. The second contribution needs special care so that it does not become an excuse for secretly retaining the erased history.

The universe need not reset all at once in this proposal. Some local histories continue while other portions return to the core. The continuing world can therefore have structures that influence later formation. This is compatible with total erasure of a returning history only if those structures do not preserve recoverable information about that history. If a photograph, memory, or trace still records it, then that particular history has not yet been totally erased in the stated sense.

There is a scope question here that cannot be avoided. The total reset of the distinctions belonging to one returning portion is not automatically the erasure of every event elsewhere in the universe. Other histories can continue. But wherever information about the returning history survives, the account must include that information within what is removed before calling the erasure complete. "Local" describes where the process occurs; it cannot weaken the meaning of "total."

This makes the proposal demanding. If the returning history has left recoverable traces elsewhere, the reset must account for their removal as well. The thesis cannot satisfy that requirement by saying the material at the centre has changed. The philosophical claim identifies the required outcome. A physical development of the model would need to explain how the outcome is achieved while other histories continue. Naming this requirement preserves the totality the proposal intends.

There is another question about variety. Suppose two resets produce exactly the same complete starting conditions, and suppose the rule governing what follows allows only one outcome from those conditions. Then the resulting development must be the same. Different

outcomes cannot be explained by an erased difference. This is a consequence of what the assumptions say, rather than an objection based on preferring a different model of Nature.

DEMU has several possible ways to describe variety, but they must be stated openly. Local renewal can occur within different continuing surroundings. The productive process might also permit more than one outcome under the same conditions. My account of freedom may be relevant to that second possibility, but freedom needs more than unexplained variation. Whichever route is taken, the variety cannot be attributed to information that the reset has removed everywhere.

The same constraint applies to cosmic learning. A local mind can learn while its experiences leave a continuing effect on it. A culture can accumulate understanding while records and practices are retained. If every relevant trace of a history is totally erased, however, that history cannot continue teaching the core through a hidden memory. The foundation's timeless character and the learning of its temporal expressions must be kept distinct.

Now consider the individual person. If the complete organization and every history-bearing distinction that supported a person are erased, the proposal cannot establish that the same person later returns merely by saying consciousness remains. Consciousness continuing as the nature of reality is one claim. The continuation of this individual, with a relation to this life, is another. The first does not automatically supply the missing relation required by the second.

An advocate of survival might propose some further principle of identity that does not depend on historical continuity. Such a principle would need an explanation of its own. It could not operate as a hidden label that identifies the former person after the reset, because that label would preserve a distinction the reset was said to erase. The account must choose consistent meanings rather than promise both total erasure and an unexplained recovery of the same biography.

This does not withdraw the possibility of longevity through renewal. As the previous section explained, an individual might remain organized while using renewed material or output. In that case the person's relevant continuity is maintained outside the reset of the returning portion. The process would be sustaining the person, not erasing the person and

later calling a new individual their continuation. The practical claim remains to be developed on its own terms.

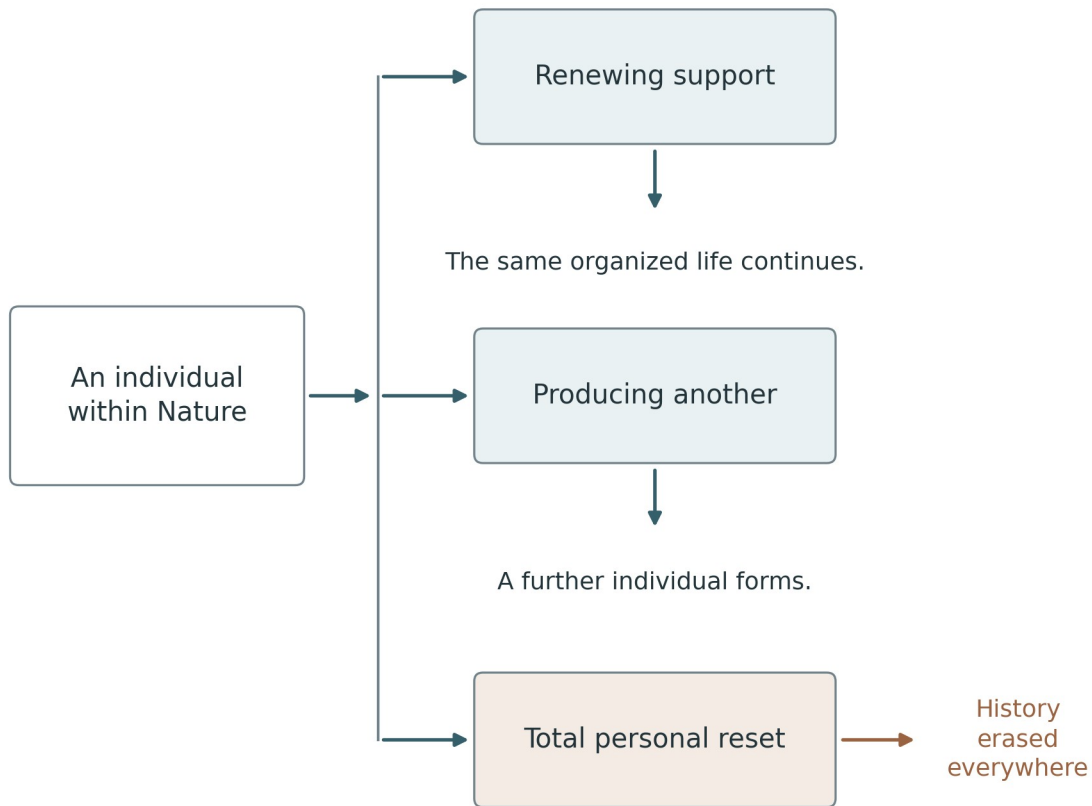
Nor should ordinary biological death automatically be identified with the complete reset described here. The two expressions name different conditions. Whether a particular death is followed by the proposed return to the core, and what happens to all related information, requires an account. The universal claim that Nature continues gives no immediate measurement of what happens to an individual at death. It cannot serve as a shortcut through the identity question.

The consequence for Nature's replicative activity is clear. Renewal can produce new centres without preserving every earlier centre. The productive capacity continues even where a particular history does not. A later life is valuable as its own life; it need not be advertised as an exact return of a former person. Eternal creativity and personal continuity can both have a place in the theory, provided the conditions for each are stated separately.

This also prevents reset from becoming a moral excuse. The prospect that a history may eventually be erased does not make its pain unreal while it is lived. A person's welfare matters within that person's experience. If the whole supports many local lives, its continuing existence is not a substitute for caring about what happens to them. The next section develops the role of consciousness in that care and in Nature's ongoing renewal.

Figure 24.3 distinguishes the three outcomes visually. Table 24.2 then states what continues in each case.

Three different outcomes within the proposal



The foundation continues in all three cases; the personal outcomes differ.

Figure 24.3. Three different outcomes for an individual in the proposed account. Renewing support can maintain the same organized life. Replication produces a further individual. Total personal reset removes every recoverable trace of that person's returning history. The continued foundation is common to the three cases; it does not make their consequences for personal identity identical.

Table 24.2. What continues under different scopes of renewal in the proposal.

Case	What continues	Consequence for identity
Renewing support	The person's organized life, while supporting material is renewed.	Personal continuity may remain.
Producing another	A parent contributes to a further organized life.	The offspring is a further individual.
Resetting a person	The foundation, with no recoverable trace of the returning personal history.	The same person's survival is not established by the foundation's persistence.
New formation	Productive conditions available after reset.	A new life need not recover an earlier biography.

24.7 Why Consciousness Matters to Renewal

Why call consciousness Nature's replicator at all? Why not say that patterns form, persist, and disappear, and leave consciousness out of the account? The answer depends on the earlier argument that experience belongs to the nature of the foundation. If that argument is accepted, the productive process is not an entirely nonconscious mechanism that somehow acquires an unrelated spectator. It is the activity of a reality whose nature includes experience.

This does not mean a separate awareness must push the circulation from outside. Such a picture would divide the account into a machine and an unexplained operator. In DEMU, the organized activity described from outside and its experiential character belong to the same reality. Consciousness is not a second supply added after the activity has already done everything. It names the inner character of what is active.

The proposal still needs to distinguish conscious Nature in this broad sense from deliberate individual action. A person can act for a reason, revise a plan, and care about the result. I do not assign all these capacities to every small structure in the model. Their development depends on the organization of particular lives. Fundamental experience and reflective human agency are connected within the proposal, but they are not identical descriptions.

At the human level, the productive role becomes especially concrete. A person imagines a possibility, tests it through action, encounters a result, and changes what they do. A builder can picture a shelter before assembling it. A musician can try a phrase, hear it, and alter it. The account must accommodate these ordinary relationships between experience, reasons, and the formation of things. It should not make a person's experience irrelevant to everything the person accomplishes.

This is not an argument that imagined objects automatically become physical objects. The builder still needs materials and appropriate action. The musician must develop skills and work through an instrument or voice. Conscious participation takes place through the conditions of a life. The relevance to DEMU is that creation within the world can be understood as organized activity whose experienced and publicly observable aspects belong together.

Replication also becomes more interesting when it is not limited to producing exact copies. A person can pass on a practice that another person understands and changes. A child can inherit a language and say something that no parent has said. These examples involve retained histories and therefore belong to the continuing temporal world. They show how local inheritance can support novelty without making novelty a return from an erased record.

The new contribution can be meaningful because it is experienced and assessed. A better shelter matters to someone who was cold. A clearer explanation matters to someone who was confused. A loving action matters to the person who receives it. The productive process is not evaluated solely by counting the number of structures it generates. What those structures make possible for conscious life is part of the account of their significance.

Here the earlier discussion of balance needs a distinction. The foundational balance concerns the condition through which reality continues without external support in DEMU. A balanced human life concerns such things as health, relationships, freedom, and the ability to respond to difficulties. These meanings are related within my interpretation, but they are not interchangeable. A stable physical arrangement can still be part of a painful life.

Consequently, an ethical conclusion needs an ethical reason. I cannot show that an institution is good merely by calling it stable, or that a choice is wrong because it changes an established pattern. Some harmful arrangements persist for a long time. My argument connects value to the lives affected: their capacity to experience, suffer, understand, and

flourish. Stability matters when it supports those lives, not simply because it resists alteration.

This gives love a definite place in the productive account. Love can be a way of attending to another centre of experience as valuable in its own right. It can help sustain that person's possibilities without absorbing them into one's own purposes. In this sense, conscious Nature becomes productive through relationships that preserve difference. Two people need not become one person for their relationship to bring something new into the world.

The idea also changes how cosmic sex should be understood at the human level. The metaphor cannot make reproduction a duty for every person. Generating and sustaining valuable forms of life includes care, teaching, friendship, discovery, and the repair of damaged relationships. These activities do not all replicate an organism, and I should not pretend they do. They express the wider claim that conscious beings can participate in Nature's productive activity in many ways.

Free will adds a further question: can an individual contribute to what happens through their own understanding and reasons? Calling Nature conscious does not settle every part of that debate. Nor would unexplained randomness by itself give a person control. The account developed in Chapter 19 must connect freedom to an organized subject who can respond to reasons. The present chapter identifies why that subject matters to the productive whole.

If local individuals were only interchangeable appearances with no significance of their own, their contribution would be difficult to distinguish from the movement of any other pattern. But if their perspectives are real and their choices belong to how the world develops, then their individuality matters. The whole is expressed through distinct lives whose encounters shape further possibilities. The replicator proposal can therefore include participation rather than merely multiplication.

This does not imply that every outcome is chosen wisely or that suffering is secretly deserved. A world containing limited perspectives can contain error, conflict, and failed cooperation. The claim that all lives share a foundation does not make each person responsible for every event in the whole. Responsibility depends on the capacities, knowledge, and opportunities of the individual concerned. A spiritual interpretation should keep these ordinary distinctions visible.

The idea of renewal then becomes practical without losing its philosophical meaning. Within a life, renewal may involve learning, recovering, repairing, or beginning a different project. These changes preserve enough history for the individual to continue. They are not the total reset of the core. Their relationship to foundational renewal is one of dependence and expression, not a claim that every fresh start erases the universe's records.

The contribution of consciousness is therefore twofold. In the metaphysical proposal, it is the inner nature of the activity through which forms arise. In developed individual lives, it also involves the experience and assessment of possibilities. These are different levels of the account. Together they explain why I describe Nature as a productive conscious reality, and why the lives it supports should be understood as more than units in a numerical increase.

24.8 What the Argument Establishes

The argument of this chapter begins where the earlier chapters leave us. Within DEMU, the foundation is timeless, active, and conscious. It does not receive its first motion from an earlier event. The universe of changing forms belongs to its ongoing expression. This chapter has asked what it means for that expression to produce further organized lives, and how those lives can relate to a foundation that resets returning histories completely.

The answer needs several connected claims. First, new local forms need not be new foundations. A further individual can arise within the same reality. Second, a continuing foundation need not preserve every form it has supported. Third, the formation of a new individual and the continuation of an existing individual are different relations. These distinctions let the proposal describe production, survival, and erasure without using one word for all three.

The claim about replication is strongest when its conditions are explicit. There must be something that contributes to the production, something further that is produced, and an account of the relevant relationship between them. If the product is described as an individual, its organization and continuity must also be explained. If it is described as conscious, the account must connect that organization with a lived perspective. A geometric resemblance alone cannot complete these tasks.

The chapter therefore proposes a sequence of dependence. The foundation supports organized activity. Under suitable conditions, that activity supports local centres that can

continue and interact. Some interactions contribute to the formation of further centres. Their experiences and relationships can then contribute to additional forms within the temporal world. This is a proposed structure of explanation, not a claim that every step has already been experimentally demonstrated.

An important question is whether eternal activity must produce many individuals. Activity alone does not settle that point. A description could contain a continuing process without showing that it ever generates a second organized centre. To reach the productive claim, DEMU must include the relationships through which further centres become possible. Chapter 15 began that task with geometry and a conditional model of persistence; the present chapter explains what successful production would have to mean.

This identifies the additional contribution of the replicator idea. It is a claim about the character of the activity, not merely a new name for activity in general. Nature's circulation is proposed to support the continuing formation of differentiated lives. The proposal becomes informative when it specifies which relations permit that formation, what allows a form to persist, and how it can contribute to further formation. Those questions can guide later development of the model.

The use of consciousness is equally specific. If I removed the earlier argument for a conscious foundation, replication alone would not restore it. A process might produce further patterns without the description explaining any experience. My claim depends on joining the productive account to the earlier account of the nature of reality. The chapters support different parts of the whole argument; none should be treated as a substitute for all the others.

The reset condition supplies a firm restriction. A new centre cannot inherit a history that has already been erased everywhere. If it receives information from a continuing parent, that parent's relevant history has been retained during the transfer. If it forms after a total reset, the explanation must concern the productive conditions then available. This is how the account can contain both ordinary inheritance and total erasure without confusing them.

The distinction also clarifies the limits of a claim about immortality. The foundation's eternity supports the possibility of continuing expression. It does not, without a further relation of identity, establish the survival of every person through the destruction of every trace of their life. The more direct route considered here is the maintenance of an individual

through renewal while their organized continuity remains. That possibility fits the model without requiring erased information to return.

There are useful consequences for future investigation. A model of local production should allow us to identify when a disturbance becomes a maintained organization and when that organization produces another. A model of individuality should explain why some interactions support one perspective while others remain relations between distinct perspectives. A model of reset should make clear which histories are erased and why no record of them remains elsewhere. These are concrete tasks rather than a request to repeat the entire argument from the beginning.

The results of such work could also require revision. If the proposed interactions never support further organizations under the specified conditions, the account of replication would need to change. If a purported reset preserves recoverable historical distinctions, it would not be the total reset described here. If the criteria for a subject classify every arbitrary collection as one mind, they would fail to explain the individuality that motivated them. A developed proposal should make these differences visible.

At the philosophical level, the chapter offers a way to bring several earlier themes into one account. Cosmic sex concerns productive relationships. Individuality concerns the local unity of experience. Replication concerns the formation of further organized expressions. Renewal concerns the continued possibility of such expression, including after histories have ended. Consciousness supplies the proposed inner character of the reality in which all these relations occur.

The phrase “Nature’s replicator” is therefore deliberately broader than the biological use of the term. It names the productive conscious reality through which particular replicating processes become possible. Its value depends on keeping that broader meaning clear. Genes can copy, organisms can reproduce, and people can create without any of them becoming independent foundations. Their different activities belong to the one world whose character DEMU seeks to explain.

Figure 24.4 maps the main connections in the replicator argument.

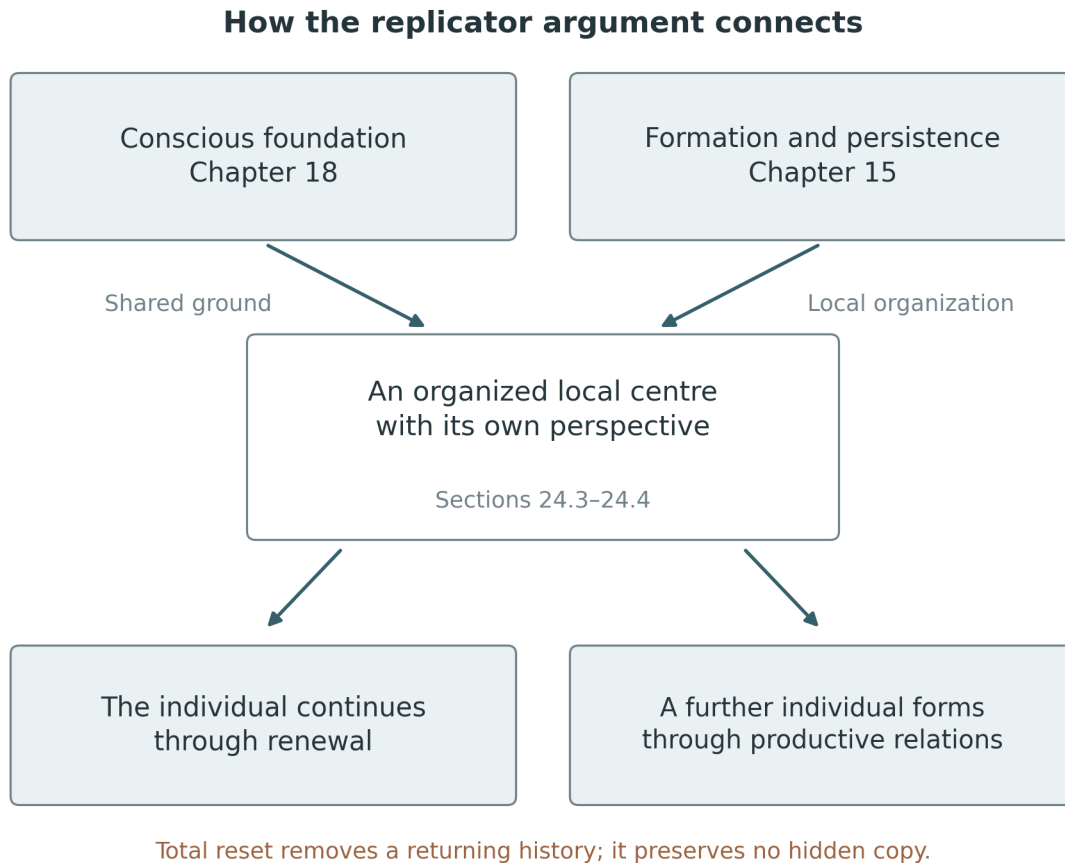


Figure 24.4. Connections in the replicator argument. The accounts of a conscious foundation and local formation meet in the proposed centre of experience. The lower branches distinguish continuing as the same individual from producing another. Arrows show proposed explanatory connections, not a timeline.

I can now return to the image of the great whirlpool and the smaller whirlpools within it. The image is useful because it allows one continuing activity and many local forms to be pictured together. The philosophical work consists in explaining their relationships: what forms, what continues, what contributes to further formation, and what is finally erased. The image becomes an account only when those questions receive distinct answers.

On the interpretation developed here, Nature's eternity is expressed through its capacity for ongoing life and formation, while each individual is a particular way that activity is lived. The total reset removes returning histories; it does not preserve them under another name. The continuing world allows lives to develop, cooperate, and contribute to further lives. This is the sense in which I propose consciousness as Nature's replicator, and the sense in which the chapter prepares the final conclusion.

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CHAPTER 25

The Conclusion

25.1 The Full Circle

I set out to answer the question: *Can a global solution to infinite regress give humanity a new source of energy?*

To answer that question, I needed to produce a complete metaphysical account of reality. By “complete,” I mean an account that allows me to stand back and understand Nature as a whole. Did I accomplish that?

Yes.

If you ask me, then yes. Let us briefly review what I mean by a complete understanding of Nature.

I understand why Nature is here. Nature is here because nonexistence is a fiction produced by language. I can see how the evolution of cave paintings led to the formation of the concept of nonexistence.

But establishing that Nature is here still does not answer the question of what it is doing.

I also understand that paradoxes arise from our manipulation of symbols. Like nonexistence, they are products of language, not things Nature can embody. Nature must therefore be coherent, which means it cannot embody infinite regress.

What Nature is and what Nature is doing therefore amount to the same thing: being coherent and maintaining coherence.

Being coherent means that Nature must achieve closure. And I understand that only consciousness can fulfil that role.

I further understand that closure means the perpetual creation and recreation of balance.

This balance must be spiritual rather than mechanical, because consciousness is sensory and cannot be reduced to atoms. Thus, the purpose of existence cannot be merely a description; it must be a lived reality. The universe is here because it has a job to do (balance), rather than simply existing as a brute fact.

Finally, I would argue that this lived reality is ultimately love. In the long run, existence is about the game of balance, and love is what accomplishes that.

I can also see how free will connects to all of this, because love must be chosen. I understand this spiritually, and I have now argued for it philosophically.

The skeptic may still disagree with me, but he cannot call my metaphysical system incomplete in the sense I mean here: I can now stand back, look at the whole of Nature, and understand my role, my own immortality through balance, and why we all had to be immortal.

I can also understand the roles of the two things I dread: suffering and death.

Death is your friend—the very thing that makes you immortal by resetting even entropy. Even entropy cannot survive death.

Suffering is an alarm telling you that you are no longer walking the path of balance. It is more than that, but you get the picture.

But if I am right, then consciousness, rather than the gene, must be Nature's replicator.

If consciousness does not play that central role, then my picture falls apart.

So yes, I can understand why we are conscious!

And if that is not a complete account, then I do not understand what "complete" means. I could be more humble, of course, and infinitely many details may be missing from my account, but I do not think my overarching picture is incorrect.

But I know what the skeptic will reply. Because he cannot defend nonexistence, paradox, or inactivity, he will turn to brute fact in order to save his worldview.

He will think he is clever and ask: "What makes a brute fact incoherent?"

I have already said what: it supplies nothing of explanatory value. A skeptic cannot simply appeal to a brute fact; he must actually answer the question rather than stop at, "It is just a brute fact."

You cannot invoke a brute fact to argue that an infinite regress is coherent and then fail to give me a reason why that regress is coherent. If that is not inconsistent, then nothing is!